

Sectoral Labor Reallocation, Inflation and Monetary Policy

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Abstract

We study how frictional labor mobility across sectors shapes inflation dynamics and monetary policy in a multi-sector New Keynesian economy, where workers' dynamic discrete-choice decisions govern sectoral mobility. We derive analytical expressions for welfare and a generalized Phillips curve and show that imperfect labor reallocation influences sectoral inflation through two sufficient statistics: a mobility elasticity and the steady-state labor transition matrix. Using a spectral decomposition, we show that the persistence of sectoral inflation is governed not by average price rigidity or aggregate labor mobility alone, but by the price rigidity of the sectors that workers move to and from. When labor reallocation across sticky sectors is imperfect, the effects of sectoral shocks on these sectors become highly persistent. In response, monetary policy balances stabilization of standard aggregate output gaps with persistent labor and price misallocation.

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1 Introduction

The multi-sector New Keynesian model is the canonical framework for studying how sectoral heterogeneity—in price rigidity, in input-output linkages, and in the shocks that hit different parts of the economy—shapes inflation and monetary policy. Yet, on the labor margin, the framework often relies on frictionless benchmarks of a unified market in which a representative worker moves freely across sectors to equalize wages, or segmented markets that abstract from reallocation entirely. In the data, sectoral labor flows exist, are small relative to employment stocks, structured by a network of bilateral reallocation flows, and slow to respond to changes in sectoral wages. Answering how monetary policy should respond to shocks requiring sectoral reallocation requires a framework in which nominal rigidity and the speed of labor adjustment are modeled jointly.

Frictional labor mobility introduces a real rigidity that operates simultaneously with the nominal friction of sticky prices. Because this nominal rigidity varies across sectors, shifts in aggregate demand affect sectoral prices unevenly. As a result, monetary policy does more than determine the aggregate price level: it shapes the relative prices that guide how workers reallocate across sectors. This generates a policy trade-off: if the central bank aggressively stabilizes inflation, it may suppress the relative price signals required to move workers toward expanding activities. Conversely, accommodating these relative price adjustments to facilitate efficient reallocation can generate persistent sectoral inflation. Which force dominates depends on the interaction of nominal rigidities and the labor flow network.

To formalize this interaction, we embed a frictional worker-reallocation block into a canonical multi-sector New Keynesian economy and ask how the sticky prices and sluggish worker flows interact to shape the transmission of sectoral shocks, the persistence of inflation, and optimal monetary policy. On the nominal side, sectors differ in price rigidity; on the real side, workers choose sectors based on forward-looking continuation values, subject to bilateral mobility costs and idiosyncratic preference shocks (Artuç et al., 2010). The model nests the standard multi-sector New Keynesian economy with integrated labor markets (e.g., Aoki, 2001) when mobility is frictionless, and the multi-sector economy with segmented labor markets when mobility costs are prohibitive. Despite the richness of the underlying micro structure, the linearized equilibrium depends on the reallocation block only through two sufficient statistics: a mobility elasticity ϕ that governs the responsiveness of worker flows to sectoral value differentials, and the steady-state transition matrix F that summarizes the pattern of gross worker flows across sector pairs.

In Section 3, we derive a generalized Phillips curve: a recursive mapping from aggregate demand and sectoral productivity to sectoral inflation, with a state tracking inherited price misalignment and employment misallocation. The eigenvalues of the associated transition matrix govern the persistence of sectoral inflation. Under homogeneous price stickiness, aggregate

demand shifts marginal costs uniformly, so the system decouples into an aggregate block that monetary policy controls and a relative block that it cannot. At the polar cases, impact and persistence move in opposite directions: segmented markets produce a larger initial inflation response but faster subsequent convergence, because the lack of marginal cost attenuation from worker inflows make firms reprice more aggressively.

In general, the eigenvalues of the transition matrix are implicit functions of the full parameter vector and have no closed-form expressions. We obtain analytical results by imposing a mild symmetry restriction on reallocation costs. Under uniform price stickiness, the system decomposes into independent components, each pairing a Phillips curve with a labor reallocation equation. The persistence of each component is governed by the interaction between nominal rigidity and the speed of labor reallocation along that dimension, with the slower-adjusting margin acting as the bottleneck.

Allowing price stickiness to vary across sectors breaks this independence but yields two key insights. First, when sectors that adjust prices slowly also sit along slow-moving reallocation dimensions, nominal and real frictions compound and adjustment stalls. Second, monetary policy has leverage over reallocation only when sticky- and flexible-price sectors load on opposite sides of a component, so that aggregate demand can tilt relative prices along it.

We formalize the optimal policy response as a dynamic Ramsey problem (Section 5). The second-order welfare loss penalizes aggregate output gaps, sectoral inflation dispersion, relative price distortions, and cross-sector employment misallocation—the last of which is new to our setting and compounds over time when labor reallocation is slow. As in any multi-sector model, the central bank faces a fundamental constraint: a single instrument can move sectoral prices only along the direction of Phillips curve slopes, which generically differs from the direction of efficient relative price adjustment.

The optimal targeting criterion nests the textbook rule when price stickiness is uniform but introduces a misallocation wedge when it is not. The wedge is governed by the interaction between the two diagnostic statistics: effective rigidity determines whether misalignment along a reallocation component is persistent enough to warrant intervention, and the demand transmission determines whether aggregate demand has leverage to correct it. When both are large along the same component, the planner actively tilts demand to redirect worker flows—tightening when efficient reallocation requires outflows from sticky sectors, loosening in the reverse case.

We bring the model to U.S. sectoral data (Section 6), estimating the worker transition matrix from industry-to-industry worker flows in the CPS and measuring sectoral price rigidity from micro price data. We evaluate the alignment between the labor flow network and the distribution of nominal rigidities to assess the quantitative relevance of these channels.

Related Literature

Our paper is most closely related to work on monetary policy during structural reallocation. [Guerrieri et al. \(2021\)](#) show in a two-sector model with downward wage rigidity that sectoral reallocation can shift optimal policy in either direction: contractionary when easing slows reallocation, expansionary when easing accelerates it. [Ferrante et al. \(2023\)](#) study the interaction of inflation and labor reallocation in a multi-sector model with convex hiring costs, showing that reallocation costs account for a large share of post-pandemic U.S. inflation; [Bengui et al. \(2025\)](#) characterize how the phase of reallocation shapes monetary policy transmission. We generalize the mechanisms and policy insights by micro-founding sectoral labor reallocation through workers' forward-looking sector choice, deriving sufficient statistics computable from worker flow data and sectoral price adjustment frequencies that determine both when reallocation frictions amplify the persistence of inflation and when they affect optimal policy.

We also contribute to the study of optimal monetary policy in multi-sector New Keynesian economies. Optimal stabilization puts more weight on sticky-price sectors ([Aoki, 2001](#); [Benigno, 2004](#)), a principle that extends to heterogeneous rigidities and production networks ([Pasten et al., 2020](#); [La'O and Tahbaz-Salehi, 2022](#); [Rubbo, 2023](#); [Afrouzi and Bhattarai, 2025](#)). In much of this literature, labor markets are either perfectly integrated or exogenously segmented.¹ We study the intermediate case of imperfect sectoral mobility and show that it generates propagation channels and policy wedges that are absent at either polar benchmark.

In our model, imperfect sectoral labor mobility is a source of real rigidity, connecting our framework to the literature on inflation dynamics and the Phillips curve. Nominal rigidities generate large macroeconomic effects when accompanied by real rigidities ([Ball and Romer, 1990](#); [Woodford, 2003](#)), with wage rigidity serving as a prominent microfoundation ([Blanchard and Galí, 2007](#)); the empirical Phillips curve is quite flat ([Hazell et al., 2022](#)), consistent with substantial real rigidity. In our setting, a different labor market friction—imperfect sectoral reallocation—generates persistent endogenous cost-push dynamics: inherited employment misallocation shifts sectoral marginal costs, and these shifts decay slowly because the same frictions that created them also delay their correction.

The magnitude of this mechanism depends on the distribution of nominal frictions across sectors. Micro evidence documents substantial heterogeneity in price adjustment frequencies ([Bils and Klenow, 2004](#); [Nakamura and Steinsson, 2008](#)), and this heterogeneity has become central to understanding inflation persistence ([Altissimo et al., 2009](#); [Pasten et al., 2020](#); [Carvalho et al., 2021](#); [Pastén et al., 2024](#)). We show that the interaction between this nominal heterogeneity and the topology of worker flows—which sectors are connected, and how quickly workers move between them—determines which dimensions of misalignment are persistent and which

¹Recent work incorporates limited labor adjustment via imperfect two-sector mobility ([Cantelmo and Melina, 2023](#)) or convex hiring costs ([Ferrante et al., 2023](#)). Instead, we model worker reallocation as an explicit dynamic transition system that shapes the Phillips curve and welfare.

matter for aggregate inflation.

Our micro-foundation for frictional labor reallocation builds on the literature on forward-looking sector choice, originating from [Lilien \(1982\)](#) and formalized via dynamic discrete choice by [Artuç et al. \(2010\)](#) with general equilibrium applications in [Caliendo et al. \(2019\)](#); [Allen et al. \(2020\)](#); [Seo and Oh \(2025\)](#); [Kim et al. \(2026\)](#). We embed this micro-foundation into a New Keynesian model and show that the reallocation block enters the Phillips curve and welfare loss through our sufficient statistics. Methodologically, we adapt spectral decomposition techniques used in spatial migration ([Kleinman et al., 2023](#)) and input-output production networks ([Liu and Tsyvinski, 2024](#)) to characterize the dynamics of cross-sector worker flows.

2 Model

We build a multi-sector New Keynesian model in which sectors differ in price rigidity and workers reallocate across sectors through a dynamic discrete choice problem. We describe the production side, household preferences, and the labor reallocation problem, then state the equilibrium and its log-linear approximation.

Time is discrete, $t = 0, 1, \dots$. The economy consists of N sectors indexed by $i, j \in \mathcal{N} = \{1, \dots, N\}$.

2.1 Production and Pricing

Final and sectoral aggregation. The economy produces a final good Y_t which is a Cobb-Douglas aggregate of sectoral bundles $\{Y_{it}\}_{i \in \mathcal{N}}$:

$$Y_t = \prod_{i \in \mathcal{N}} (Y_{it} / \alpha_i)^{\alpha_i}, \quad \sum_{i \in \mathcal{N}} \alpha_i = 1, \quad (1)$$

with associated consumer price index $P_t^C = \prod_i P_{it}^{\alpha_i}$, where P_{it} is the price index of sector i . Each sectoral bundle is a CES aggregate of differentiated varieties $f \in [0, 1]$ with elasticity of substitution $\epsilon_i > 1$:

$$Y_{it} = \left(\int_0^1 Y_{it}(f)^{\frac{\epsilon_i-1}{\epsilon_i}} df \right)^{\frac{\epsilon_i}{\epsilon_i-1}},$$

yielding demand for variety f and the sectoral price index:

$$Y_{it}(f) = \left(\frac{P_{it}(f)}{P_{it}} \right)^{-\epsilon_i} Y_{it}, \quad P_{it} = \left(\int_0^1 P_{it}(f)^{1-\epsilon_i} df \right)^{\frac{1}{1-\epsilon_i}}.$$

Technology. Each firm f in sector i operates a linear production technology:

$$Y_{it}(f) = A_{it}L_{it}(f), \quad (2)$$

where A_{it} is sector-specific productivity and $L_{it}(f)$ is labor input. Nominal marginal cost is $MC_{it} = W_{it}/A_{it}$, where W_{it} is the nominal wage in sector i .

Pricing. Firms set prices subject to Calvo frictions. In each period, a fraction $1 - \delta_i \in (0, 1)$ of firms in sector i can reset their prices. The government provides a revenue subsidy of rate $\frac{1}{\epsilon_i - 1}$ to each sector, financed by lump-sum taxes, that offsets the monopolistic markup and ensures the flexible-price steady state is efficient. Under this subsidy, a re-optimizing firm chooses $P_{it}^*(f)$ to maximize:

$$\max_{P_{it}^*(f)} \mathbb{E}_t \sum_{s=0}^{\infty} \delta_i^s Q_{t,t+s} \left(\frac{P_{it}^*(f)}{P_{it+s}} \right)^{-\epsilon_i} Y_{it+s} \left[\frac{\epsilon_i}{\epsilon_i - 1} P_{it}^*(f) - MC_{it+s} \right], \quad (3)$$

where $Q_{t,t+s} = \beta^s (C_{t+s}/C_t)^{-\gamma} (P_t^C/P_{t+s}^C)$ is the stochastic discount factor.

2.2 Households

A representative family consists of a unit mass of workers distributed across sectors. Let N_{it} denote the fraction of workers in sector i at date t , with $\sum_{i \in \mathcal{N}} N_{it} = 1$. Each worker in sector i supplies L_{it} hours during period t . The distribution $\{N_{it}\}$ evolves through a reallocation process described in Section 2.3.

Preferences. The family maximizes:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \sum_{i \in \mathcal{N}} N_{it} \zeta_i \frac{L_{it}^{1+\varphi}}{1+\varphi} - \Psi_t \right], \quad (4)$$

where $\beta \in (0, 1)$ is the discount factor, $\gamma > 0$ is the inverse of the elasticity of intertemporal substitution, $\varphi > 0$ is the inverse Frisch elasticity, $\zeta_i > 0$ are sector-specific disutility shifters, and Ψ_t is the cost of labor reallocation incurred in period t (defined in Section 2.3).

Budget constraint. The family faces the nominal budget constraint:

$$P_t^C C_t + B_t \leq (1 + i_{t-1})B_{t-1} + \sum_{i \in \mathcal{N}} W_{it} L_{it} N_{it} + \Pi_t + T_t, \quad (5)$$

where B_t are one-period nominal bonds paying gross interest $1 + i_t$ set by the central bank, Π_t are aggregate profits, and T_t are net lump-sum transfers from the government.

Optimality conditions. Maximizing (4) subject to (5), taking $\{N_{it}\}$ as given, yields the intratemporal labor supply condition for each sector i :

$$\frac{W_{it}}{P_t^C} = \zeta_i L_{it}^\varphi C_t^\gamma, \quad (6)$$

and the intertemporal Euler equation:

$$1 = \beta(1 + i_t) \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+1}^C} \right]. \quad (7)$$

2.3 Labor Reallocation

We model labor mobility as a family-level choice: at the start of period t , the representative household reallocates workers across sectors subject to reallocation costs Ψ_t .²

Reallocation cost. In each period t , the family incurs a labor reallocation cost

$$\Psi_t = \sum_{i \in \mathcal{N}} N_{i,t-1} \psi_i(\{\mu_t^{ij}\}_{j \in \mathcal{N}}),$$

where $N_{i,t-1}$ is the inherited worker distribution and μ_t^{ij} are the chosen transition probabilities from i to j satisfying $N_{jt} = \sum_i N_{i,t-1} \mu_t^{ij}$. We adopt the following functional form for ψ_i , standard in the dynamic discrete choice literature (Anderson et al., 1988; Artuç et al., 2010; Caliendo et al., 2019):

$$\psi_i(\{\mu_t^{ij}\}_j) = \sum_{j \in \mathcal{N}} \mu_t^{ij} \tau^{ij} + \frac{1}{\theta} \sum_{j \in \mathcal{N}} \mu_t^{ij} \log \mu_t^{ij},$$

where $\tau^{ij} \geq 0$ are bilateral reallocation costs ($\tau^{ii} = 0$). This penalty is equivalent to a setting in which individual workers draw i.i.d. Type-I Extreme Value preference shocks with dispersion $1/\theta$ (Anderson et al., 1992). The parameter $\theta > 0$ governs the elasticity of labor reallocation: as $\theta \rightarrow 0$, labor markets become segmented; conversely, as $\theta \rightarrow \infty$, workers move frictionlessly to equalize wages across sectors, recovering a canonical one-labor-type New Keynesian economy.

Value functions and optimal flows. Let V_{it} denote the marginal continuation value to the family of an additional worker located in sector i at date t . The family optimally sets transition rates to equate the marginal cost of moving workers to the realized value differentials $\{V_{jt}\}$.

²The timing within period t is as follows: (1) the period begins with predetermined employment $\{N_{i,t-1}\}$ and prices $\{P_{i,t-1}\}$; (2) aggregate and sectoral shocks at time t are realized; (3) all period- t choices (labor reallocation $\{N_{it}\}$, Calvo price setting, labor supply L_{it} , and consumption) are determined contemporaneously.

The cost function ψ_i yields transition probabilities of the logit form:

$$\mu_t^{ij} = \frac{\exp(\theta[V_{jt} - \tau^{ij}])}{\sum_{j' \in \mathcal{N}} \exp(\theta[V_{jt} - \tau^{ij'}])}. \quad (8)$$

Substituting these optimal flows back into the family's objective yields the Bellman equation:

$$V_{it} = U_{it}^{\text{flow}} + \beta \mathbb{E}_t \left[\frac{1}{\theta} \log \left(\sum_{j \in \mathcal{N}} \exp(\theta[V_{j,t+1} - \tau^{ij}]) \right) \right], \quad (9)$$

where the flow utility from working in sector j at time t is $U_{jt}^{\text{flow}} \equiv \frac{W_{jt}}{P_t^C} C_t^{-\gamma} L_{jt} - \zeta_j \frac{L_{jt}^{1+\varphi}}{1+\varphi}$. Using the intratemporal labor supply condition (6), this simplifies to:

$$U_{jt}^{\text{flow}} = \zeta_j \frac{\varphi}{1+\varphi} L_{jt}^{1+\varphi}.$$

Steady-state worker flows. Let F denote the *forward* transition matrix, where $F_{ij} = \bar{\mu}^{ij}$ is the steady-state probability that a worker in sector i moves to sector j . The stationary employment distribution satisfies $\sum_i \bar{N}_i F_{ij} = \bar{N}_j$. We define the *backward* transition matrix B by

$$B_{ji} \equiv \frac{\bar{N}_i}{\bar{N}_j} F_{ij}. \quad (10)$$

While F describes where workers go, B describes where they came from: B_{ji} gives the fraction of workers currently in sector j who arrived from sector i . Both matrices are row-stochastic and share the same stationary distribution $\bar{\mathbf{N}}$. Because $\bar{\mathbf{N}}$ is itself determined by F , the backward matrix B is fully pinned down by F .

2.4 Equilibrium

Market clearing. In equilibrium, the goods market for each variety and the aggregate economy must clear:

$$Y_{it}(f) = C_{it}(f), \quad Y_t = C_t, \quad (11)$$

where C_t is household consumption. Combined with Cobb–Douglas demand, this implies that nominal sectoral output equals the expenditure share of total nominal consumption:

$$P_{it} Y_{it} = \alpha_i P_t^C C_t. \quad (12)$$

Labor market clearing in each sector requires:

$$\int_0^1 L_{it}(f) df = N_{it} L_{it}, \quad (13)$$

where N_{it} is the post-reallocation employment distribution. The government finances the output subsidies through lump-sum taxes, balancing its budget each period:

$$T_t + \sum_{i \in \mathcal{N}} \frac{1}{\epsilon_i - 1} P_{it} Y_{it} = 0. \quad (14)$$

The central bank sets the nominal interest rate i_t according to a policy rule specified below.

Definition 1 (Equilibrium). *Given initial conditions $(\{P_{i,-1}\}, \{N_{i,-1}\})$ and a sequence of productivity shocks $\{A_{it}\}$, an equilibrium is a collection of prices $\{P_{it}(f), P_{it}, P_t^C, W_{it}, i_t\}$, quantities $\{C_t, Y_{it}, L_{it}(f), L_{it}, B_t\}$, employment shares $\{N_{it}\}$, transition probabilities $\{\mu_t^{ij}\}$, and value functions $\{V_{it}\}$ such that: (i) households optimize consumption, savings, labor supply, and reallocation; (ii) firms optimize pricing subject to Calvo constraints; (iii) all markets clear and the government budget constraint (14) holds; and (iv) the central bank follows its policy rule.*

Steady state. We impose that steady-state flow surplus is equalized across sectors. This ensures an interior steady state: without it, the frictionless limit ($\theta \rightarrow \infty$) would concentrate all workers in the highest-surplus sector. We achieve this by normalizing $\bar{L}_i = \bar{L}$ and $\zeta_i = \zeta$ for all i .³ Combining steady-state labor supply (6) with goods market clearing (12) gives $\alpha_i \bar{C}^{1-\gamma} = \zeta \bar{N}_i \bar{L}^{1+\varphi}$, so $\bar{N}_i = \alpha_i$: steady-state employment shares equal expenditure shares.

2.5 Linearized Equilibrium

Log-linearization of the equilibrium conditions around a zero-shock steady state gives a system that organizes into three groups: cost determination, pricing, and demand. The pricing and demand blocks are standard; the cost determination block is where frictional labor reallocation enters the system.

Notation. Throughout the analysis, we compare the sticky-price and flexible-price equilibria using the following notation. Vectors are set in bold. Lowercase letters denote log-deviations from the steady state ($x_t \equiv \log X_t - \log \bar{X}$), except for interest rates (i_t, r_t^{nat}) which are in levels. The superscript nat denotes flexible-price natural levels (Lemma 1), and tildes denote gaps from this benchmark ($\tilde{x}_t \equiv x_t - x_t^{\text{nat}}$). For sectoral variables, hats denote log-deviations relative

³The normalization $\bar{L}_i = \bar{L}$ is without loss of generality: N_{it} measures the mass of workers in sector i , and L_{it} can be interpreted as either per-worker hours or the fraction of workers who supply labor (Galí, 2011). The condition $\zeta_i = \zeta$ then ensures equal surplus.

to the aggregate economy. For example, relative prices and wages are $\hat{p}_{it} \equiv p_{it} - p_t^C$ and $\hat{w}_{it} \equiv w_{it} - p_t^C$, while relative productivity is $\hat{a}_{it} \equiv a_{it} - \sum_j \alpha_j a_{jt}$. Finally, $\stackrel{(1)}{=}$ and $\stackrel{(2)}{=}$ denote first- and second-order approximations, and $=$ denotes an exact identity.

2.5.1 Real Marginal Cost and Labor Reallocation

Relative prices and labor reallocation jointly determine real marginal cost. Relative prices follow the law of motion:

$$\hat{p}_{it} = \hat{p}_{it-1} + \pi_{it} - \pi_t^C. \quad (15)$$

Log-linearizing nominal marginal cost $MC_{it} = W_{it}/A_{it}$ and substituting the labor supply condition (6), the production function (2), and Cobb–Douglas sectoral demand,

$$y_{it} - y_t = -\hat{p}_{it}, \quad (16)$$

we get the real marginal cost for sector i :

$$mc_{it} - p_{it} \stackrel{(1)}{=} (\gamma + \varphi) y_t - (1 + \varphi)(\hat{p}_{it} + a_{it}) - \varphi n_{it}. \quad (17)$$

Three terms drive sectoral cost pressures: aggregate demand y_t , the relative price–productivity gap $\hat{p}_{it} + a_{it}$, and current employment n_{it} . The third is the novel reallocation channel: an overstaffed sector ($n_{it} > 0$) has lower hours per worker and hence lower marginal cost, while understaffing creates upward cost pressure.

The employment state n_{it} is governed by the labor reallocation block. Normalizing value and utility deviations by $\varphi\zeta\bar{L}^{1+\varphi}$:

$$\mathbf{n}_t \stackrel{(1)}{=} B \mathbf{n}_{t-1} + \phi (I - BF) \mathbf{v}_t, \quad (18)$$

$$\mathbf{v}_t \stackrel{(1)}{=} \mathbf{u}_t + \beta F \mathbb{E}_t[\mathbf{v}_{t+1}], \quad (19)$$

$$u_{it} \stackrel{(1)}{=} y_t - \hat{p}_{it} - a_{it} - n_{it}, \quad (20)$$

where B, F are the steady-state transition matrices, and $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$ is the *mobility elasticity*. The ratio ϕ/φ isolates the elasticity of transition probabilities to destination wages.⁴

Equation (18) is the law of motion for employment, where the matrix B governs mean reversions, while current value $\mathbf{v}_t$ induce flows to sectors. In the frictionless limit $\phi \rightarrow \infty$, $(I - BF)\mathbf{v}_t \rightarrow \mathbf{0}$ so value differentials vanish, giving $n_{it} = -(\hat{p}_{it} + \hat{a}_{it})$. Equation (19) is the

⁴Normalized deviations are $v_{it} \equiv (V_{it} - \bar{V}_i)/(\varphi\zeta\bar{L}^{1+\varphi})$ and $u_{it} \equiv (U_{it}^{\text{flow}} - \bar{U}^{\text{flow}})/(\varphi\zeta\bar{L}^{1+\varphi})$. Combining steady-state conditions gives $\phi = \varphi\theta\bar{C}^{1-\gamma}$. Appendix D.2 maps this to the dispersion parameter ν in Artuç et al. (2010); Caliendo et al. (2019), giving $\phi = \varphi/\nu$.

Bellman recursion: iterating forward,

$$\mathbf{v}_t \stackrel{(1)}{=} \sum_{s=0}^{\infty} \beta^s F^s \mathbb{E}_t[\mathbf{u}_{t+s}],$$

so sectoral continuation values are the discounted, reallocation-aware present value of the flow utilities.

2.5.2 Pricing

Given real marginal costs, we log-linearize the firms' optimal reset price under Calvo pricing to obtain the New Keynesian Phillips curve:

$$\pi_{it} \stackrel{(1)}{=} \kappa_i [mc_{it} - p_{it}] + \beta \mathbb{E}_t \pi_{it+1}, \quad (21)$$

where $\kappa_i \equiv (1 - \delta_i)(1 - \beta\delta_i)/\delta_i$ is the slope of sector i 's Phillips curve. The derivation is standard and identical to the one-sector benchmark (Woodford, 2003).

2.5.3 Demand

The IS curve is:

$$\tilde{y}_t \stackrel{(1)}{=} \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma} \left(i_t - \mathbb{E}_t \pi_{t+1}^C - r_t^{\text{nat}} \right), \quad (22)$$

where $\pi_t^C = \sum_i \alpha_i \pi_{it}$ is CPI inflation and $r_t^{\text{nat}} = -\log \beta + \gamma(\mathbb{E}_t y_{t+1}^{\text{nat}} - y_t^{\text{nat}})$ is the natural real rate. The central bank sets the nominal interest rate according to:

$$i_t = \psi_\pi \pi_t(\omega) + \psi_y \tilde{y}_t, \quad (23)$$

for some inflation index $\pi_t(\omega) = \sum_i \omega_i \pi_{it}$ with weights summing to one. Sectoral demand y_{it} follows from Cobb–Douglas preferences, giving Equation (16) used above.

The IS curve (22) and Taylor rule (23) are standard. Richer demand structures (heterogeneous agents, nonhomothetic preferences) would modify (22) and (16) but leave the rest of the cost determination block unchanged.

3 Dynamics of Inflation and Labor Reallocation

This section characterizes the joint dynamics of sectoral inflation and labor reallocation. Section 3.1 shows that the linearized reallocation block depends on only two sufficient statistics: a mobility elasticity and the steady-state worker transition matrix. Section 3.2 uses these to

derive a generalized Phillips curve in which employment misallocation acts as an endogenous cost-push force.

We then characterize the persistence of these dynamics (Section 3.3), showing that the persistence of a sectoral shock depends on its exposure to the labor-flow network rather than the shocked sector's own price rigidity. Finally, to isolate the role of mobility frictions, Section 3.4 contrasts the polar cases of segmented and integrated labor markets, demonstrating an inverse relationship between impact inflation and inflation persistence.

Throughout this section, we condition on a given path of aggregate demand $\{y_t\}$; Section 5 studies the optimal choice of that path.

3.1 Sufficient Statistics and the State

Two features of the reallocation block govern the linearized dynamics: how strongly workers respond to value differentials, and which sectors they flow between. Proposition 1 shows that these are the only features that matter.

Proposition 1 (Sufficient Statistics). *The linearized system depends on the labor reallocation structure only through:*

- (i) *the mobility elasticity $\phi = \varphi\theta\zeta\bar{L}^{1+\varphi}$, and*
- (ii) *the forward transition matrix F (and the backward matrix B derived from it via (10)).*

The limiting cases are: $\phi \rightarrow 0$ (segmented markets), in which $\mathbf{n}_t = \mathbf{0}$ for all t ; and $\phi \rightarrow \infty$ (free mobility), in which continuation values equalize across sectors ($\mathbf{v}_t \propto \mathbf{1}$) and n_{it} offsets the relative price–productivity gap ($n_{it} = -(\hat{p}_{it} + \hat{a}_{it})$) at every date $t \geq 0$.

Proof. See Appendix C.1. □

This parallels findings in dynamic spatial models, where the linearized system similarly reduces to a transition matrix and a mobility elasticity (Kleinman et al., 2023; Seo and Oh, 2025). The $\phi \rightarrow \infty$ limit recovers the standard multi-sector NK model with a single labor type.

Next, we identify the state variables of the linearized system. We first characterize the flexible-price benchmark. We define the natural level at date t as the solution to the flex-price equilibrium conditional on past variables $(\hat{\mathbf{p}}_{t-1}, \mathbf{n}_{t-1})$, with prices flexible from t onward and the reallocation friction held fixed.⁵

Lemma 1 (Natural Allocation). *The flexible-price allocation satisfies:*

⁵Following Woodford (2003), we re-anchor this counterfactual each period. When computing y_{t+1}^{nat} , the predetermined input is the realized sticky-price state $\mathbf{n}_t$. Consequently, the sequence $\{y_t^{\text{nat}}, \mathbf{n}_t^{\text{nat}}, \dots\}_{t \geq 0}$ captures the period-by-period flex-price response to the actual history.

(i) Aggregate natural output and the natural real rate depend only on aggregate productivity:

$$y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \sum_i \alpha_i a_{it}, \quad r_t^{\text{nat}} = -\log \beta + \gamma \mathbb{E}_t(y_{t+1}^{\text{nat}} - y_t^{\text{nat}}),$$

independent of the predetermined labor distribution $\mathbf{n}_{t-1}$.

(ii) Sectoral natural quantities are state-dependent: the predetermined $\mathbf{n}_{t-1}$ yields an endogenous natural employment distribution $\mathbf{n}_t^{\text{nat}}$, which determines sectoral hours and natural relative prices:

$$l_{it}^{\text{nat}} = \frac{1-\gamma}{1+\varphi} y_t^{\text{nat}} - \frac{1}{1+\varphi} n_{it}^{\text{nat}}, \quad (24)$$

$$\hat{p}_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat},0} - \frac{\varphi}{1+\varphi} n_{it}^{\text{nat}}, \quad (25)$$

where $\hat{p}_{it}^{\text{nat},0} = \sum_j \alpha_j a_{jt} - a_{it}$ is the natural relative price when $\mathbf{n}_t^{\text{nat}} = \mathbf{0}$. A naturally-overstaffed sector ($n_{it}^{\text{nat}} > 0$) has lower hours and a lower natural relative price.

Proof. See Appendix C.1. □

Parts (i) and (ii) establish that while aggregate natural output depends only on aggregate productivity, yielding a standard one-sector IS curve (22), sectoral natural quantities depend on the endogenous employment distribution $\mathbf{n}_t^{\text{nat}}$. Labor reallocation therefore enters the system entirely through cost determination.

To characterize the joint price and employment dynamics, we define the *relative price distortion* as the deviation of relative prices from their steady-state benchmark:

$$\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}. \quad (26)$$

Because $\hat{p}_{it}$ and $\hat{a}_{it}$ always appear together in the marginal cost and flow utility, χ_{it} acts as a sufficient statistic for pricing dynamics.⁶ Substituting this into (17), marginal cost becomes:

$$mc_{it} - p_{it} = (\gamma + \varphi) \tilde{y}_t - (1 + \varphi) \chi_{it} - \varphi n_{it}, \quad (27)$$

and from (15), the distortion evolves as

$$\chi_t = \chi_{t-1} + \hat{\pi}_t + \Delta \hat{\mathbf{a}}_t. \quad (28)$$

Thus, two variables summarize the pricing and labor allocation dynamics: price misalignment (χ_t) and employment misallocation ($\mathbf{n}_t$). Because both adjust contemporaneously, their inherited

⁶Equation (25) gives $\hat{p}_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat},0} - \frac{\varphi}{1+\varphi} n_{it}^{\text{nat}}$, so $\hat{p}_{it} - \hat{p}_{it}^{\text{nat}} = \chi_{it} + \frac{\varphi}{1+\varphi} n_{it}^{\text{nat}}$. Defining χ_{it} relative to the steady-state benchmark keeps it a function of observable sticky-price quantities and makes χ_{it} and n_{it} enter additively in the marginal cost equation.

components form the predetermined state, $\mathbf{s}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$. These two margins interact: sticky prices distort the relative-price signals that drive worker transitions, while the resulting misallocation feeds back into marginal cost. The next subsection characterizes the dynamics of this joint adjustment.

3.2 Transition Dynamics and the Generalized Phillips Curve

The marginal cost equation (27), the sectoral pricing rule (21), and the reallocation block (18)–(20) jointly define a *generalized Phillips curve*: a mapping from the paths of aggregate demand and productivity into sectoral inflation,

$$\{\pi_{it}\}_{i,t} = \mathcal{K}(\{y_t\}_t, \{a_{it}\}_{i,t}). \quad (29)$$

This mapping admits a recursive representation in the state $\mathbf{s}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$: price misalignment and employment misallocation inherited from the previous period.⁷ We write $\text{Cov}_\alpha(x_i, y_i) \equiv \sum_i \alpha_i (x_i - \bar{x})(y_i - \bar{y})$, with $\bar{x} = \sum_i \alpha_i x_i$, for the α -weighted covariance.

Proposition 2 (Generalized Phillips Curve). *Under determinacy, the mapping $\mathcal{K}$ in (29) admits a recursive representation:*

(i) **Transition.** *The state evolves as*

$$\mathbf{s}_{t+1} = \mathbf{P} \mathbf{s}_t + \mathbf{b}_t^y + \mathbf{b}_t^a, \quad (30)$$

with $\mathbf{b}_t^y = \sum_{j \geq 0} \mathbf{H}_j^y \mathbb{E}_t[\tilde{y}_{t+j}]$, $\mathbf{b}_t^a = \sum_{j \geq 0} \mathbf{H}_j^a \mathbb{E}_t[\Delta \hat{\mathbf{a}}_{t+j}]$, and $\rho(\mathbf{P}) < 1$.

(ii) **Relative inflation.** *Current relative inflation $\hat{\pi}_{it} = \pi_{it} - \pi_t^C$ satisfies*

$$\hat{\pi}_t = \Theta_{11} \chi_{t-1} + \Theta_{12} \mathbf{n}_{t-1} + \sum_{j \geq 0} \Pi_j^y \mathbb{E}_t[\tilde{y}_{t+j}] + \sum_{j \geq 0} \Pi_j^a \mathbb{E}_t[\Delta \hat{\mathbf{a}}_{t+j}], \quad (31)$$

where all coefficient sequences decay exponentially. The matrix Θ_{12} vanishes in the free-mobility limit; under finite mobility, it is the channel through which inherited employment misallocation enters sectoral inflation.

(iii) **Aggregate inflation.** *CPI inflation satisfies*

$$\pi_t^C = \bar{\kappa}(\gamma + \varphi) \tilde{y}_t - (1 + \varphi) \text{Cov}_\alpha(\kappa_i, \chi_{it}) - \varphi \text{Cov}_\alpha(\kappa_i, n_{it}) + \beta \mathbb{E}_t \pi_{t+1}^C, \quad (32)$$

⁷Both χ_{t-1} and $\mathbf{n}_{t-1}$ satisfy $\sum_i \alpha_i \chi_{i,t-1} = 0$ and $\sum_i \alpha_i n_{i,t-1} = 0$, so the state is $2(N-1)$ -dimensional. Let $\mathcal{R} \equiv \{x \in \mathbb{R}^N : \alpha^\top x = 0\}$ denote the relative subspace; the state lives in $\mathcal{R} \times \mathcal{R}$.

with $\bar{\kappa} \equiv \sum_i \alpha_i \kappa_i$. Sectoral inflation is recovered as $\pi_t = \hat{\pi}_t + \pi_t^C \mathbf{1}$. Sectoral price and employment distortions affect aggregate inflation only through their covariances with the sectoral Phillips slopes κ_i .

Proof. See Appendix C.1. Part (iii) follows from weighting (21) by α_i and substituting (27). $\square$

Given the inherited state $\mathbf{s}_t$ and the expected paths of aggregate demand and productivity, equations (30)–(32) fully summarize the path and persistence of inflation and labor allocation.

Equation (32) isolates how sectoral distortions generate endogenous cost-push inflation. We can rewrite the two covariance terms as a single cost-push shifter u_t^π :

$$u_t^\pi = \sum_i \alpha_i (\bar{\kappa} - \kappa_i) [(1 + \phi) \chi_{it} + \phi n_{it}]. \quad (33)$$

Aggregate inflation rises when sticky-price sectors ($\kappa_i < \bar{\kappa}$) face higher relative marginal costs. From (27), the bracketed term equals the *negative* of sector i 's relative marginal cost gap: it is large when the sector is overpriced ($\chi_{it} > 0$) or overstaffed ($n_{it} > 0$). So $u_t^\pi > 0$ when the flexible sectors have excess marginal costs—they respond by raising prices rapidly, while sticky sectors fail to offset this with price cuts. Price-rigidity heterogeneity determines whether a given sectoral misalignment *passes through* to CPI; the reallocation block determines how *persistent* that misalignment is.

Under free mobility ($\phi \rightarrow \infty$), Proposition 1 gives $n_{it} = -\chi_{it}$, and the two covariances collapse to the single cost-push term $\text{Cov}_\alpha(\kappa_i, \chi_{it})$ familiar from multi-sector NK models. Away from this limit, the two covariances are independent: $\text{Cov}_\alpha(\kappa_i, n_{it})$ is the novel channel, operating even when relative prices are aligned.

Under uniform stickiness, both covariances vanish and the relative and aggregate blocks decouple entirely:

Corollary 1 (Block Separation). *If $\kappa_i = \kappa$ for all i , the aggregate and relative blocks decouple. The aggregate economy reduces to a textbook one-sector New Keynesian system,*

$$\pi_t^C = \kappa(\gamma + \phi) \tilde{y}_t + \beta \mathbb{E}_t \pi_{t+1}^C,$$

while the relative block evolves independently of monetary policy: $\mathbf{H}_j^y = \mathbf{0}$ and $\Pi_j^y = \mathbf{0}$ for all j , so $\tilde{y}_t$ drops out of the transition equation (30) and the relative-inflation equation (31).

Proof. See Appendix C.1. $\square$

When $\kappa_i = \bar{\kappa}$, monetary policy can stabilize the aggregate economy but cannot influence sectoral reallocation: aggregate demand shifts all sectoral marginal costs equally, leaving relative prices

and reallocation incentives unchanged.⁸ Under heterogeneous stickiness, the covariances are generically nonzero, and the relative block spills over to the aggregate Phillips curve whenever misalignment is correlated with the distribution of nominal rigidities.

Persistence and the transition matrix. The persistence of these cost-push dynamics is governed by the eigenvalues of $\mathbf{P}$. Under free mobility, $\mathbf{P}$ collapses to a single block in χ_{t-1} with eigenvalues determined by pricing alone. Finite mobility adds a second block tracking $\mathbf{n}_{t-1}$, with dynamics shaped by the reallocation matrices F and B . The two blocks couple through price rigidity $\mathbf{K}$, and their interaction can produce eigenvalues more persistent than either block in isolation. The next subsection makes this precise.

3.3 Spectral Analysis

To analytically characterize the persistence of misalignment and the dynamic propagation of shocks, we eigendecompose the transition matrix $\mathbf{P}$.⁹ Write $\mathbf{P} = \mathbf{U} \mathbf{\Lambda} \mathbf{V}$, where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_{2(N-1)})$ contains the eigenvalues in decreasing order of absolute value, $\mathbf{U}$ collects the right eigenvectors $\mathbf{u}_h$ as columns, and $\mathbf{V} = \mathbf{U}^{-1}$ collects the left eigenvectors $\mathbf{v}_h^\top$ as rows. Since $\rho(\mathbf{P}) < 1$, all eigenvalues satisfy $|\lambda_h| < 1$. Each right eigenvector $\mathbf{u}_h = (\mathbf{u}_h^\chi, \mathbf{u}_h^n)$ identifies a *corridor*: a joint pattern of inherited price misalignment and employment misallocation that decays by λ_h each period, orthogonal to other corridors. Table 1 collects these and related objects for reference.

To characterize how each corridor translates into inflation, we separate its contribution into price-misalignment and employment-misallocation channels. For corridor h , define the inflation response

$$\mathbf{q}_h \equiv \Theta \mathbf{u}_h = \Theta_{11} \mathbf{u}_h^\chi + \Theta_{12} \mathbf{u}_h^n, \quad (34)$$

so that q_{ih} is sector i 's relative inflation response associated with corridor h . The first term is inflation generated by inherited relative-price misalignment; the second is the labor-reallocation channel, introducing a cost-push pressure from inherited employment misallocation. It vanishes under free mobility ($\Theta_{12} = \mathbf{0}$).

We focus on a permanent unit productivity shock to sector j : this is a one-time shock that shifts frictionless relative prices, forcing the economy to transition to a new steady state. Aggregate demand $\tilde{y}_t$ depends on monetary policy—for a clean baseline, we set $\tilde{y}_t = 0$; the

⁸When κ_i varies across sectors, flexible sectors respond more to aggregate demand than sticky ones, generating relative price movements that tilt reallocation incentives. Section 4 characterizes when this channel is quantitatively important.

⁹This adapts the spectral approach to spatial transition matrices in Kleinman et al. (2023) and to input-output networks in Liu and Tsyvinski (2024) to the joint dynamics of prices and employment in a New Keynesian setting.

corridor decomposition extends to any path $\{\tilde{y}_t\}$ or a Taylor rule, as the quantitative analysis in Section 6 shows.

We define the *shock exposure* $c_h^{(j)}$ as the projection of the shock's initial impact onto corridor h . Combined with the eigenvalues and corridor-specific inflation patterns, these exposures yield a closed-form representation of the entire transition path as a weighted sum of corridor contributions:

Proposition 3 (Persistence of Relative Prices). *Consider a one-time permanent unit productivity shock to sector j at $t = 0$, starting from steady state, with $\tilde{y}_t = 0$ for all t .*

(i) **Shock exposure.** *The initial impact of the shock on corridor h is*

$$c_h^{(j)} = \mathbf{v}_h^\top \mathbf{H}_0^a (\mathbf{e}_j - \alpha_j \mathbf{1}), \quad (35)$$

determined jointly by the shocked sector's position in the labor-flow network and the sectoral distribution of price rigidities.

(ii) **Transition path.** *For $t \geq 1$, relative inflation decomposes into corridor contributions:*

$$\hat{\pi}_t^{(j)} = \sum_{h=1}^{2(N-1)} \lambda_h^{t-1} c_h^{(j)} \mathbf{q}_h. \quad (36)$$

(iii) **Half-life.** *Corridor h has half-life $\tau_h^{1/2} = \left\lceil \frac{\log(1/2)}{\log|\lambda_h|} \right\rceil$ governed by $|\lambda_h|$ alone, independent of the shock.*

(iv) **Long-run neutrality.** *The cumulative relative inflation in sector i is*

$$\sum_{t=0}^{\infty} \hat{\pi}_{it}^{(j)} = \alpha_j - \delta_{ij}. \quad (37)$$

Frictions reshape the transition path but do not alter the long-run relative price.

Proof. See Appendix C.1. □

Proposition 3 characterizes the transition path via the shock exposure ($c_h^{(j)}$), persistence of the corridor (λ_h), and sectoral incidence (q_{ih}). Part (iv) establishes a long-run neutrality result: while nominal rigidities and reallocation frictions reshape the timing of relative price adjustments, they do not alter the cumulative relative inflation required to reach the flexible-price equilibrium.

The labor-flow network plays two distinct roles. The shock exposure $c_h^{(j)}$ and the inflation pattern $\mathbf{q}_h$ capture the *incidence* of the shock: $c_h^{(j)}$ determines which corridors the shock activates, and $\mathbf{q}_h$ determines how an active corridor distributes inflation across sectors. The eigenvalues $\{\lambda_h\}$ govern *dynamic propagation*: the rate at which corridors decay. Importantly, inflation

Object	Definition	Role
λ_h	eigenvalue of $\mathbf{P}$	persistence of corridor h
$\mathbf{u}_h = (\mathbf{u}_h^\chi, \mathbf{u}_h^n)$	right eigenvector	joint (χ, n) pattern that decays at rate λ_h
$\mathbf{v}_h^\top$	left eigenvector	projects state (χ, n) onto corridor h
$\mathbf{q}_h \equiv \Theta \mathbf{u}_h$	inflation pattern	sectoral relative inflation along corridor h
$c_h^{(j)}$	shock exposure	loading of a sector- j shock onto corridor h
$q_h^C \equiv \sum_i \alpha_i q_{ih}$	aggregate inflation response	contribution of corridor h to CPI

Table 1: Corridor objects. Each corridor $h \in \{1, \dots, 2(N-1)\}$ is a joint price–employment misalignment pattern, indexed by an eigenvalue λ_h of the transition matrix $\mathbf{P}$.

persistence following a shock to sector j is not a function of sector j 's own price rigidity in isolation. Rather, it depends on the rigidities of the sectors with which j is linked through labor flows. A flexible-price sector can exhibit highly persistent inflation if it loads onto a slow corridor anchored by sticky-price neighbors, and vice versa.

Note that long-run neutrality is a statement about relative prices. It does not carry over to the aggregate price level: the aggregate Phillips curve has no force that mechanically reverses past inflation, so any transitory cost-push sequence cumulates into aggregate inflation unless policy explicitly targets it.

Proposition 4 (Cumulative Aggregate Inflation). *Following the sector- j productivity shock in Proposition 3, the cumulative aggregate inflation is*

$$p_\infty^{C,(j)} - p_{-1}^C = \sum_{t=0}^{\infty} \pi_t^C = \pi_0^C + \sum_{h=1}^{2(N-1)} \frac{q_h^C c_h^{(j)}}{(1 - \lambda_h)(1 - \beta \lambda_h)}, \quad (38)$$

where $q_h^C \equiv \sum_i \alpha_i q_{ih}$ is the $t = 0$ aggregate inflation response of corridor h .

(i) Under uniform stickiness, $q_h^C = 0$ for all h and $p_\infty^{C,(j)} - p_{-1}^C = 0$.

(ii) Under heterogeneous stickiness, $p_\infty^{C,(j)} - p_{-1}^C \neq 0$ generically.

Proof. See Appendix C.1. □

Proposition 4 is the aggregate counterpart to Proposition 3: relative prices fully revert, but aggregate inflation along the transition path cumulates into a permanent shift in the price level.¹⁰

A corridor's contribution to the price-level shift depends on three factors: whether the shock loads on it ($c_h^{(j)}$), whether its inflation pattern is misaligned with the distribution of sectoral rigidities (q_h^C), and how slowly it decays. The persistence factor $1/[(1 - \lambda_h)(1 - \beta \lambda_h)]$ is sharply

¹⁰This is conditional on the path $\{\tilde{y}_t\}$. Monetary policy can undo the price-level change by engineering an appropriate output gap; Section 5 formalizes this tradeoff.

convex in λ_h : a corridor with $\lambda_h = 0.9$ contributes roughly 50 times more than the shock's impact effect; at $\lambda_h = 0.95$, roughly 300 times more. A shock that looks modest at impact can therefore generate a price-level effect far exceeding what the shocked sector's own rigidity would suggest, provided it loads onto a slow corridor. Given observable labor-flow data (F) and the sectoral distribution of price rigidities ($\{\kappa_i\}$), the decomposition is computable; Section 6 quantifies this channel for the U.S. economy.

The same decomposition extends beyond the one-shot productivity shock of Proposition 3 to general shocks and monetary policy. Recall from Proposition 2 that the state evolves as $\mathbf{s}_{t+1} = \mathbf{P} \mathbf{s}_t + \mathbf{b}_t^y + \mathbf{b}_t^a$, where $\mathbf{b}_t^y$ and $\mathbf{b}_t^a$ summarize the expected paths of aggregate demand and productivity. Projecting onto corridor h ,

$$s_{ht+1}^* = \lambda_h s_{ht}^* + \mathbf{v}_h^\top (\mathbf{b}_t^y + \mathbf{b}_t^a), \quad s_{ht}^* \equiv \mathbf{v}_h^\top \mathbf{s}_t, \quad (39)$$

shows that each corridor evolves as an independent AR(1) with persistence λ_h . The corridor-specific demand loading $\mathbf{v}_h^\top \mathbf{b}_t^y$ measures monetary policy's transmission to corridor h : under uniform stickiness it vanishes, while under heterogeneous stickiness it is generically nonzero, allowing aggregate demand to activate corridors differentially. Section 5 examines this leverage.

3.4 Segmented versus Integrated Markets

To clarify how labor reallocation affects the persistence of the system, we examine the eigenvalues of $\mathbf{P}$ at the two polar extremes of labor mobility.

When labor markets are segmented ($\phi = 0$), employment never moves ($\mathbf{n}_t = \mathbf{0}$). The shock-generated dynamics reduce to a single-variable system in χ_{t-1} , and firms face the full intensive-margin slope $\kappa_i(1 + \phi)$ in marginal cost. Under uniform stickiness, the restriction of $\mathbf{P}$ to this subspace has stable root $\lambda_{\text{seg}} \in (0, 1)$ solving

$$\beta\lambda^2 - [\kappa(1 + \phi) + \beta + 1]\lambda + 1 = 0. \quad (40)$$

When labor is perfectly mobile ($\phi \rightarrow \infty$), employment offsets price distortions contemporaneously: $n_{it} = -\chi_{it}$. Reallocation absorbs the intensive-margin cost pressure, and the effective slope drops to κ_i . Under uniform stickiness, the stable root $\lambda_{\text{int}} \in (0, 1)$ solves

$$\beta\lambda^2 - (\kappa + \beta + 1)\lambda + 1 = 0. \quad (41)$$

The two equations differ only in the coefficient on λ : $\kappa(1 + \phi) + \beta + 1$ versus $\kappa + \beta + 1$. The segmented-market coefficient is larger, which pushes the stable root down: $\lambda_{\text{seg}} < \lambda_{\text{int}}$.

Proposition 5 (Segmented vs. Integrated Markets). *Under uniform stickiness ($\kappa_i = \kappa$), consider a one-time permanent productivity shock with $\tilde{y}_t = 0$:*

- (i) *Segmented labor markets ($\phi = 0$) produce a larger initial inflation response and faster subsequent decay at rate λ_{seg} .*
- (ii) *Integrated labor markets ($\phi \rightarrow \infty$) produce a smaller initial inflation response and slower subsequent decay at rate $\lambda_{\text{int}} > \lambda_{\text{seg}}$.*
- (iii) *Cumulative relative inflation is $\sum_{t=0}^{\infty} \hat{\pi}_{it} = -\hat{a}_i$ in both cases.*

Proof. See Appendix C.1. □

The mechanism is the margin of labor market adjustment. Under segmentation, sectoral employment is fixed, so sectoral wages clear the market. Upward sloping labor supply converts productivity shocks into relative wage movements, steepening the marginal cost gap to $mc_{it} - p_{it} = -(1 + \phi) \chi_{it}$, and the Phillips curve slope to $\kappa(1 + \phi)$. Under integration, free mobility equalized labor market values ($n_{it} = -\chi_{it}$), and the marginal cost gap reduces to $mc_{it} - p_{it} = -\chi_{it}$, and the slope is κ . The slope differential explains both halves of Proposition 5: the steep schedule under segmentation amplifies impact inflation but forces rapid mean-reversion; the flat schedule under integration mutes impact but slows convergence.

For intermediate ϕ , the two margins interact; Section 4 characterizes the resulting eigenvalues under a mild assumption.

4 Closed-Form Results under Quasi-symmetry

The eigenvalues of $\mathbf{P}$ are implicit functions of the full parameter vector $(\varphi, \beta, \phi, F, B, \mathbf{K})$ and have no closed-form expressions in general. In this section, we analytically characterize them by restricting bilateral reallocation costs to be quasi-symmetric:

Assumption 1 (Quasi-Symmetric Reallocation Costs). *The bilateral reallocation costs satisfy $\tau^{ii} = 0$ and $\tau^{ij} = \tau_i^{\text{out}} + \tau_j^{\text{in}} + \hat{\tau}^{ij}$ for $i \neq j$, where $\hat{\tau}^{ij} = \hat{\tau}^{ji}$ for all i, j .*

This is standard in the dynamic discrete choice literature (Artuç et al., 2010; Allen et al., 2020; Seo and Oh, 2025), nesting symmetric costs and additive origin-destination costs. Its key implication is that F is reversible ($B = F$), with real eigenvalues and an orthogonal eigenbasis:

Lemma 2 (Diagonalization of F). *Under Assumption 1:*

- (i) *The steady-state transition matrix satisfies $\bar{N}_i F_{ij} = \bar{N}_j F_{ji}$, so $B = F$.*
- (ii) *The matrix $Q \equiv A^{1/2} F A^{-1/2}$, where $A = \text{diag}(\alpha_i)$, is real symmetric with eigenvalues $1 = \rho_1 > \rho_2 \geq \dots \geq \rho_N > -1$ and orthonormal eigenvectors $\{s_k\}$.*

Proof. See Appendix C.2. □

U.S. labor flows closely approximate reversibility: the asymmetry measure $\|Q - Q^\top\|/\|Q\|$ is just 8×10^{-4} (Appendix D.1). Under uniform stickiness (Section 4.1), the eigenvectors of F partition the full transition path into $N - 1$ independent components, each pairing a Phillips curve with a reallocation equation. Section 4.2 relaxes this uniform benchmark, deriving two diagnostic statistics: when reallocation frictions amplify persistence, and when aggregate demand moves relative prices.¹¹

Table 2 collects the key labor-component objects of this section.

Object	Definition	Role
ρ_k	eigenvalue of F	labor flow persistence of component k
s_k	eigenvector of $Q = A^{1/2}FA^{-1/2}$	sectoral loading of component k
$\Delta\kappa_k^{\text{eff}}$	$\sum_i \alpha_i \Delta\kappa_i v_{k,i}^2$	whether component k connects sticky sectors
d_k	$\sum_i \alpha_i \Delta\kappa_i v_{k,i}$	demand transmission to relative prices in component k

Table 2: Component objects. Each component $k \in \{2, \dots, N\}$ is a labor-flow pattern in the eigenbasis of F (Lemma 2).

4.1 Independent Components under Uniform Stickiness

Under uniform stickiness ($\kappa_i = \kappa$), block separation (Corollary 1) removes aggregate demand from the relative dynamics, and the diagonalization of F (Lemma 2) together reduce the relative block to $N - 1$ independent component-level systems, each governed by a single eigenvalue λ_k^* that solves a quartic in $(\kappa, \beta, \varphi, \phi, \rho_k)$.

Lemma 3 (Decoupling and Persistence). *Under Assumption 1 and uniform stickiness ($\kappa_i = \kappa$):*

- (i) **Decoupling.** *The aggregate economy evolves as a one-sector New Keynesian system in $(\pi_t^C, \tilde{y}_t)$ (Corollary 1). The coordinate transformation $x_{kt}^* = s_k^\top A^{1/2} \hat{\mathbf{x}}_t$ for $x \in \{\chi, n, v, \hat{a}\}$ decomposes the relative block into $N - 1$ independent three-variable systems, one for each component $k \geq 2$:*

$$\begin{aligned}
[\kappa(1 + \varphi) + \beta + 1] \chi_{kt}^* &= \beta \mathbb{E}_t[\chi_{kt+1}^*] + \chi_{kt-1}^* - \kappa\varphi n_{kt}^* + \Delta\hat{a}_{kt}^* - \beta \mathbb{E}_t[\Delta\hat{a}_{kt+1}^*], \\
n_{kt}^* &= \rho_k n_{k,t-1}^* + \phi(1 - \rho_k^2) v_{kt}^*, \\
v_{kt}^* &= -\chi_{kt}^* - n_{kt}^* + \beta\rho_k \mathbb{E}_t[v_{kt+1}^*].
\end{aligned}$$

¹¹Throughout, *corridor* (index h) refers to eigenvectors of $\mathbf{P}$, which are joint price-employment patterns; *component* (index k) refers to eigenvectors of F , which are pure labor-flow patterns.

(ii) **Component persistence.** The eigenvalues of $\mathbf{P}$ associated with component k solve the quartic

$$\begin{aligned} & \left[(\lambda - \rho_k)(1 - \beta\rho_k\lambda) + \phi(1 - \rho_k^2)\lambda \right] \left[\beta\lambda^2 - (\kappa(1 + \phi) + \beta + 1)\lambda + 1 \right] \\ & + \kappa\phi\phi(1 - \rho_k^2)\lambda^2 = 0. \end{aligned} \quad (42)$$

Exactly two roots satisfy $|\lambda| < 1$; let λ_k^* denote the larger in absolute value. Then λ_k^* is continuous in ϕ , with

$$\lim_{\phi \rightarrow 0} \lambda_k^* = \max\{\lambda_{\text{seg}}, |\rho_k|\}, \quad \lim_{\phi \rightarrow \infty} \lambda_k^* = \lambda_{\text{int}},$$

where λ_{seg} and λ_{int} are the stable roots of (40) and (41).

Proof. See Appendix C.2. □

Each component k pairs a Phillips curve with a reallocation block. The Phillips curve is a second-order difference equation in χ_{kt}^* , with employment misallocation n_{kt}^* entering as an endogenous cost-push shifter. The reallocation block governs how workers respond: inherited employment $n_{k,t-1}^*$ mean-reverts at rate ρ_k , while contemporaneous value differentials v_{kt}^* pull employment toward $-\chi_{kt}^*$ with elasticity ϕ . Under uniform stickiness, F enters component k 's dynamics only through ρ_k .

At $\phi = 0$, the quartic factors into the price and labor margins separately, and persistence is the slower of the two: $\max\{\lambda_{\text{seg}}, |\rho_k|\}$. Positive ϕ couples them, and the limiting cases formalize the tradeoff from Section 3.4: moving from segmented to integrated markets raises persistence when prices are the slower margin ($\lambda_{\text{seg}} > |\rho_k|$) and lowers it when labor is ($|\rho_k| > \lambda_{\text{int}}$).

4.2 Heterogeneous Rigidity: Persistence and Policy Transmission

When κ_i varies across sectors, the independent-component structure of Section 4.1 breaks down: eigenvalues shift from their uniform- κ baselines, and the demand channel $\mathbf{b}_t^y$ becomes active. We first characterize the first-order effect of heterogeneous stickiness on persistence using a perturbation argument.

Proposition 6 (Effective Rigidity). Write $\kappa_i = \bar{\kappa} + \varepsilon \Delta\kappa_i$ with $\sum_i \alpha_i \Delta\kappa_i = 0$ and $|\varepsilon|$ small. For each component $k \geq 2$, define the effective rigidity

$$\Delta\kappa_k^{\text{eff}} \equiv \sum_i \alpha_i \Delta\kappa_i v_{k,i}^2, \quad (43)$$

where $v_{k,i} = \alpha_i^{-1/2} s_{k,i}$ is sector i 's loading on component k . To first order in ε , the dynamics along component k behave as if governed by a uniform slope $\bar{\kappa} + \varepsilon \Delta\kappa_k^{\text{eff}}$, and the dominant eigenvalue of $\mathbf{P}$ along

component k shifts to

$$\lambda_k^*(\varepsilon) = \lambda_k^* + \varepsilon \Delta\kappa_k^{\text{eff}} \cdot \Gamma_k + O(\varepsilon^2),$$

where λ_k^* is the baseline eigenvalue under uniform $\bar{\kappa}$ (Lemma 3) and $\Gamma_k < 0$ is a constant. A negative $\Delta\kappa_k^{\text{eff}}$ strictly increases persistence.

Proof. See Appendix C.2. □

The effective rigidity $\Delta\kappa_k^{\text{eff}}$ is a weighted average of sectoral stickiness deviations, with weights $v_{k,i}^2$ proportional to each sector's squared loading on component k . It is negative when the sectors that load heavily on component k also tend to have stickier prices. When $\Delta\kappa_k^{\text{eff}} < 0$, nominal and real frictions compound: sluggish price adjustment in the sectors connected by component k dampens the wage differentials that drive reallocation, while slow labor flows sustain the marginal-cost gaps that drive inflation. The effect is strongest for slow-mixing components (high ρ_k), where labor frictions are already the bottleneck.

Heterogeneous κ_i also breaks the block separation of Corollary 1: aggregate demand spills into relative dynamics.

Proposition 7 (Demand Transmission). *A unit change in $\tilde{y}_t$ shifts relative inflation in component $k \geq 2$ by $(\gamma + \varphi) d_k$, where*

$$d_k \equiv \sum_i \alpha_i \Delta\kappa_i v_{k,i} = \text{Cov}_\alpha(\Delta\kappa_i, v_{k,i}). \quad (44)$$

Under uniform κ , $d_k = 0$ for all k , recovering block separation.

Proof. Project the demand term $(\gamma + \varphi) \mathbf{MK1} \tilde{y}_t$ from the demeaned Phillips curve (C.5) onto the Q -eigenbasis. The k -th component receives $(\gamma + \varphi) s_k^\top A^{1/2} \mathbf{MK1} \tilde{y}_t = (\gamma + \varphi) d_k \tilde{y}_t$, since $s_k^\top A^{1/2} \mathbf{1} = 0$ for $k \geq 2$ implies $s_k^\top A^{1/2} M = s_k^\top A^{1/2}$. □

The transmission coefficient d_k is exact for arbitrary heterogeneity in κ_i . It is large when sticky and flexible sectors sit on opposite sides of a component: aggregate demand moves their prices at different speeds, tilting reallocation incentives. For example, if component k separates a flexible sector (positive loading) from a sticky sector (negative loading), then $d_k > 0$: expansionary policy raises the flexible sector's relative price, drawing workers toward the sticky sector.

The two statistics capture distinct channels. The effective rigidity $\Delta\kappa_k^{\text{eff}}$ weights stickiness by the squared loading $v_{k,i}^2$ and governs persistence of misallocation along the component. The transmission coefficient d_k weights stickiness by the signed loading $v_{k,i}$ and governs how aggregate demand moves relative prices along the component. The two are independent: in a symmetric two-sector economy with $\kappa_1 \neq \kappa_2$, demand transmission is maximal while effective rigidity is exactly zero.

Both statistics are computable from the labor transition matrix F and sectoral price adjustment frequencies. The effective rigidity is a first-order approximation; whether it correctly identifies the most persistent corridors when dispersion in κ_i is large is a quantitative question addressed in Section 6. The transmission coefficient is exact. Together, they determine when and how monetary policy interacts with sectoral reallocation (Section 5).

4.3 Example: A Three-Sector Illustration

We illustrate how the alignment of price rigidity with the labor-flow network governs inflation persistence and monetary-policy leverage. The example compares two economies with identical average mobility, identical labor-flow eigenvalues, and the same shock. They differ only in which sectors are connected by the slowest labor-flow component.

Setup. Consider three equal-sized sectors ($\alpha_i = 1/3$). Sectors A and B are sticky ($\kappa_A = \kappa_B < \bar{\kappa}$), while sector C is flexible ($\kappa_C > \bar{\kappa}$). A permanent negative productivity shock hits sector A .

We compare two labor networks with transition probabilities m and ℓ , where $0 < m < \ell \ll \frac{1}{2}$. In network AB , workers move slowly between A and B ; in network AC , between A and C :

$$F^{AB} = \begin{bmatrix} 1 - m - \ell & m & \ell \\ m & 1 - m - \ell & \ell \\ \ell & \ell & 1 - 2\ell \end{bmatrix}, \quad F^{AC} = \begin{bmatrix} 1 - m - \ell & \ell & m \\ \ell & 1 - 2\ell & \ell \\ m & \ell & 1 - m - \ell \end{bmatrix}.$$

One matrix is a permutation of the other, so they share the same eigenvalues:

$$\rho_1 = 1, \quad \rho_2 = 1 - 2m - \ell, \quad \rho_3 = 1 - 3\ell.$$

Since $m < \ell$, component 2 is the slow labor-mixing bottleneck ($\rho_2 > \rho_3$) in both economies. If persistence were determined by average mobility or sectoral stickiness per se, the two economies would behave identically.

Different bottlenecks. What differs is which sectors the slow component connects. The eigenvectors of the slow mode are

$$s_2^{AB} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad s_2^{AC} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Under F^{AB} , the slow mode separates the two sticky sectors A and B . Under F^{AC} , it separates the sticky sector A from the flexible sector C .

Sufficient statistics. We evaluate $\Delta\kappa_k^{\text{eff}}$ and d_k for the slow component. With equal shares, sector i 's loading simplifies to $v_{k,i} = \sqrt{3} s_{k,i}$. Write $\tilde{\kappa}_i \equiv \kappa_i - \bar{\kappa}$, so $\tilde{\kappa}_A = \tilde{\kappa}_B = -\varepsilon$ and $\tilde{\kappa}_C = 2\varepsilon$.

Under network AB , the slow mode loads on A and B with $v_{2,A} = \sqrt{3/2}$, $v_{2,B} = -\sqrt{3/2}$, $v_{2,C} = 0$. The signed terms cancel while the squared terms reinforce:

$$\begin{aligned}\Delta\kappa_2^{\text{eff},AB} &= \sum_i \alpha_i \tilde{\kappa}_i v_{2,i}^2 = \frac{1}{3} [(-\varepsilon)\frac{3}{2} + (-\varepsilon)\frac{3}{2} + 0] = -\varepsilon, \\ d_2^{AB} &= \sum_i \alpha_i \tilde{\kappa}_i v_{2,i} = \frac{1}{3} [(-\varepsilon)\sqrt{\frac{3}{2}} + (-\varepsilon)(-\sqrt{\frac{3}{2}}) + 0] = 0.\end{aligned}$$

Under network AC , the slow mode loads on A and C with $v_{2,A} = -\sqrt{3/2}$, $v_{2,B} = 0$, $v_{2,C} = \sqrt{3/2}$. Both statistics are nonzero:

$$\begin{aligned}\Delta\kappa_2^{\text{eff},AC} &= \sum_i \alpha_i \tilde{\kappa}_i v_{2,i}^2 = \frac{1}{3} [(-\varepsilon)\frac{3}{2} + 0 + 2\varepsilon \cdot \frac{3}{2}] = +\frac{\varepsilon}{2}, \\ d_2^{AC} &= \sum_i \alpha_i \tilde{\kappa}_i v_{2,i} = \frac{1}{3} [(-\varepsilon)(-\sqrt{\frac{3}{2}}) + 0 + 2\varepsilon\sqrt{\frac{3}{2}}] = \sqrt{\frac{3}{2}} \varepsilon.\end{aligned}$$

	Network AB	Network AC
Effective rigidity $\Delta\kappa_2^{\text{eff}}$	$-\varepsilon$	$+\varepsilon/2$
Demand transmission d_2	0	$\sqrt{3/2} \varepsilon$

Interpretation. Under F^{AB} , the slow component connects two equally sticky sectors. The effective rigidity is negative: neither sector adjusts prices quickly, muting the wage signal that would redirect workers out of A —so persistence rises above the uniform- κ baseline ($\lambda_{2,AB}^* > \lambda_2^*$). The demand transmission is zero: aggregate demand moves A 's and B 's prices symmetrically, leaving the central bank no leverage over reallocation along this component.

Under F^{AC} , the slow component connects a sticky sector to a flexible one. Sector C 's rapid price adjustment generates a sharp wage differential that accelerates reallocation out of A : the effective rigidity is positive and persistence falls below the baseline ($\lambda_{2,AC}^* < \lambda_2^*$). The demand transmission is positive: expansionary policy raises C 's price faster than A 's, tilting reallocation along the slow component.

Sector A is equally sticky and faces the same shock in both economies. What differs is its network position, which determines both inflation persistence and the central bank's leverage over reallocation.

5 Optimal Monetary Policy

Sections 3–4 characterized the generalized Phillips curve $\mathcal{K}$ (29) and transition dynamics conditional on the path of aggregate demand. We now consider the optimal choice of that path. This section derives the welfare loss function and the Ramsey problem (Section 5.1), characterizes the optimal targeting criterion (Section 5.2), and identifies conditions under which optimal policy deviates from the standard rule (Section 5.4).

5.1 The Ramsey Problem

We adopt the timeless perspective of Woodford (2003). The efficient steady-state subsidy eliminates first-order terms, so the problem admits a linear-quadratic approximation.

Proposition 8 (Ramsey Problem under Commitment). *Under commitment in the timeless perspective, the central bank chooses a sequence $\{\tilde{y}_t, \pi_t, \mathbf{n}_t, \chi_t, \mathbf{v}_t\}_{t=0}^{\infty}$ to minimize the intertemporal welfare loss*

$$\min \quad \mathcal{W} = \sum_{t=0}^{\infty} \beta^t \mathcal{L}_t, \quad (45)$$

where the per-period loss is

$$\mathcal{L}_t \stackrel{(2)}{=} \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1}{2} \chi_{it}^2 + \frac{\varphi}{2} (\chi_{it} + n_{it})^2 \right], \quad (46)$$

subject to, for each sector i and date t :

$$\pi_{it} = \kappa_i [(\gamma + \varphi) \tilde{y}_t - (1 + \varphi) \chi_{it} - \varphi n_{it}] + \beta \mathbb{E}_t \pi_{it+1}, \quad (47)$$

$$\chi_{it} = \chi_{it-1} + \pi_{it} - \pi_{it}^C + \Delta \hat{a}_{it}, \quad (48)$$

$$\mathbf{n}_t = B \mathbf{n}_{t-1} + \phi(I - BF) \mathbf{v}_t, \quad (49)$$

$$\mathbf{v}_t = -\chi_t - \mathbf{n}_t + \beta F \mathbb{E}_t [\mathbf{v}_{t+1}], \quad (50)$$

taking as given initial conditions $(\hat{\mathbf{p}}_{-1}, \mathbf{n}_{-1})$ and the exogenous productivity process $\{\mathbf{a}_t\}$.

Proof. The per-period loss (46) follows from a second-order expansion of household welfare around the efficient steady state (Appendix B). The constraints restate the marginal cost (27) and NKPC (21), relative price identity, and reallocation block from Section 2.5. $\square$

The output gap and sectoral inflation welfare loss terms are standard. The first misallocation term, $\frac{1}{2} \chi_{it}^2$, penalizes relative price distortions as in the multisector New Keynesian optimal policy literature (Rubbo, 2023). The novel term, $\frac{\varphi}{2} (\chi_{it} + n_{it})^2$, penalizes deviations from the full-mobility benchmark $n_{it} = -\chi_{it}$; it captures the welfare cost of labor-market frictions and

vanishes as $\phi \rightarrow \infty$. Through these constraints, the planner may seek to influence the state $\mathbf{s}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$ along the generalized Phillips curve (29).

5.2 The Targeting Criterion

The first-order conditions of the Ramsey problem involve multipliers λ_t (Phillips curves), η_t (relative price identities, weighted by α_i), ψ_t (employment law of motion), and ξ_t (continuation values); the full system is in Appendix C.3.1. These collapse into a targeting criterion:

Proposition 9 (Monetary Policy Target). *Under commitment, optimal monetary policy is characterized by the targeting rule*

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \pi_{it} = \sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it}. \quad (51)$$

where η_t is determined by the system of first-order conditions in Appendix C.3.1.

Proof. See Appendix C.3.1. □

The left-hand side is the standard targeting criterion. The right-hand side is a *misallocation wedge*: it is the covariance between the shadow value of relative-price misalignment ($\alpha_i \eta_{it}$) and Phillips curve slopes (κ_i). When all sectors share the same slope, the wedge vanishes and the criterion reduces to

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \pi_{it} = 0, \quad (52)$$

which, if additionally $\epsilon_i = \epsilon$, becomes the textbook rule $\tilde{y}_t - \tilde{y}_{t-1} + \epsilon \pi_t^C = 0$.

The misallocation wedge. The shadow value of raising sector i 's relative price is $\alpha_i \eta_{it}$. Rearranging the FOC (F_χ) and iterating forward gives

$$A\eta_t = - \sum_{s=0}^{\infty} \beta^s \mathbb{E}_t \mathbf{g}_{t+s}, \quad (53)$$

$$\mathbf{g}_t \equiv \underbrace{A[(1+\varphi)\chi_t + \varphi \mathbf{n}_t]}_{\text{cross-sector misallocation}} + \underbrace{(1+\varphi)K\lambda_t}_{\text{within-sector misallocation}} + \underbrace{\xi_t}_{\text{future reallocation}}.$$

All three terms in $\mathbf{g}_t$ are forms of misallocation: relative-price and employment gaps across sectors, within-sector Calvo price dispersion driven by sectoral inflation, and the shadow value of distorting future reallocation through continuation values. Persistent corridors of $\mathbf{P}$ generate slowly-decaying paths of $\mathbf{g}$, which the discounted sum amplifies into a large $|\eta_{it}|$ —and thus a large wedge in monetary policy. Section 5.4 makes this precise.

The sign of the wedge. Aggregate demand moves flexible-sector prices more than sticky-sector prices. The planner uses this asymmetry to tilt relative prices toward sectors with the largest welfare gain from a price change. Consider a negative productivity shock to a sticky sector that requires labor outflows. The planner wants to raise its relative price ($\eta_{it} > 0$): a higher relative price draws workers out of the unproductive sector and accelerates reallocation. Since the sector has $\kappa_i < \bar{\kappa}$, the wedge $\sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it}$ is negative, and optimal policy tightens: contraction lowers flexible-sector prices more than sticky-sector prices, delivering the desired relative-price movement. The planner accepts a negative output gap to obtain a relative-price movement that sticky prices would otherwise delay. The logic reverses when workers must leave flexible sectors, or when other components of η_t —within-sector dispersion, future continuation values—dominate.

5.3 Benchmarks and the Discretionary Policy

The targeting criterion (51) differs from benchmarks in ways that are clearest in two limit cases: policy that stabilizes the output gap, and the static discretionary problem in which the central bank chooses only today's output gap.

The output-gap index. Following Rubbo (2023), define

$$\pi_t^R \equiv \sum_i \omega_i^R \pi_{it}, \quad \omega_i^R = \frac{\alpha_i / \kappa_i}{\sum_j \alpha_j / \kappa_j}.$$

The weights tilt the index toward sticky-price sectors. Summing the sectoral Phillips curves (21) with these weights and using $\sum_i \alpha_i n_{it} = 0$ gives

$$\pi_t^R = (\gamma + \varphi) \tilde{y}_t + \beta \mathbb{E}_t \pi_{t+1}^R, \quad (54)$$

independent of $\mathbf{n}_t$, χ_t , or F . Stabilizing $\pi_t^R = 0$ at all dates therefore closes the output gap. This is suboptimal whenever the misallocation wedge $\sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it}$ is nonzero: the planner accepts output-gap costs to obtain relative-price movements.

The discretionary problem. The discretionary benchmark—equivalent to a static one-period problem—isolates the within-period force behind the dynamic wedge. Suppose the central bank chooses $\tilde{y}_t$ taking inherited states $(\chi_{t-1}, \mathbf{n}_{t-1})$ and expected continuation terms $\mathbb{E}_t \pi_{i,t+1}$ as given. Substituting contemporaneous worker flows into (27), the effective marginal cost is

$$mc_{it} - p_{it} = (\gamma + \varphi) \tilde{y}_t - \sigma \chi_{it}, \quad \sigma \equiv 1 + \frac{\varphi}{1 + \phi},$$

where σ summarizes the within-period reallocation margin: $\sigma = 1 + \phi$ at $\phi = 0$ (segmentation) and $\sigma \rightarrow 1$ as $\phi \rightarrow \infty$ (full mobility). The impact Phillips curve takes the form

$$\pi_{it} = b_i \tilde{y}_t + u_{it},$$

with b_i summarizing how aggregate demand transmits to sector i 's inflation and u_{it} collecting inherited states, productivity shocks, and expected continuation inflation. Letting $b^C \equiv \sum_i \alpha_i b_i$ denote the CPI response, a unit increase in $\tilde{y}_t$ raises sector i 's inflation by b_i and changes its relative price by $b_i - b^C$. Appendix C.3.2 gives explicit forms for b_i and u_{it} .

The discretionary welfare loss specializes (46) by treating continuation terms as exogenous:

$$\mathcal{L}_t^{\text{disc}} = \frac{\gamma + \phi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \frac{\sigma}{2} \sum_i \alpha_i \chi_{it}^2.$$

Differentiating with respect to $\tilde{y}_t$ and using $d\pi_{it}/d\tilde{y}_t = b_i$ and $d\chi_{it}/d\tilde{y}_t = b_i - b^C$ yields

$$0 = (\gamma + \phi) \tilde{y}_t + \sum_i \alpha_i \omega_i^{\text{disc}} \pi_{it} + c_t, \quad \omega_i^{\text{disc}} = \underbrace{\frac{\epsilon_i}{\kappa_i} b_i}_{\text{within-sector dispersion}} + \underbrace{\sigma (b_i - b^C)}_{\text{relative-price transmission}},$$

with c_t collecting inherited states (Appendix C.3.2). The first weight is the standard Calvo motive: sticky-sector inflation creates within-sector price dispersion. The second is the reallocation motive: aggregate demand is useful insofar as it moves *relative* prices, with sectors satisfying $b_i > b^C$ being the ones the central bank can move more than the CPI. The discretionary rule therefore already balances sticky-sector dispersion costs against relative-price transmission to reallocation.

Discretion as inflation targeting. The discretionary rule can be re-expressed as inflation targeting. The output-gap index π_t^R from (54) satisfies $\pi_t^R = (\gamma + \phi) \tilde{y}_t$ when continuation terms are treated as given, so substituting into the FOC yields a target on a linear combination of sectoral inflation rates. The result is a discretion-implementable inflation-targeting rule whose weights combine the output-gap weights with the reallocation-transmission weights $\sigma(b_i - b^C)$. Under commitment, the rule no longer reduces to a zero target on a fixed inflation index: the planner internalizes that today's $\tilde{y}_t$ shifts future relative prices, employment, and continuation values, and the static transmission term is replaced by the present-value shadow value η_t . The Ramsey criterion targets the same standard object as the discretionary rule, but the target is set to a state-dependent value—the misallocation wedge in (51).

5.4 A Spectral Targeting Rule

The misallocation wedge $\sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it}$ in (51) is large when relative-price misalignment is both persistent and exposed to aggregate demand. By Proposition 3, the predetermined state $(\chi_{t-1}, \mathbf{n}_{t-1})$ decomposes into corridors evolving as AR(1) processes with persistence λ_h . The wedge inherits this corridor structure: it decomposes into per-corridor contributions, each given by a demand exposure times a shadow value.

Proposition 10 (Spectral Targeting Rule). *Under the assumptions of Proposition 3, the targeting criterion (51) decomposes as*

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \pi_{it} = \sum_h d_h^P \eta_{ht}^P, \quad (55)$$

where d_h^P is the projection of $A(\kappa - \bar{\kappa}\mathbf{1})$ onto the left eigenvector $\mathbf{v}_h$ of $\mathbf{P}$, and η_{ht}^P is the projection of η_t onto the right eigenvector $\mathbf{u}_h$.

Proof. See Appendix C.3.3. □

The exposure d_h^P captures how aggregate demand transmits along corridor h : because monetary policy moves flexible-sector prices more than sticky-sector prices, the relative-price difference it induces is governed by how κ_i varies across sectors. The shadow value η_{ht}^P is the welfare gain the planner would obtain from tilting relative prices along corridor h today. Substituting the present-value representation (53) shows that persistent corridors with large λ_h accumulate marginal welfare costs over time, amplifying $|\eta_{ht}^P|$ by roughly $1/(1 - \beta\lambda_h)$.

The wedge contribution $d_h^P \eta_{ht}^P$ along each corridor therefore combines two economic factors: how much welfare a relative-price change would gain, and how much aggregate demand can move relative prices in that direction. A persistent corridor with little dispersion in κ_i along its direction matters for welfare but cannot be moved by monetary policy. A corridor that monetary policy can move but along which welfare costs are small is reachable but not worth acting on. Only corridors that are both costly and movable by aggregate demand contribute meaningfully to the wedge.

Under quasi-symmetry (Assumption 1), the corridor statistics d_h^P, η_{ht}^P reduce to the F -component statistics from Section 4: d_h^P specializes to d_k (Proposition 7), and the shadow value projection becomes

$$\eta_{kt}^* \equiv \sum_i \alpha_i v_{k,i} \eta_{it},$$

so that (55) reduces to

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \pi_{it} = \sum_{k=2}^N d_k \eta_{kt}^*. \quad (56)$$

The persistence governing η_{kt}^* is the component eigenvalue λ_k^* , shifted by $\Delta\kappa_k^{\text{eff}} \cdot \Gamma_k$ under heterogeneous stickiness (Proposition 6). When $d_k \approx 0$ for all k , the wedge vanishes to first order. With near-reversibility, (56) approximates (55).

The spectral form makes the wedge a function of two measurable objects: the demand-transmission statistics $\{d_k\}$ and the persistence-amplified shadow values $\{\eta_{kt}^*\}$. Section 6 measures these for the U.S. economy and asks how large the wedge is in practice.

6 Quantitative Evidence

We bring the model to U.S. sectoral data to assess whether the analytical channels are quantitatively relevant. We estimate the transition matrix F, B from worker flows, measure sectoral price rigidity from micro price data, and characterize the generalized Phillips curve and assess the quantitative importance of reallocation frictions for inflation persistence and monetary policy.

6.1 Data and Calibration

We set the model period to one quarter. We set $\gamma = 1$ (log utility over consumption), $\varphi = 2$, and $\beta = 0.99$ (quarterly discount factor), following standard calibrations in the New Keynesian literature (e.g., [Rubbo, 2023](#)). Sectoral price-adjustment frequencies come from [Pasten et al. \(2020\)](#), converted to quarterly Calvo slopes κ_i . The sectoral transition matrix F is estimated from industry-to-industry worker flows in the March CPS, converted to quarterly frequency following [Artuç et al. \(2010\)](#).¹² The expenditure shares $\{\alpha_i\}$ equal the stationary distribution of F , and the backward matrix B is constructed via (10). The mobility elasticity ϕ is mapped to the migration dispersion estimate of [Caliendo et al. \(2019\)](#) as $\phi = \varphi/\nu$ (Appendix D.2), giving $\phi \approx 0.37$.

Industries are aggregated into 14 sectors, chosen to balance granularity with reliable CPS cell sizes.¹³ Table 3 summarizes the sector-level inputs. Price rigidity varies by nearly two orders of magnitude across sectors, with an expenditure-weighted mean of $\bar{\kappa} = 0.112$. The estimated transition matrix implies a quarterly switching rate of 6.75% across sectors, with detailed balance holding to within 0.08%—quasi-symmetry is an empirically tight approximation (Appendix D.1). Eigenvalues of F range from $\rho_2 = 0.955$ to $\rho_{13} = 0.898$, corresponding to half-lives of 15 and

¹²Respondents report their current industry and the industry of their longest-held job in the previous year. Recollection bias implies an effective horizon of approximately five months ([Artuç et al., 2010](#)); we treat the raw matrix as F_{5m} and convert via $F = F_{5m}^{3/5}$.

¹³The 14 sectors are: Agriculture, Mining & Extraction (AGM); Construction (CON); Nondurable Manufacturing (NDM); Durable & Advanced Manufacturing (DUR); Utilities (UTL); Trade, Wholesale & Retail (TRD); Transportation & Warehousing (TRN); Leisure, Hospitality & Other Services (LSR); Information & Communications (INF); Finance, Insurance & Real Estate (FIR); Professional & Business Services (PBS); Government (GOV); Education (EDU); and Healthcare (HLT).

Code	Sector	α_i	κ_i	Duration (q)
PBS	Professional & Business Services	0.091	0.008	12.4
HLT	Healthcare	0.118	0.011	10.7
INF	Information & Communications	0.041	0.012	9.9
DUR	Durable & Advanced Manufacturing	0.085	0.030	6.4
LSR	Leisure, Hospitality & Other Services	0.061	0.039	5.7
EDU	Education	0.096	0.041	5.6
GOV	Government	0.053	0.056	4.8
TRN	Transportation & Warehousing	0.048	0.059	4.7
FIR	Finance, Insurance & Real Estate	0.101	0.093	3.9
NDM	Nondurable Manufacturing	0.025	0.157	3.1
UTL	Utilities	0.012	0.194	2.8
TRD	Trade (Wholesale & Retail)	0.180	0.235	2.6
CON	Construction	0.067	0.304	2.4
AGM	Agriculture, Mining & Extraction	0.021	0.758	1.8
Expenditure-weighted mean		—	0.112	4.8

Table 3: Sectoral price rigidity and expenditure shares

Notes. Phillips curve slopes $\kappa_i = (1 - \delta_i)(1 - \beta\delta_i)/\delta_i$ computed on a quarterly horizon. Duration is median price duration in quarters. Sectors sorted by κ_i ascending (stickiest first).

6 quarters respectively. Even the fastest-mixing component takes more than a year to decay halfway: labor reallocation is slow at every horizon relevant to monetary policy.

6.2 The Generalized Phillips Curve

Proposition 2 characterizes the generalized Phillips curve $\mathcal{K}$ as a recursive mapping from aggregate demand $\{\tilde{y}_t\}$ and sectoral productivity growth $\{\Delta\hat{\mathbf{a}}_t\}$ into sectoral inflation, with state $\mathbf{s}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$. We now study its Jacobian.

6.2.1 A Two-Sided Sequence-Space Phillips Curve

The Jacobian is two-sided: firms are forward-looking, and inherited price and employment misalignment are predetermined. Current inflation thus depends on past inputs through $\mathbf{s}_t$ and on expected future inputs through today's reset-price decisions.

Sectoral inflation $\boldsymbol{\pi}_t = (\pi_{1t}, \dots, \pi_{Nt})'$ is the main object: relative inflation $\hat{\boldsymbol{\pi}}_t$ and CPI inflation $\pi_t^C = \boldsymbol{\alpha}^\top \boldsymbol{\pi}_t$ are linear functions of it. Define the sectoral demand and supply Jacobians at lag $\tau \in \mathbb{Z}$,

$$\mathbf{J}_{\pi y}(\tau) \equiv \left. \frac{\partial \boldsymbol{\pi}_t}{\partial \tilde{\mathbf{y}}_{t-\tau}} \right|_{\text{ss}} \in \mathbb{R}^N, \quad \mathbf{J}_{\pi a}(\tau) \equiv \left. \frac{\partial \boldsymbol{\pi}_t}{\partial (\Delta\hat{\mathbf{a}}_{t-\tau})'} \right|_{\text{ss}} \in \mathbb{R}^{N \times N}, \quad (57)$$

which depend on $\tau = t - s$ alone. CPI and relative-inflation Jacobians follow by linear aggrega-

tion:

$$\mathcal{J}_{\pi z}^C(\tau) = \alpha^\top \mathbf{J}_{\pi z}(\tau), \quad \hat{\mathbf{J}}_{\pi z}(\tau) = \mathbf{J}_{\pi z}(\tau) - \mathcal{J}_{\pi z}^C(\tau) \mathbf{1}, \quad z \in \{y, a\}. \quad (58)$$

Any scalar object reported below is a linear combination of $\mathbf{J}_{\pi z}(\tau)$.

The sign of τ separates three forces. For $\tau > 0$, the derivative captures *propagation* of past shocks through inherited $(\chi, \mathbf{n})$. For $\tau = 0$, it is *impact*. For $\tau < 0$, it captures *anticipation* of future demand and productivity, transmitted through the terms $\mathbf{b}_t^y$ and $\mathbf{b}_t^a$ in the state transition.

6.2.2 The Demand Jacobian

We characterize the response of inflation to aggregate demand,

$$\mathbf{J}_y(h) \equiv \left. \frac{\partial \pi_{t+h}}{\partial \tilde{y}_t} \right|_{ss}, \quad (59)$$

together with its CPI version $J_y^C(\tau) = \alpha^\top \mathbf{J}_y(\tau)$. Figure 1 plots both for the calibrated economy.

Panel (a) shows the two-sided structure. On the propagation side ($\tau > 0$), flexible sectors front-load: AGM jumps to 0.73 at impact and overshoots to -0.42 one quarter later, as the cost pressure passes and reset prices correct downward. Sticky sectors respond modestly at impact but continue adjusting for many quarters. On the anticipation side ($\tau < 0$), the ordering reverses, with flexible sectors building up most aggressively toward $\tau = 0$. The CPI response inherits the sharp impact spike and short-lived overshoot.

Panel (b) compares cumulative pass-through across counterfactuals. Heterogeneity in κ_i is what makes the Jacobian dynamic: under *Uniform* κ , block separation (Corollary 1) flattens S_H^y at $\bar{\kappa}(\gamma + \phi) = 0.335$, while the calibrated heterogeneity produces the steeply declining *Data* profile. The mobility counterfactuals isolate the timing role of reallocation: the *Integrated* benchmark delivers the largest impact and fastest decay, as contemporaneous reallocation flattens the effective supply slope from $\kappa_i(1 + \phi)$ to κ_i (Proposition 5). The calibrated economy sits close to *Segmented*, consistent with $\phi = 0.37$. All three configurations converge to the same long-run pass-through $S_\infty^y \approx 0.082$: mobility reshapes *when* demand passes through into prices, not how much.

6.2.3 The Supply Jacobian

The supply Jacobian gives the response of inflation to a permanent unit shock to sector j 's productivity at $t = 0$. A permanent level shock implies $\Delta \hat{a}_{j,t} \neq 0$ only at impact, matching the

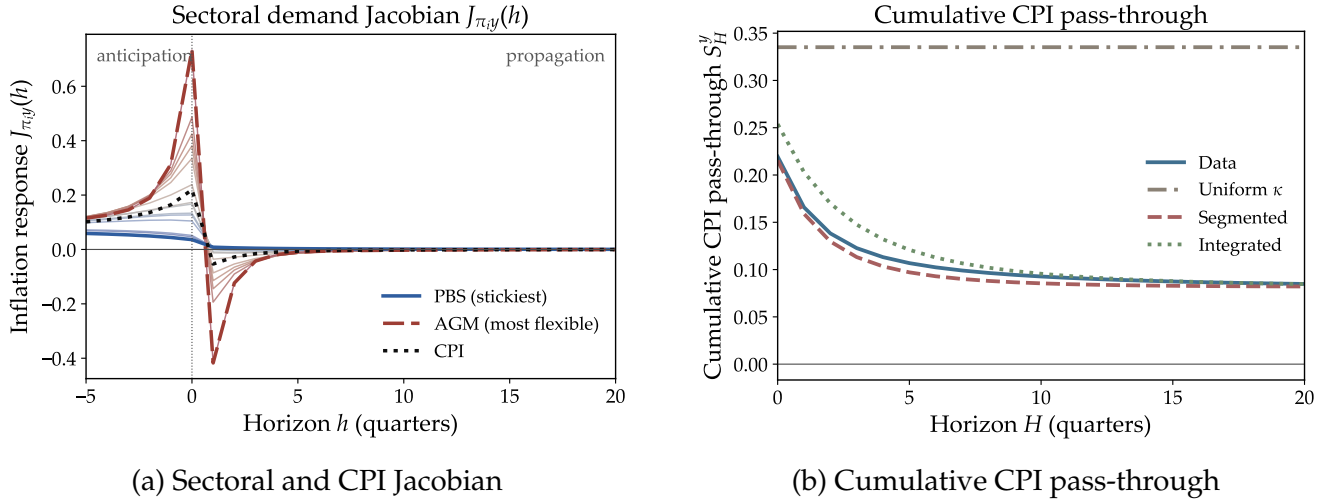


Figure 1: The demand Jacobian in the calibrated economy.

Note. Panel (a): sectoral inflation responses $J_{y,i}(\tau)$ across both anticipation ($\tau < 0$) and propagation ($\tau > 0$) horizons, ordered from blue (stickiest, PBS) to red (most flexible, AGM); dashed black is the CPI response $J_y^C(\tau)$. Panel (b): cumulative pass-through $S_H^y \equiv \sum_{h=0}^H J_y^C(h)$ across rigidity and mobility counterfactuals.

productivity term in $\mathcal{K}$. Define

$$\mathbf{J}_{a,j}(\tau) \equiv \left. \frac{\partial \pi_t}{\partial a_{j,t-\tau}} \right|_{ss}, \quad \tau \in \mathbb{Z}, \quad (60)$$

with CPI version $J_{a,j}^C(\tau) = \boldsymbol{\alpha}^\top \mathbf{J}_{a,j}(\tau)$. We set $\tilde{y}_t = 0$ at every date, consistent with the partial derivative notation to isolate the cost-push side of $\mathcal{K}$.

Figure 2 contrasts the stickiest sector (PBS, $\kappa = 0.008$) with the most flexible (AGM, $\kappa = 0.758$). An AGM shock is absorbed almost entirely by AGM's own price, which falls 0.74 on impact and dissipates within two quarters; cross-sector and CPI responses are small. A PBS shock generates the opposite pattern: PBS's own price falls only 0.089, while positive cross-sector responses dominate, yielding $J_{a,\text{PBS}}^C(0) = 0.047$ on impact even as the shocked sector's own inflation falls. The shock is much more persistent, remaining even after 20 quarters.

Same stickiness, different network position: HLT and INF. Stickiness is not the only thing that matters. HLT and INF have nearly identical rigidity ($\kappa_{\text{HLT}} = 0.011$, $\kappa_{\text{INF}} = 0.012$), but their CPI Jacobians differ by a factor of three: HLT's impact and eight-quarter cumulative responses are 0.054 and 0.166, against 0.016 and 0.047 for INF (Figure 3). HLT is 2.9 times larger in expenditure share, but even after scaling, HLT's eight-quarter CPI response is still 22% larger. The residual reflects network position: HLT's own persistence $F_{\text{HLT},\text{HLT}} = 0.953$ exceeds INF's 0.911, and HLT loads more heavily on slow labor-flow components ($\rho_h \approx 0.955, 0.948, 0.941$ versus 0.909, 0.904, 0.898), giving a heavier tail—8.8% of HLT's first-eight-quarter CPI mass

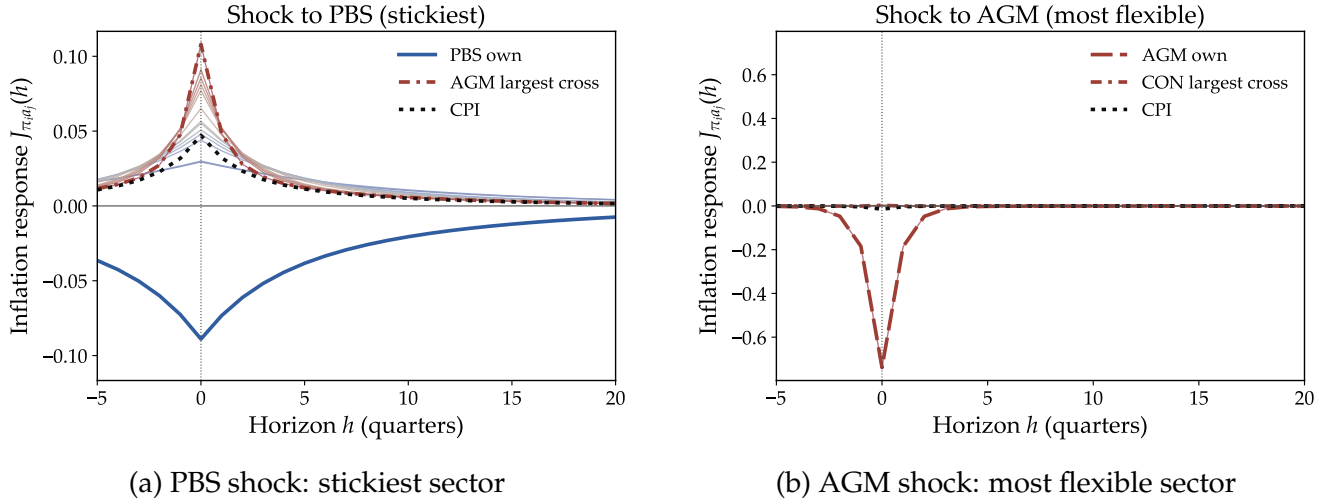


Figure 2: Supply Jacobians: sticky versus flexible shocks.

Note. Sectoral inflation responses $J_{\pi_{aj}}(h) = \partial \pi_{i,t+h} / \partial \Delta \hat{a}_{j,t}$, with the shocked sector's own response, the largest cross-sector response, and CPI highlighted.

remains over $h = 9, \dots, 20$, against 7.4% for INF. Stickiness pins down how a sector absorbs its own shock; size and network position pin down how it spreads and persists.

Cross-section under three mobility configurations. Figure 4 reports $J_{a,j}^C(h)$ for all 14 shocks under *Data*, *Segmented*, and *Integrated*. Sticky-sector shocks generate the largest positive CPI responses: HLT peaks at impact, PBS at the eight-quarter cumulative. The largest negative response comes from TRD, not AGM, because TRD combines high flexibility with a much larger expenditure weight. As with the demand Jacobian, the data economy sits close to *Segmented*—over $h = 0, \dots, 20$, the data-segmented distance is about one third of the data-integrated distance.

6.3 Inflation Persistence

We now turn from the shape of the supply Jacobian to the persistence a policymaker observes after a sectoral shock. We distinguish three nested objects: the supply-shock tail (the CPI path following a sectoral productivity innovation, assuming zero output gap); endogenous propagation through the price–employment corridors; and policy closure, which reshapes the observed path once shocks are persistent and the policy rule feeds back through aggregate demand. We find that they all need not coincide.

Sectoral tails and look-through. Figure 5 plots the impact CPI response to a unit sectoral shock against the cumulative CPI response remaining after one year. PBS and HLT generate both a positive impact and a large positive tail; TRD has a large impact but little tail; EDU,

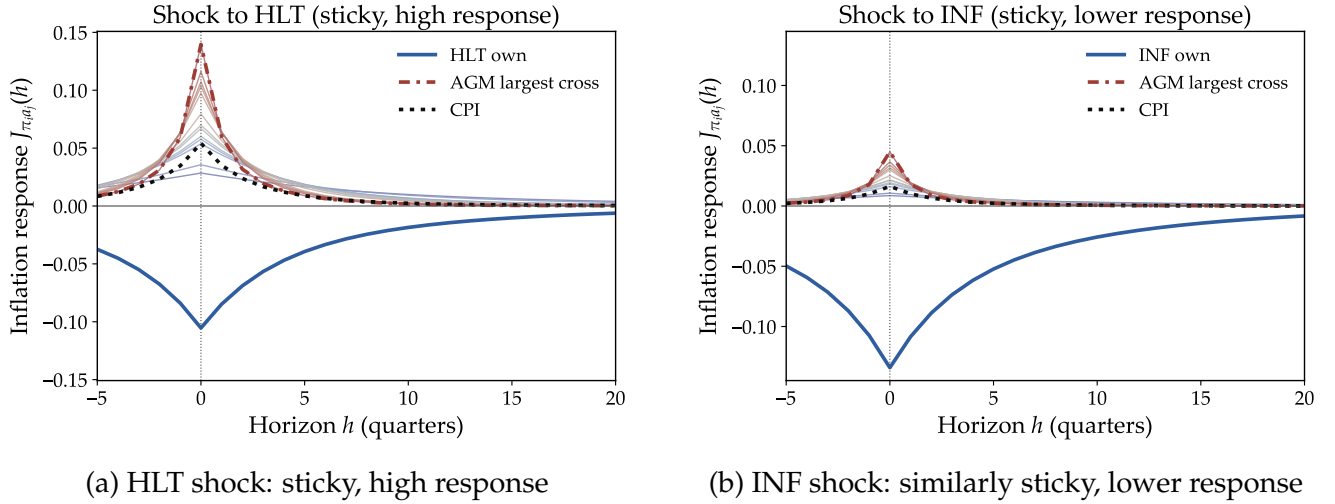


Figure 3: Supply Jacobians: similarly sticky shocks.

Note. Unit productivity shock to HLT and INF, that have similar price rigidity but differ in sector size and labor-flow exposure.

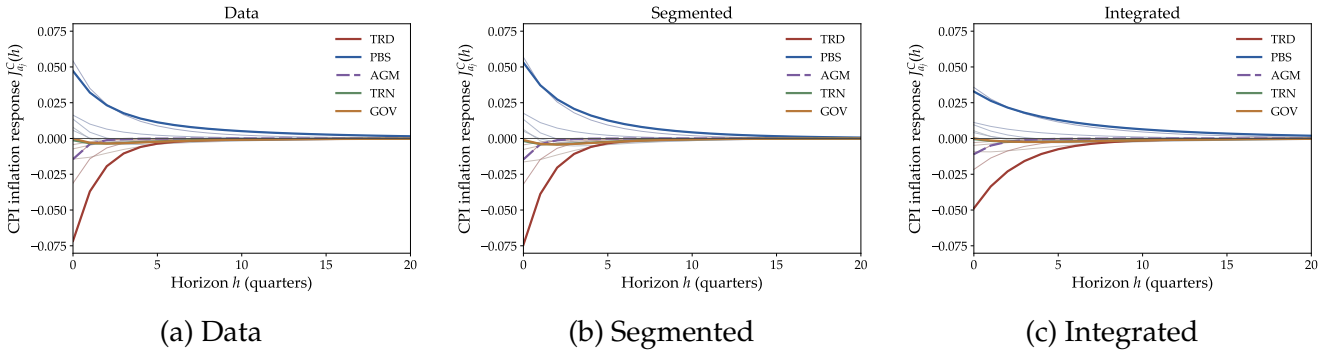


Figure 4: CPI responses to sectoral productivity shocks.

Note. Cumulative CPI inflation response $J_{a_j}^C(h) = \partial \pi_{t+h}^C / \partial \Delta \hat{a}_{j,t}$ to a permanent unit shock in sector j , under strict output-gap targeting. Gray: all shock sectors; colored: selected sectors.

LSR, and DUR have modest impacts but visible negative tails. Persistence is not pinned down by sector size or stickiness alone—it is the interaction of rigidity, labor-market position, and expenditure weight.

The conclusion sharpens once we fix impact inflation: given a unit CPI shock caused by a specific sector, how much remains? Figure 6 rescales each sectoral shock by its own impact CPI response under zero-gap policy. Among well-conditioned sectors ($|\pi_0^C| \geq 0.005$), the future mass from $h = 1, \dots, 20$ ranges from +3.84 impact units for FIR to -4.71 for EDU. Headline inflation at impact masks significant heterogeneity in how much pressure remains to be absorbed.

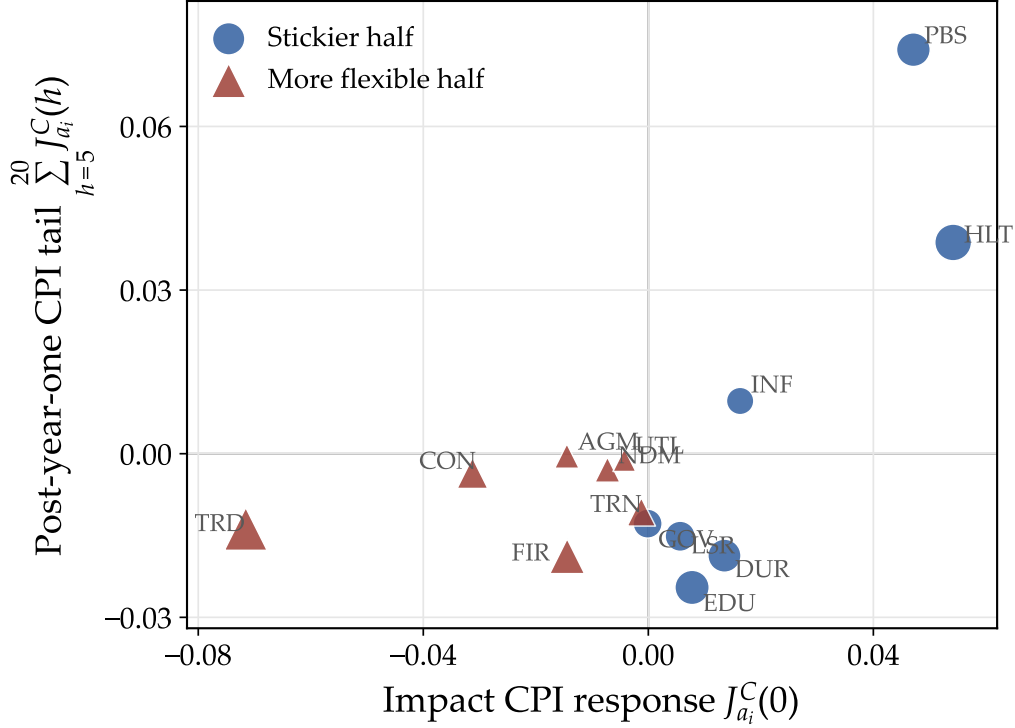


Figure 5: Sectoral map of CPI persistence.

Note. Each point is a permanent unit productivity shock to the labeled sector. Horizontal axis: $J_{a_i}^C(0)$; vertical axis: $\sum_{h=5}^{20} J_{a_i}^C(h)$. Marker size is proportional to α_i . Circles: stickier half; triangles: more flexible half.

6.3.1 Persistent Corridors

Because sectoral shocks are one-time impulses, the broad dispersion in CPI tails must reflect endogenous propagation through the price–employment system. The spectral decomposition (Section 3.3) makes this explicit. For any post-impact observable o_t linear in the state:

$$o_t^{(j)} = \sum_h \lambda_h^{t-1} c_h^{(j)} q_h^o, \quad t \geq 1, \quad (61)$$

where $c_h^{(j)}$ is the initial incidence of shock j on corridor h , and q_h^o is the transmission of that corridor to observable o . We order corridors by decreasing persistence ($|\lambda_h|$). CPI, sectoral inflation, relative prices, and employment are simply different projections of these same underlying dynamics. We use this decomposition to characterize the slow corridors, identify which macroeconomic observables actually reflect them, and explain the equal-impact dispersion.

Slow corridors are mixed. Table 4 reports the six corridors with the largest $|\lambda_h|$. Employment misallocation drives roughly 50–66% of these channels, with price misalignment accounting for the rest. The most persistent sectors—healthcare, professional services, information, education,

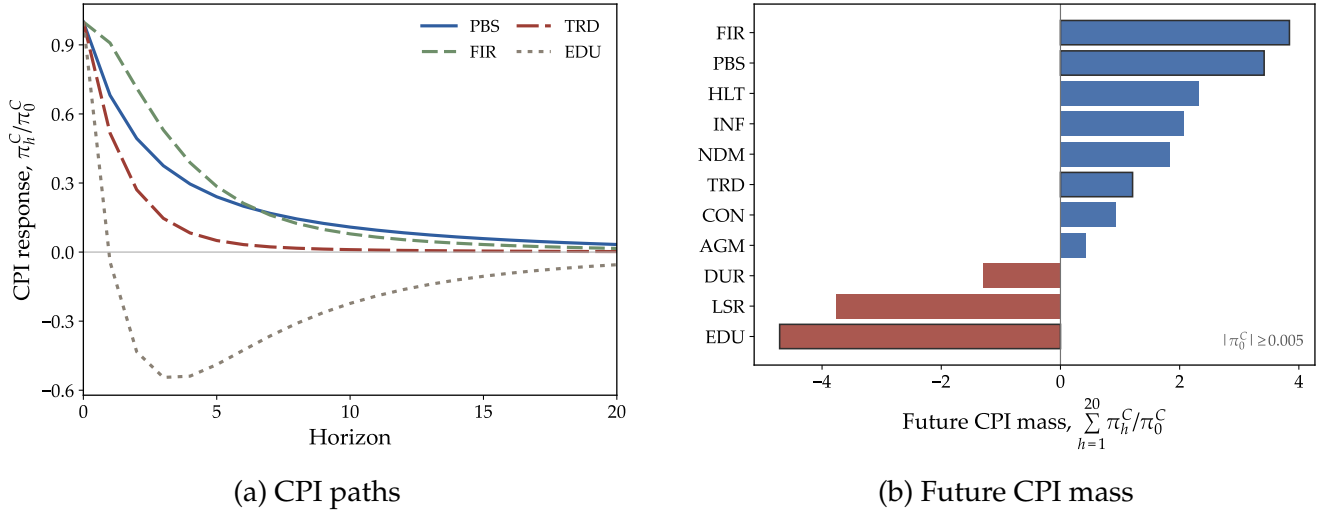


Figure 6: Same impact inflation, different tails.

Note. Panel (a) plots CPI inflation responses to selected sectoral productivity shocks, normalized so that $\pi_0^C = 1$. Panel (b) reports $\sum_{h=1}^{20} \pi_h^C/\pi_0^C$ across sectors with $|\pi_0^C| \geq 0.005$ under a unit shock.

Table 4: Anatomy of the leading persistent corridors.

Corridor	$ \lambda_h $	Half-life	Employment share	Leading sectors
$h = 1$	0.918	8.1	0.55	HLT, PBS, INF
$h = 2$	0.905	7.0	0.51	INF, PBS, HLT
$h = 3$	0.898	6.5	0.64	EDU, CON, DUR
$h = 4$	0.892	6.1	0.58	INF, DUR, PBS/CON
$h = 5$	0.880	5.4	0.66	CON, DUR, GOV
$h = 6$	0.878	5.3	0.64	GOV, DUR, CON

Note. Corridors are ordered by decreasing $|\lambda_h|$. Employment share is the fraction of the eigenvector norm in the employment block. Leading sectors are the three largest absolute loadings across the price and employment blocks; for $h = 4$, PBS is among the top price loadings and CON among the top employment loadings.

construction, and durables—reflect this mix. Pure pricing or pure labor-flow models miss these dynamics; persistence comes directly from their interaction.

Pure labor flows exhibit long half-lives (6.5–15.1 quarters) relative to pure price adjustment (1.8–4.4 quarters under segmentation). The calibrated economy’s leading corridors fall strictly between these bounds (5.3–8.1 quarters). Therefore, the slow dynamics relevant for inflation are not simply labor-market frictions, but joint price–employment phenomena.

Intermediate mobility thickens the spectrum. Figure 7 plots the persistence spectrum against mobility elasticity (ϕ). The limits each produce a thin slow spectrum: segmented markets feature only one corridor with a half-life exceeding four quarters, while integrated markets yield four. Intermediate mobility does not smoothly interpolate between these extremes. Instead, a small positive ϕ (including the calibrated 0.37) activates the employment margin without accelerating

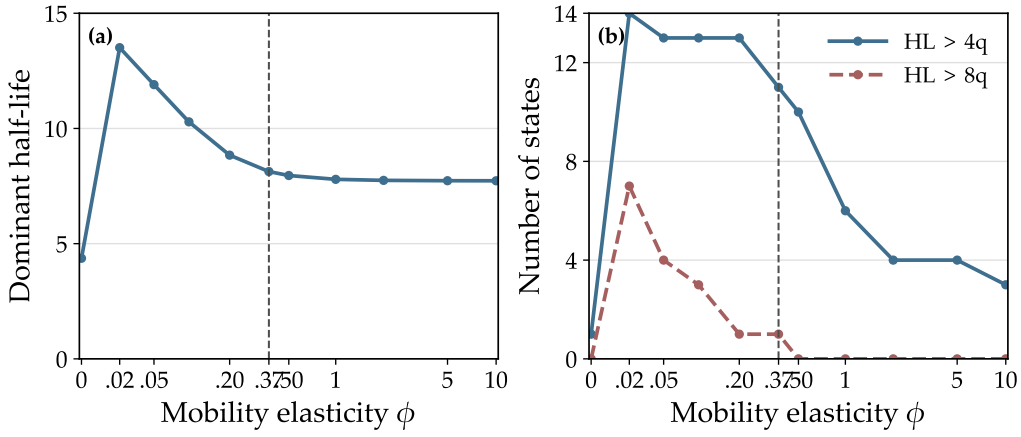


Figure 7: Mobility elasticity and the persistence spectrum.

Note. Dominant half-life and counts of corridors exceeding half-life thresholds as ϕ varies, computed from the $2(N - 1) = 26$ eigenvalues of $\mathbf{P}$. The dashed line marks the calibrated value $\phi = 0.37$.

overall reallocation, generating highly persistent joint corridors absent in either polar case. As ϕ increases further, reallocation accelerates to absorb price gaps, and the spectrum thins toward the integrated limit.

Equal-impact look-through. The corridor decomposition explains the equal-impact dispersion of Section 6.3. Normalizing a sectoral shock by its initial CPI impact, the cumulative CPI response through horizon H is:

$$\sum_{t=1}^H \frac{\pi_t^{C,(j)}}{\pi_0^{C,(j)}} = \sum_h \frac{c_h^{(j)}}{\pi_0^{C,(j)}} q_h^C \frac{1 - \lambda_h^H}{1 - \lambda_h} \quad (62)$$

where q_h^C is the transmission of corridor h to the CPI. Overall persistence depends on the shock's initial incidence across corridors, their transmission to aggregate inflation, and their respective decay rates (λ_h). Figure 8 compares PBS and EDU—sectors with similar expenditure shares but opposite inflation tails.

PBS generates a normalized CPI tail of +3.42, propagating through highly persistent channels ($h = 2$ and $h = 4$, with half-lives of 7.0 and 6.1 quarters). While PBS is sticky, this persistence comes from its slow price adjustment interacting with sluggish labor reallocation through multiple corridors. By contrast, EDU generates a tail of −4.71. Under segmented labor markets, EDU's own price persistence would have a half-life of just 2.0 quarters. Instead, its persistent disinflationary tail is a network effect: an EDU shock induces a negative response along $h = 4$, the same 6.1-quarter corridor that propagates PBS inflation.

Identical headline CPI impacts can mask vastly different inflationary pressures over time. Accurately measuring persistence requires looking through the aggregate index to identify the

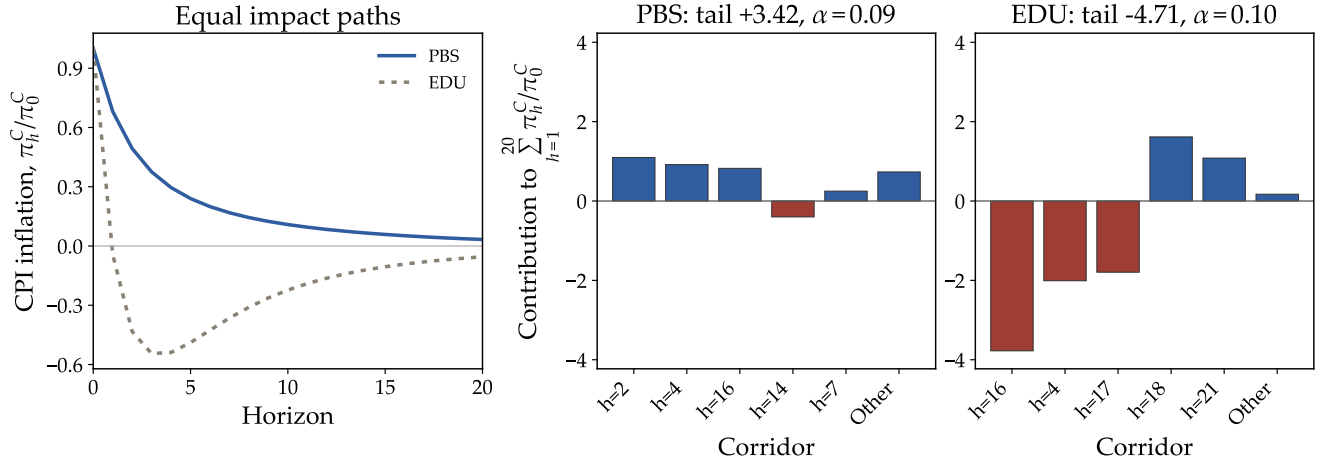


Figure 8: Corridor contributions to equal-impact CPI tails.

Note. The left panel plots CPI responses normalized by impact CPI inflation. The middle and right panels decompose $\sum_{h=1}^{20} \pi_h^C / \pi_0^C$ into the five largest signed corridor contributions plus the remaining corridors, using (62). The corridors h are sorted by persistence ($|\lambda_h|$).

specific price-reallocation networks a shock disrupts.

Persistent and demand-transmitting components. The corridors describe propagation but does not automatically imply demand transmission. We evaluate the quasi-symmetric F^* components using two diagnostics. First, effective rigidity ($\Delta\kappa_k^{\text{eff}}$) asks whether component k connects stickier-than-average sectors, which compounds nominal and real frictions (Proposition 6). Second, demand transmission (d_k) measures whether the component's sectors have enough rigidity dispersion for aggregate demand to tilt relative prices (Proposition 7).

Table 5 reports these diagnostics for the leading data corridors and their closest labor components. Four of the top six corridors ($h = 2, 3, 4, 6$) have negative $\Delta\kappa_k^{\text{eff}}$. By connecting predominantly sticky service sectors (e.g., healthcare, professional services, education), their data half-lives exceed the uniform- κ baseline by 1–3 quarters, confirming first-order amplification. Because these sectors share similar rigidities, their demand transmission is weak ($|d_k| \in [0.005, 0.023]$). In contrast, the two corridors with positive $\Delta\kappa_k^{\text{eff}}$ ($h = 1, 5$) mix sticky and flexible sectors, yielding demand transmissions an order of magnitude larger ($|d_k| = 0.075, 0.030$).

In short, corridors amplified by heterogeneous rigidity connect similarly sticky sectors, while highly transmissive corridors connect mixed-stickiness pairs. The Ramsey problem (Section 6.4) weighs this precise tradeoff between persistence and policy reachability.

Table 5: Corridor matching and effective rigidity.

Data corridor	Matched comp.	Overlap	F^* half-life	Quartic half-life	Data half-life	$\Delta\kappa_k^{\text{eff}}$	$ d_k $
$h = 1$	$k = 2$	0.54	15.1	6.3	8.1	0.031	0.075
$h = 2$	$k = 14$	0.75	6.5	3.9	7.0	-0.087	0.005
$h = 3$	$k = 3$	0.75	12.9	5.8	6.5	-0.077	0.018
$h = 4$	$k = 10$	0.64	7.2	4.2	6.1	-0.070	0.023
$h = 5$	$k = 4$	0.82	11.4	5.4	5.4	0.098	0.030
$h = 6$	$k = 5$	0.92	10.1	5.1	5.3	-0.044	0.016

Note. The matched component is the non-aggregate F^* component whose uniform- κ reallocation eigenvector has the largest absolute overlap with the data corridor. The quartic half-life is the larger stable root of (42) under uniform κ for the matched component. Negative $\Delta\kappa_k^{\text{eff}}$ means the component loads disproportionately on stickier-than-average sectors; by Proposition 6, this raises persistence to first order. $|d_k|$ measures aggregate-demand reachability.

6.4 Monetary Policy and the Misallocation Wedge

6.4.1 Welfare across Policy Closures

We now evaluate the policy problem. The shocks are one-percent negative productivity innovations. We compare the Ramsey allocation from Appendix D.3 with four implementable closures: zero output gap, the baseline Taylor rule, an inflation-only Taylor rule, and strict CPI inflation targeting.

Table 6 reports losses averaged across the fourteen single-sector shocks using expenditure weights. Three patterns stand out. First, the welfare ranking inverts the standard intuition: the more aggressively a closure fights aggregate inflation, the worse it performs. Strict IT is 23% above Ramsey, the inflation-only Taylor rule is 9% above, the standard Taylor rule is 4% above, and zero-gap policy comes within 1.4% of Ramsey. Second, the differences are driven primarily by within-sector inflation costs $\mathcal{W}_\pi$. Strict IT delivers $\pi_t^C = 0$ by construction, but does so through opposing sectoral inflation movements that explode within-sector dispersion ($\mathcal{W}_\pi$ rises from 1.077 under Ramsey to 1.342). The relative-price and employment misallocation components, $\mathcal{W}_\chi$ and $\mathcal{W}_r$, are essentially invariant across closures.

The third pattern is more subtle. Ramsey beats zero-gap by reducing $\mathcal{W}_\pi$ from 1.101 to 1.077, accepting a small output gap ($\mathcal{W}_y$ rises from 0 to 0.003). The misallocation components do not move. Optimal policy uses the misallocation wedge of (51) not to reduce misallocation directly, but to align the path of sectoral inflations so that within-sector dispersion is smaller. The reallocation friction shapes the welfare landscape—it makes $\mathcal{W}_\chi + \mathcal{W}_r$ a meaningful fraction of total loss—but the policy lever acts on the standard multi-sector NK margin.

Table 6: Welfare losses across policy closures.

Policy	$\mathcal{W}$	$\mathcal{W}_\pi$	$\mathcal{W}_y$	$\mathcal{W}_\chi$	$\mathcal{W}_r$
Ramsey	1.191	1.077	0.003	0.057	0.055
Zero-gap	1.208	1.101	0.000	0.055	0.053
Taylor	1.239	1.128	0.002	0.055	0.054
Infl. Taylor	1.297	1.183	0.004	0.056	0.054
Strict IT	1.465	1.342	0.011	0.057	0.056

Notes. Losses are scaled by 10^4 and averaged across single-sector negative productivity shocks using expenditure weights. Components are Calvo price-dispersion loss ($\mathcal{W}_\pi$), output-gap loss ($\mathcal{W}_y$), relative-price distortion ($\mathcal{W}_\chi$), and employment misallocation ($\mathcal{W}_r$).

6.4.2 Sectoral Case Studies

The average masks a systematic sectoral pattern in how Ramsey deviates from zero-gap policy. The right organizing object is the impact wedge in the targeting criterion,

$$m_0^{(j)} \equiv \sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{i0}^{(j)},$$

which under timeless-perspective initialization satisfies $m_0^{(j)} = \tilde{y}_0^{(j)} + \epsilon \pi_0^{C,*(j)}$. The wedge sign tells which side of the standard targeting rule Ramsey deviates to.

Table 7 reports three shocks: PBS, the stickiest sector; TRD, a large flexible sector; and AGM, the most flexible sector but with small expenditure weight. PBS has a negative wedge: under zero-gap policy the shock generates CPI deflation, and Ramsey expands to lean against it, though by less than strict CPI targeting would require. TRD and AGM have positive wedges: under zero-gap they generate CPI inflation, and Ramsey contracts. In every case Ramsey moderates the zero-gap CPI response without reversing it, choosing an interior point between zero-gap and strict CPI stabilization.

The pattern is systematic. Across all fourteen sectors, the rank correlation between κ_j and the impact Ramsey wedge is +0.952: flexible sectors generate positive wedges and trigger contractionary deviations, sticky sectors generate negative wedges and trigger expansionary deviations. Twelve of the fourteen sectors preserve the sign of impact CPI between zero-gap and Ramsey, with Ramsey moderating the zero-gap response; only GOV and TRN come close to sign flips, and both have near-zero wedges.

6.4.3 Spectral Decomposition of the Ramsey Wedge

The cross-sectional wedge pattern of Section 6.4.2 projects onto the F^* components through Proposition 10: along the component basis, the wedge is a sum $\sum_k d_k \eta_{kt}^*$ of demand-transmission coefficients times shadow values. We use this decomposition to identify which labor components

Table 7: Sectoral case studies of optimal policy response.

Sector	Policy	π_0^C	$\tilde{y}_0$	m_0	$\mathcal{W}$
PBS ($\kappa = 0.008$)	Ramsey	-0.028	+0.031	-0.191	2.061
PBS	Zero-gap	-0.047	0	-	2.107
PBS	Taylor	-0.025	+0.025	-	2.228
PBS	Strict IT	0	+0.056	-	2.656
TRD ($\kappa = 0.235$)	Ramsey	+0.051	-0.056	+0.354	1.025
TRD	Zero-gap	+0.072	0	-	1.057
TRD	Taylor	+0.048	-0.048	-	1.094
TRD	Strict IT	0	-0.128	-	1.708
AGM ($\kappa = 0.758$)	Ramsey	+0.012	-0.009	+0.091	0.068
AGM	Zero-gap	+0.014	0	-	0.068
AGM	Taylor	+0.011	-0.011	-	0.068
AGM	Strict IT	0	-0.039	-	0.085

Notes. Impact responses and the Ramsey wedge m_0 are scaled by 100 for a one-percent negative productivity shock; welfare losses are scaled by 10^4 . The wedge $m_0 = \tilde{y}_0 + \epsilon \pi_0^{C,*}$ is reported only for the Ramsey allocation.

carry the wedge in the calibrated economy.

Table 8 ranks the F^* components by their average absolute share in the present-value Ramsey wedge across the fourteen single-sector shocks. The wedge is concentrated: two components (5 and 0) account for the bulk of the wedge mass in every shock, with mean absolute shares of 7.83 and 6.80, against 2.33 or below for the next four. These two components have the largest d_k values among the dominant contributors (+0.086 and +0.075), together with high F^* persistence ($\rho_k = 0.926$ and 0.955 ; half-lives 9.0 and 15.1 quarters). Optimal policy acts through the components that combine high persistence with high demand transmission, exactly as the spectral targeting rule (56) predicts.

6.5 Robustness and Counterfactuals

[Appendix, in construction]

7 Conclusion

We have developed a multi-sector New Keynesian model in which frictional labor reallocation, summarized by a mobility elasticity and a steady-state transition matrix, reshapes the transmission of sectoral shocks and the design of optimal monetary policy. Three results stand out. First, labor mobility creates countervailing effects on sectoral inflation: it dampens impact but slows convergence. Second, the persistence of shocks depends not on average mobility or average stickiness alone, but on whether displaced workers flow into flexible or sticky sectors. Third,

Table 8: Spectral decomposition of the Ramsey wedge.

Component	ρ_k	Half-life (q)	d_k	Mean $ s_k $	Top-3 count
5	0.926	9.0	+0.086	7.83	9
0	0.955	15.1	+0.075	6.80	11
8	0.909	7.2	+0.023	2.33	5
2	0.941	11.4	−0.030	2.10	2
7	0.917	8.0	+0.046	1.37	7
10	0.904	6.8	−0.031	1.26	3

Notes. The six F^* components with the largest mean absolute share of the present-value Ramsey wedge across the fourteen single-sector shocks. d_k is the demand-transmission coefficient (Proposition 7). Mean $|s_k|$ is the cross-sector average absolute share; signed shares can offset, so absolute shares can sum to more than one. “Top-3 count” is the number of shocked sectors for which the component is among the three largest absolute contributors to the wedge.

optimal policy under commitment deviates from standard sticky-price targeting precisely when price rigidity aligns with the slowest-mixing component of the reallocation network, as the planner exploits flexible sectors’ price responsiveness to redirect worker flows.

Several extensions are natural. Our model assumes that aggregate demand affects demand for all sectors symmetrically: in practice, interest rate changes have heterogeneous incidence across sectors—durables respond more than services, for instance—which would change the incidence of aggregate demand onto sectoral demand. Incorporating production networks alongside the labor network would allow both input-output and worker-flow linkages to shape persistence and policy simultaneously.

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Appendix

A Log-Linearization Derivations

This appendix details the log-linearization of the model around the steady state. Because all output is consumed, we impose goods market clearing $y_t = c_t$ and $y_{it} = c_{it}$ throughout, using y to denote both output and consumption.

Notation. For any variable X_t , we write $x_t \equiv \log X_t - \log \bar{X}$ for its log-deviation from steady state. We use the following conventions throughout:

- *Natural (flexible-price) levels:* x_t^{nat} denotes the value under flexible prices, defined conditional on the actual predetermined state at t ; Lemma 1 gives the explicit construction.
- *Gaps:* $\tilde{x}_t \equiv x_t - x_t^{\text{nat}}$ denotes the deviation from the natural level.
- *Relative prices:* $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the log sectoral price relative to the CPI.
- *Relative price distortion:* $\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}$. Equivalently, $\chi_{it} = \hat{p}_{it} - \hat{p}_{it}^{\text{nat},0}$ where $\hat{p}_{it}^{\text{nat},0} \equiv -\hat{a}_{it}$ is the relative price when prices are flexible and inherited employment is at its steady-state distribution (Lemma 1).

We write $\stackrel{(1)}{=}$ for equalities that hold up to a first-order approximation, and $\stackrel{(2)}{=}$ for second-order. Plain $=$ denotes exact relationships or definitions.

A.1 The Macro Block

A.1.1 Sectoral Marginal Cost

Real marginal cost in sector i (in log-deviations) is given by $mc_{it} - p_{it} = w_{it} - a_{it} - p_{it}$. Using the intratemporal labor supply condition (6) and substituting aggregate consumption with output ($c_t = y_t$), the sectoral wage satisfies:

$$w_{it} - p_t^C = \phi l_{it} + \gamma y_t. \quad (\text{A.1})$$

Substituting this into the marginal cost expression yields:

$$mc_{it} - p_{it} = \gamma y_t + \phi l_{it} - a_{it} - \hat{p}_{it},$$

where $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the relative sectoral price. To eliminate l_{it} , we use the log-linearized production function ($l_{it} = y_{it} - a_{it} - n_{it}$) and substitute out sectoral output y_{it} using the demand

curve ($y_{it} = y_t - \hat{p}_{it}$). Note that n_{it} here is the *contemporaneous*, post-reallocation employment share in sector i — this is the mass of workers physically producing in sector i at date t , and hence the relevant margin for marginal cost. Grouping terms directly yields Equation (17) from the main text:

$$mc_{it} - p_{it} \stackrel{(1)}{=} (\gamma + \varphi)y_t - (1 + \varphi)(\hat{p}_{it} + a_{it}) - \varphi n_{it}. \quad (\text{A.2})$$

A.1.2 Phillips Curve

Standard log-linearization of the Calvo pricing problem (3) yields the sectoral New Keynesian Phillips curve. A re-optimizing firm in sector i sets its price as a markup over a weighted average of current and expected future marginal costs. Log-linearizing the optimal reset price around the zero-inflation steady state, and combining with the Calvo price index evolution $P_{it} = [\delta_i P_{it-1}^{1-\epsilon_i} + (1 - \delta_i)(P_{it}^*)^{1-\epsilon_i}]^{1/(1-\epsilon_i)}$, yields:

$$\pi_{it} \stackrel{(1)}{=} \kappa_i(mc_{it} - p_{it}) + \beta \mathbb{E}_t \pi_{it+1}, \quad (\text{A.3})$$

where $\kappa_i \equiv (1 - \delta_i)(1 - \beta\delta_i)/\delta_i$. This is Equation (21).

A.1.3 The IS Curve and Natural Real Rate

Note on convention: As is standard, we make an exception to the log-deviation convention for interest rates. We let i_t and r_t^{nat} denote the levels of the net nominal and natural real interest rates, respectively.

The Euler equation (7) is

$$1 = \beta(1 + i_t) \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+1}^C} \right].$$

Taking logs and using the first-order approximation $\log(1 + i_t) \stackrel{(1)}{=} i_t$:

$$0 \stackrel{(1)}{=} \log \beta + i_t - \gamma \mathbb{E}_t(y_{t+1} - y_t) - \mathbb{E}_t \pi_{t+1}^C,$$

where we have substituted $c_t = y_t$ (goods market clearing) and $\pi_{t+1}^C = p_{t+1}^C - p_t^C$. Rearranging yields:

$$y_t \stackrel{(1)}{=} \mathbb{E}_t y_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t \pi_{t+1}^C + \log \beta). \quad (\text{A.4})$$

The same equation holds under flexible prices with y_t^{nat} in place of y_t and i_t^{nat} in place of i_t (and

$\pi_t^C = 0$ under flexible prices). We define the natural real rate (in levels) as:

$$r_t^{\text{nat}} \equiv -\log \beta + \gamma(\mathbb{E}_t y_{t+1}^{\text{nat}} - y_t^{\text{nat}}).$$

Subtracting the flexible-price Euler equation from (A.4) gives the IS curve, Equation (22):

$$\tilde{y}_t \stackrel{(1)}{=} \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma} \left(i_t - \mathbb{E}_t \pi_{t+1}^C - r_t^{\text{nat}} \right). \quad (\text{A.5})$$

A.1.4 Goods Market Clearing and Relative Prices

Cobb–Douglas demand gives $P_{it} C_{it} = \alpha_i P_t^C C_t$, so sectoral output satisfies $P_{it} Y_{it} = \alpha_i P_t^C C_t$. In logs:

$$y_{it} - y_t = -\hat{p}_{it}. \quad (\text{A.6})$$

Log aggregate output is $y_t = \sum_i \alpha_i a_{it} + \sum_i \alpha_i (n_{it} + l_{it})$, using $y_{it} = a_{it} + n_{it} + l_{it}$ (from the production function and labor market clearing) weighted by consumption shares.

The CPI definition $P_t^C = \prod_i P_{it}^{\alpha_i}$ implies $p_t^C = \sum_i \alpha_i p_{it}$, so CPI inflation is $\pi_t^C = \sum_i \alpha_i \pi_{it}$. Relative prices evolve as:

$$\hat{p}_{it} - \hat{p}_{it-1} = (p_{it} - p_{it-1}) - (p_t^C - p_{t-1}^C) = \pi_{it} - \pi_t^C, \quad (\text{A.7})$$

which is Equation (15).

A.2 Dynamic Labor Reallocation Block

The dynamic reallocation block consists of the flow utility, the Bellman equation, and the employment evolution equation.

A.2.1 Flow Utility

Using the intratemporal labor supply condition (6), flow utility simplifies to $U_{it}^{\text{flow}} = \zeta \frac{\varphi}{1+\varphi} L_{it}^{1+\varphi}$. Define the normalized deviation $u_{it} \equiv (U_{it}^{\text{flow}} - \bar{U}^{\text{flow}}) / (\varphi \zeta \bar{L}^{1+\varphi})$:

$$\frac{U_{it}^{\text{flow}} - \bar{U}^{\text{flow}}}{\varphi \zeta \bar{L}^{1+\varphi}} = \frac{\zeta \frac{\varphi}{1+\varphi} (L_{it}^{1+\varphi} - \bar{L}^{1+\varphi})}{\varphi \zeta \bar{L}^{1+\varphi}} = \frac{L_{it}^{1+\varphi} - \bar{L}^{1+\varphi}}{(1+\varphi) \bar{L}^{1+\varphi}}.$$

A first-order expansion of $L_{it}^{1+\varphi}$ around $\bar{L}$ gives $L_{it}^{1+\varphi} \approx \bar{L}^{1+\varphi}(1 + (1 + \varphi)l_{it})$, so $u_{it} \stackrel{(1)}{=} l_{it}$. Substituting $l_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$ yields Equation (20):

$$u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}. \quad (\text{A.8})$$

A.2.2 Value Function Recursion

The Bellman equation (9) is:

$$V_{it} = U_{it}^{\text{flow}} + \beta \mathbb{E}_t \left[\frac{1}{\theta} \log \left(\sum_{j \in \mathcal{N}} \exp(\theta[V_{j,t+1} - \tau^{ij}]) \right) \right].$$

In steady state, $\bar{V}_i = \bar{U}^{\text{flow}} + \beta \Lambda_i(\bar{\mathbf{z}})$ where $\Lambda_i(\mathbf{z}) \equiv \frac{1}{\theta} \log \left(\sum_j \exp(\theta z_j) \right)$ and $\bar{z}_j = \bar{V}_j - \tau^{ij}$. The first-order Taylor expansion of Λ_i around $\bar{\mathbf{z}}$ uses the standard logit property:

$$\left. \frac{\partial \Lambda_i}{\partial z_j} \right|_{\bar{\mathbf{z}}} = \frac{\exp(\theta \bar{z}_j)}{\sum_{j'} \exp(\theta \bar{z}_{j'})} = F_{ij},$$

where $F_{ij} = \bar{\mu}^{ij}$ is the steady-state transition probability from sector i to sector j (Equation (8) evaluated at steady state). Therefore, with $z_{j,t+1} = V_{j,t+1} - \tau^{ij}$:

$$\Lambda_i(\mathbf{z}_{t+1}) - \Lambda_i(\bar{\mathbf{z}}) \stackrel{(1)}{=} \sum_j F_{ij} (V_{j,t+1} - \bar{V}_j).$$

Subtracting the steady-state Bellman from the time- t Bellman, taking $\mathbb{E}_t$ through the linear term, and normalizing by $\varphi \bar{L}^{1+\varphi}$:

$$v_{it} = u_{it} + \beta \sum_j F_{ij} \mathbb{E}_t[v_{j,t+1}].$$

In matrix notation, this is Equation (19):

$$\mathbf{v}_t = \mathbf{u}_t + \beta F \mathbb{E}_t[\mathbf{v}_{t+1}]. \quad (\text{A.9})$$

A.2.3 Employment Evolution

Employment evolves according to $N_{jt} = \sum_i \mu_t^{ij} N_{i,t-1}$, where the transition probabilities are given by the logit form (8):

$$\mu_t^{ij} = \frac{\exp(\theta[V_{jt} - \tau^{ij}])}{\sum_{j'} \exp(\theta[V_{j't} - \tau^{ij'}])}.$$

The transition probability μ_t^{ij} has the logit form, so its first-order expansion around the steady state $(F_{ij}, \bar{V}_j)$ is:

$$\begin{aligned}\mu_t^{ij} - F_{ij} &\stackrel{(1)}{=} F_{ij} \theta \left((V_{jt} - \bar{V}_j) - \sum_{j'} F_{ij'} (V_{j't} - \bar{V}_{j'}) \right) \\ &= F_{ij} \phi \left(v_{jt} - \sum_{j'} F_{ij'} v_{j't} \right),\end{aligned}$$

where the second line uses the normalization $v_{jt} = (V_{jt} - \bar{V}_j) / (\phi \zeta \bar{L}^{1+\varphi})$ and $\phi \equiv \varphi \theta \zeta \bar{L}^{1+\varphi}$, so that $\theta(V_{jt} - \bar{V}_j) = \phi v_{jt}$. That is:

$$\mu_t^{ij} - F_{ij} \stackrel{(1)}{=} \phi F_{ij} (v_{jt} - [F \mathbf{v}_t]_i). \quad (\text{A.10})$$

Deriving the employment law of motion. Log-linearize $N_{jt} = \sum_i \mu_t^{ij} N_{i,t-1}$ around $\bar{N}_j = \alpha_j$. Since $\bar{N}_j = \sum_i F_{ij} \bar{N}_i$ (stationarity of $\bar{\mathbf{N}}$ under F), taking total differentials:

$$\bar{N}_j n_{jt} \stackrel{(1)}{=} \sum_i F_{ij} \bar{N}_i n_{i,t-1} + \sum_i \bar{N}_i (\mu_t^{ij} - F_{ij}).$$

We handle the two terms in turn. For the first term, the definition of the backward matrix $B_{ji} \equiv \bar{N}_i F_{ij} / \bar{N}_j$ (Equation (10)) gives $\bar{N}_i F_{ij} = \bar{N}_j B_{ji}$, so:

$$\frac{1}{\bar{N}_j} \sum_i F_{ij} \bar{N}_i n_{i,t-1} = \sum_i B_{ji} n_{i,t-1} = [B \mathbf{n}_{t-1}]_j.$$

For the second term, substituting (A.10):

$$\sum_i \bar{N}_i (\mu_t^{ij} - F_{ij}) = \phi \sum_i \bar{N}_i F_{ij} (v_{jt} - [F \mathbf{v}_t]_i).$$

Applying $\bar{N}_i F_{ij} = \bar{N}_j B_{ji}$ and dividing by $\bar{N}_j$:

$$\begin{aligned}(\text{second term}) / \bar{N}_j &= \phi \sum_i B_{ji} (v_{jt} - [F \mathbf{v}_t]_i) \\ &= \phi (v_{jt} - [BF \mathbf{v}_t]_j),\end{aligned}$$

where the second line uses $\sum_i B_{ji} = 1$ (row-stochasticity of B).

Combining both terms:

$$n_{jt} = [B \mathbf{n}_{t-1}]_j + \phi [(I - BF) \mathbf{v}_t]_j.$$

In vector form, this is Equation (18):

$$\mathbf{n}_t = B \mathbf{n}_{t-1} + \phi(I - BF) \mathbf{v}_t. \quad (\text{A.11})$$

Limiting cases. When $\phi = 0$ (segmented markets), the second term vanishes and $\mathbf{n}_t = B \mathbf{n}_{t-1}$. Starting from $\mathbf{n}_{-1} = \mathbf{0}$, we have $\mathbf{n}_t = \mathbf{0}$ for all $t \geq 0$: no reallocation occurs.

When $\phi \rightarrow \infty$, boundedness of $\mathbf{n}_t$ requires $(I - BF) \mathbf{v}_t \rightarrow \mathbf{0}$. Since BF is row-stochastic and primitive, $\ker(I - BF) = \text{span}\{\mathbf{1}\}$, so $\mathbf{v}_t \rightarrow c_t \mathbf{1}$: continuation values equalize across sectors *within each period*. Substituting into the Bellman recursion (19) and using $F \mathbf{1} = \mathbf{1}$ gives $\mathbf{u}_t = m_t \mathbf{1}$, where $m_t \equiv c_t - \beta \mathbb{E}_t[c_{t+1}]$. By (20), $u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$, so $n_{it} = y_t - \hat{p}_{it} - a_{it} - m_t$. Imposing $\sum_i \alpha_i n_{it} = 0$ and $\sum_i \alpha_i \hat{p}_{it} = 0$ yields $m_t = y_t - \sum_j \alpha_j a_{jt}$, and substituting back gives $n_{it} = -(\hat{p}_{it} + \hat{a}_{it}) = -\chi_{it}$. This identity holds for *all* $t \geq 0$ because reallocation is contemporaneous.

B Second-Order Expansion Details

This appendix derives the quadratic welfare loss function in 8. We expand household utility around the efficient (natural) allocation to second order, term by term.

B.1 Proposition 8: Dynamic LQ

We derive the per-period welfare loss under the timeless perspective:

$$\mathcal{L}_t \stackrel{(2)}{=} \frac{\gamma + \varphi}{2} \hat{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1 + \varphi}{2} \chi_{it}^2 + \varphi \chi_{it} n_{it} + \frac{\varphi}{2} n_{it}^2 \right].$$

The derivation proceeds in two steps. Step 1 expands the consumption-minus-labor-disutility component $u(C_t) - \sum_i N_{it} v(L_{it})$ to second order in steady-state deviations. Step 2 shows that the mobility cost Ψ_t is a term whose discounted sum telescopes into a boundary term that enters the Ramsey Lagrangian only through the date-0 first-order conditions. Under the timeless perspective, the per-period loss is therefore the second-order expansion of $u - \sum Nv$ alone.

Notation. All variables denote log-deviations from the efficient steady state, as defined in Appendix A. We write $h_{it} \equiv l_{it} - l_t^\alpha$ for cross-sector hours and normalize by $U_C \bar{C} = \bar{C}^{1-\gamma} = \zeta \bar{L}^{1+\varphi}$. Three identities are used throughout:

- (i) *Production:* $y_t = \bar{a}_t + l_t^\alpha + \sum_i \alpha_i n_{it} + \sum_i \alpha_i \zeta_{it}$, where $\zeta_{it} \equiv -\log \Delta_{it} \leq 0$ is the within-sector price-dispersion wedge. This is exact under Cobb-Douglas utility.
- (ii) *Cross-section:* $h_{it} \stackrel{(1)}{=} -(\chi_{it} + n_{it})$, from goods market clearing and the production function.

(iii) *Adding-up*: $\sum_i \alpha_i n_{it} \stackrel{(2)}{=} -\frac{1}{2} \sum_i \alpha_i n_{it}^2$, from $\sum_i \alpha_i e^{n_{it}} = 1$.

B.1.1 Second-Order Expansion of $u(C_t) - \sum_i N_{it} v(L_{it})$

A standard expansion of $u(C)$ gives:

$$\frac{u(C_t) - u(\bar{C})}{U_C \bar{C}} \stackrel{(2)}{=} y_t + \frac{1-\gamma}{2} y_t^2. \quad (\text{B.1})$$

For labor disutility, expanding $N_{it} v(L_{it}) = \frac{\alpha_i \bar{L}^{1+\varphi}}{1+\varphi} \exp(n_{it} + (1+\varphi)l_{it})$ to second order, summing over i , dividing by $U_C \bar{C}$, and using identity (iii) to cancel the n_{it} and n_{it}^2 terms:

$$\frac{\sum_i N_{it} v(L_{it}) - \sum_i \alpha_i v(\bar{L})}{U_C \bar{C}} \stackrel{(2)}{=} l_t^\alpha + \sum_i \alpha_i n_{it} h_{it} + \frac{1+\varphi}{2} [(l_t^\alpha)^2 + \mathcal{M}_t], \quad (\text{B.2})$$

where $\mathcal{M}_t \equiv \sum_i \alpha_i h_{it}^2$. Subtracting (B.2) from (B.1):

$$\frac{[u(C_t) - \sum_i N_{it} v(L_{it})] - \text{SS}}{U_C \bar{C}} \stackrel{(2)}{=} \underbrace{(y_t - l_t^\alpha)}_{\text{first-order}} + \underbrace{\frac{1-\gamma}{2} y_t^2 - \frac{1+\varphi}{2} (l_t^\alpha)^2}_{\text{aggregate quadratic}} - \underbrace{\frac{1+\varphi}{2} \mathcal{M}_t + \sum_i \alpha_i n_{it} h_{it}}_{\text{cross-sector}}. \quad (\text{B.3})$$

We evaluate each group of terms in turn.

Aggregate quadratic. Since $l_t^\alpha \stackrel{(1)}{=} y_t - \bar{a}_t$ (from identity (i)), the aggregate quadratic in (B.3) is:

$$\frac{1-\gamma}{2} y_t^2 - \frac{1+\varphi}{2} (y_t - \bar{a}_t)^2 = -\frac{\gamma+\varphi}{2} y_t^2 + (1+\varphi) y_t \bar{a}_t - \frac{1+\varphi}{2} \bar{a}_t^2. \quad (\text{B.4})$$

Substituting $y_t = y_t^{\text{nat}} + \tilde{y}_t$ with $y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \bar{a}_t$, the coefficient of $\tilde{y}_t$ is $-(\gamma+\varphi) y_t^{\text{nat}} + (1+\varphi) \bar{a}_t = 0$. Hence:

$$(\text{B.4}) = -\frac{\gamma+\varphi}{2} \tilde{y}_t^2 + (\text{t.i.p}), \quad (\text{B.5})$$

with t.i.p denoting terms independent of policy.¹⁴

¹⁴The vanishing of the cross-term is the multi-sector counterpart of the standard result: the efficient-subsidy condition ensures that the first variation of $u - \sum Nv$ with respect to aggregate output vanishes at the natural level.

Cross-sector terms. Using identity (ii), the cross-sector contribution in (B.3) is, for each sector i (with the minus sign from the subtraction):

$$-\frac{1+\varphi}{2}(\chi_{it} + n_{it})^2 + n_{it}(\chi_{it} + n_{it}) = -\frac{1+\varphi}{2}\chi_{it}^2 - \varphi\chi_{it}n_{it} - \frac{\varphi-1}{2}n_{it}^2. \quad (\text{B.6})$$

Output–labor gap. The first-order term $y_t - l_t^\alpha$ in (B.3) has a nonzero second-order component. From the exact production identity (i) and the exogeneity of $\bar{a}_t$:

$$y_t - l_t^\alpha = \bar{a}_t + \sum_i \alpha_i n_{it} + \sum_i \alpha_i \xi_{it} \stackrel{(2)}{=} \text{t.i.p.} - \frac{1}{2} \sum_i \alpha_i n_{it}^2 + \sum_i \alpha_i \xi_{it}, \quad (\text{B.7})$$

using identity (iii). The Calvo price-dispersion recursion implies the present-value equivalence (assuming $\xi_{i,-1} = 0$):

$$\sum_{t=0}^{\infty} \beta^t \sum_i \alpha_i \xi_{it} \stackrel{(2)}{=} - \sum_{t=0}^{\infty} \beta^t \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2. \quad (\text{B.8})$$

Assembly. Substituting (B.5), (B.6), and (B.7) into (B.3), the period return relative to steady state is, in discounted-sum equivalence:

$$\begin{aligned} \frac{[u - \sum Nv] - \text{SS}}{U_C \bar{C}} &\stackrel{(2)}{=} \text{t.i.p.} - \frac{\gamma + \varphi}{2} \tilde{y}_t^2 - \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 \\ &\quad - \sum_i \alpha_i \left[\frac{1+\varphi}{2} \chi_{it}^2 + \varphi \chi_{it} n_{it} + \frac{\varphi}{2} n_{it}^2 \right], \end{aligned} \quad (\text{B.9})$$

where the coefficient $\frac{\varphi}{2}$ on n_{it}^2 combines $\frac{\varphi-1}{2}$ from the cross-sector terms (B.6) and $\frac{1}{2}$ from the output–labor gap (B.7). The former reflects the convexity of the disutility of hours; the latter reflects the concavity of the Cobb–Douglas aggregator, which penalizes uneven employment even holding total labor input fixed.

Negating (B.9) and dropping terms independent of policy:

$$\mathcal{L}_t^{u-Nv} \stackrel{(2)}{=} \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1+\varphi}{2} \chi_{it}^2 + \varphi \chi_{it} n_{it} + \frac{\varphi}{2} n_{it}^2 \right], \quad (\text{B.10})$$

in discounted-sum equivalence.¹⁵

¹⁵Because the π_{it}^2 term arises from the present-value equivalence (B.8), equation (B.10) is the per-period term whose discounted sum equals the normalized welfare loss to second order.

B.1.2 Mobility Cost Telescopes into a Startup Term

Step 1 expanded $u(C_t) - \sum_i N_{it} v(L_{it})$. The full per-period utility also includes the mobility cost $-\Psi_t$, incurred at the start of period t on the inherited mass $N_{i,t-1}$. We now construct a sector-specific net continuation value for the *pre-reallocation* stage that strips out future flow utility, yielding a boundary term weighted by the predetermined initial distribution $\{N_{i,-1}\}$.

Pre-reallocation value function. Let

$$V_{it}^{pre} \equiv \frac{1}{\theta} \log \sum_j \exp(\theta[V_{jt} - \tau^{ij}]) \quad (\text{B.11})$$

denote the value of a worker labeled in sector i at the start of period t , before drawing a preference shock and reallocating. By the envelope theorem applied to the logit optimization,

$$V_{it}^{pre} = \sum_j \mu_t^{ij} V_{jt} - \psi_i(\{\mu_t^{ij}\}_j). \quad (\text{B.12})$$

The Bellman equation (9) for post-reallocation V_{jt} reads $V_{jt} = U_{jt}^{flow} + \beta \mathbb{E}_t V_{j,t+1}^{pre}$.

Sector-specific net continuation value. Define the flow-utility continuation at the pre-reallocation stage

$$F_{it}^{pre} \equiv \sum_j \mu_t^{ij} [U_{jt}^{flow} + \beta \mathbb{E}_t F_{j,t+1}^{pre}], \quad (\text{B.13})$$

and the net pre-reallocation continuation value

$$\check{V}_{it}^{pre} \equiv V_{it}^{pre} - F_{it}^{pre}. \quad (\text{B.14})$$

F_{it}^{pre} captures the expected discounted flow utility a pre-reallocation worker in sector i would earn along the realized transition path; $\check{V}_{it}^{pre}$ isolates the option value of reallocation net of mobility costs.

Lemma B.1. Let $\check{\mathcal{B}}_t \equiv \sum_i N_{i,t-1} \check{V}_{it}^{pre}$. Then:

$$-\Psi_t = \check{\mathcal{B}}_t - \beta \mathbb{E}_t \check{\mathcal{B}}_{t+1}. \quad (\text{B.15})$$

Consequently, if $\lim_{T \rightarrow \infty} \beta^T \mathbb{E}_0 |\check{\mathcal{B}}_T| = 0$, then

$$\mathcal{W}_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[u(C_t) - \sum_i N_{it} v(L_{it}) \right] + \check{\mathcal{B}}_0. \quad (\text{B.16})$$

Proof. Combining the envelope (B.12) with $V_{jt} = U_{jt}^{\text{flow}} + \beta \mathbb{E}_t V_{j,t+1}^{\text{pre}}$:

$$V_{it}^{\text{pre}} = \sum_j \mu_t^{ij} \left[U_{jt}^{\text{flow}} + \beta \mathbb{E}_t V_{j,t+1}^{\text{pre}} \right] - \psi_i(\{\mu_t^{ij}\}_j).$$

Subtracting the F_{it}^{pre} recursion (B.13) and using $\check{V} = V - F$ term by term:

$$\check{V}_{it}^{\text{pre}} = \beta \sum_j \mu_t^{ij} \mathbb{E}_t \check{V}_{j,t+1}^{\text{pre}} - \psi_i(\{\mu_t^{ij}\}_j). \quad (\text{B.17})$$

Multiplying by $N_{i,t-1}$, summing over i , and using the flow identity $\sum_i N_{i,t-1} \mu_t^{ij} = N_{jt}$:

$$\sum_i N_{i,t-1} \check{V}_{it}^{\text{pre}} = \beta \mathbb{E}_t \sum_j N_{jt} \check{V}_{j,t+1}^{\text{pre}} - \sum_i N_{i,t-1} \psi_i(\{\mu_t^{ij}\}_j).$$

The left-hand side is $\check{\mathcal{B}}_t$; the first right-hand-side term is $\beta \mathbb{E}_t \check{\mathcal{B}}_{t+1}$ since $N_{jt} = N_{j,(t+1)-1}$; the second is Ψ_t by definition. This yields (B.15). Multiplying by β^t , summing, and applying the transversality condition telescopes the mobility cost into $-\check{\mathcal{B}}_0$. Substituting into $\mathcal{W}_0 = \mathbb{E}_0 \sum_{t \geq 0} \beta^t [u - \sum N v - \Psi_t]$ yields (B.16). $\square$

Equation (B.16) is exact: the entire contribution of the mobility block to lifetime welfare is captured by $\check{\mathcal{B}}_0 = \sum_i N_{i,-1} \check{V}_{i0}^{\text{pre}}$. The key property of this construction is that $\check{\mathcal{B}}_0$ is a *pure startup term*: it depends on the future allocation only through the initial pre-reallocation net continuation values $\{\check{V}_{i0}^{\text{pre}}\}$, which are themselves determined by the recursion (B.17), weighted by the predetermined initial distribution $\{N_{i,-1}\}$. In the Ramsey problem with $\{\check{V}_{it}^{\text{pre}}\}$ treated as auxiliary variables subject to this recursion, $\check{\mathcal{B}}_0$ affects only the date-0 first-order conditions—it is the source of the well-known tension between the optimal long-run policy rule and the incentive to deviate at the initial date.

Under the timeless perspective (Woodford, 2003), the planner commits to a policy rule that would have been optimal had it been adopted far in the past. This means requiring that the same targeting criterion hold at every date, including $t = 0$, rather than exploiting the special initial conditions. $\check{\mathcal{B}}_0$ affects only the exceptional startup conditions of the date-0 Ramsey problem. Under the timeless perspective, these startup conditions are replaced by the same self-consistent targeting conditions that apply for $t \geq 1$, and the per-period loss for the Ramsey problem under the timeless perspective is therefore $\mathcal{L}_t^{u-Nv}$ from (B.10).

Combining Steps 1 and 2 and dropping terms independent of policy, the per-period loss for the Ramsey problem is:

$$\mathcal{L}_t \stackrel{(2)}{=} \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2 \kappa_i} \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1 + \varphi}{2} \chi_{it}^2 + \varphi \chi_{it} n_{it} + \frac{\varphi}{2} n_{it}^2 \right], \quad (\text{B.18})$$

in discounted-sum equivalence, which is equation (46). $\square$

C Proofs of Propositions and Lemmas

C.1 Proofs for Section 3

Proof of Proposition 1. Inspection of (18)–(20) shows that the transition rates $\{\mu^{ij}\}$ enter only through F , B , and ϕ .

When $\phi = 0$, (18) gives $\mathbf{n}_t = B \mathbf{n}_{t-1}$ with $\mathbf{n}_{-1} = \mathbf{0}$, so $\mathbf{n}_t = \mathbf{0}$ for all $t \geq 0$.

When $\phi \rightarrow \infty$, boundedness of $\mathbf{n}_t$ in (18) requires $(I - BF) \mathbf{v}_t \rightarrow \mathbf{0}$. Since BF is row-stochastic and primitive, $\ker(I - BF) = \text{span}\{\mathbf{1}\}$, so $\mathbf{v}_t = c_t \mathbf{1}$ for some scalar c_t . Substituting into the Bellman recursion (19) and using $F \mathbf{1} = \mathbf{1}$ gives $\mathbf{u}_t = m_t \mathbf{1}$, where $m_t \equiv c_t - \beta \mathbb{E}_t[c_{t+1}]$. By (20), $u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$, so $n_{it} = y_t - \hat{p}_{it} - a_{it} - m_t$. Imposing the aggregate constraints $\sum_i \alpha_i n_{it} = 0$ and $\sum_i \alpha_i \hat{p}_{it} = 0$ yields $m_t = y_t - \sum_j \alpha_j a_{jt}$. Substituting back:

$$n_{it} = -\hat{p}_{it} - (a_{it} - \sum_j \alpha_j a_{jt}) = -(\hat{p}_{it} + \hat{a}_{it})$$

for all $t \geq 0$. $\square$

Proof of Lemma 1. Under flexible prices with the efficient subsidy, $mc_{it}^{\text{nat}} = p_{it}^{\text{nat}}$, so the log-linearized marginal cost (17) evaluated at the natural allocation gives

$$(1 + \varphi)(\hat{p}_{it}^{\text{nat}} + a_{it}) = (\gamma + \varphi) y_t^{\text{nat}} - \varphi n_{it}^{\text{nat}}. \quad (\text{C.1})$$

Part (i). Weighting (C.1) by α_i and summing: the left-hand side becomes $(1 + \varphi) \sum_i \alpha_i a_{it}$ (using $\sum_i \alpha_i \hat{p}_{it}^{\text{nat}} = 0$), and the right-hand side becomes $(\gamma + \varphi) y_t^{\text{nat}}$ (using $\sum_i \alpha_i n_{it}^{\text{nat}} = 0$). Solving gives

$$y_t^{\text{nat}} = \frac{1 + \varphi}{\gamma + \varphi} \sum_i \alpha_i a_{it}.$$

This depends only on aggregate productivity: neither $\mathbf{n}_{t-1}$ nor $\mathbf{n}_t^{\text{nat}}$ appears. The natural real rate follows from the Euler equation (7) evaluated at the natural allocation.

Part (ii). Solving (C.1) for $\hat{p}_{it}^{\text{nat}}$ and substituting y_t^{nat} :

$$\hat{p}_{it}^{\text{nat}} = \sum_j \alpha_j a_{jt} - a_{it} - \frac{\varphi}{1 + \varphi} n_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat},0} - \frac{\varphi}{1 + \varphi} n_{it}^{\text{nat}}.$$

Sectoral hours follow from the production function and goods market clearing ($l_{it}^{\text{nat}} = y_t^{\text{nat}} -$

$\hat{p}_{it}^{\text{nat}} - a_{it} - n_{it}^{\text{nat}})$:

$$l_{it}^{\text{nat}} = \frac{1 - \gamma}{1 + \varphi} y_t^{\text{nat}} - \frac{1}{1 + \varphi} n_{it}^{\text{nat}}.$$

It remains to characterize $\mathbf{n}_t^{\text{nat}}$. The flex-price flow utility, from (20) evaluated at the natural allocation, is $u_{it}^{\text{nat}} = y_t^{\text{nat}} - \hat{p}_{it}^{\text{nat}} - a_{it} - n_{it}^{\text{nat}}$, which coincides with l_{it}^{nat} above. The Bellman recursion (19) evaluated at the natural allocation gives

$$\mathbf{v}_t^{\text{nat}} = \mathbf{u}_t^{\text{nat}} + \beta F \mathbb{E}_t \mathbf{v}_{t+1}^{\text{nat}},$$

and the law of motion (18) gives

$$\mathbf{n}_t^{\text{nat}} = B \mathbf{n}_{t-1} + \phi(I - BF) \mathbf{v}_t^{\text{nat}},$$

with $\mathbf{n}_{t-1}$ the actual sticky-price predetermined distribution. Together, these determine $\mathbf{n}_t^{\text{nat}}$ given $\mathbf{n}_{t-1}$ and the path of productivity. $\square$

Proof of Proposition 2. We write the conditional supply block as a linear rational-expectations system, solve it via Blanchard and Kahn (1980), and reduce it to a law of motion for the predetermined state.

All variables are N -vectors constrained to the relative subspace $\mathcal{R} \equiv \{x \in \mathbb{R}^N : \boldsymbol{\alpha}^\top x = 0\}$; equivalently, after choosing a basis for $\mathcal{R}$, all matrices below act on $\mathcal{R}$ -valued vectors. Let $M \equiv I - \mathbf{1}\boldsymbol{\alpha}^\top$ and $\mathbf{K} \equiv \text{diag}(\kappa_i)$.

First, let's reduce $\mathbf{n}_t$. Under the contemporaneous reallocation timing, the employment law of motion (18) gives

$$\mathbf{n}_t = B \mathbf{n}_{t-1} + \phi(I - BF) \mathbf{v}_t, \tag{C.2}$$

and the value recursion (19) gives

$$\mathbf{v}_t = \mathbf{u}_t + \beta F \mathbb{E}_t[\mathbf{v}_{t+1}]. \tag{C.3}$$

Equation (C.2) pins down $\mathbf{n}_t$ as a static function of the predetermined $\mathbf{n}_{t-1}$ and the contemporaneous forward-looking $\mathbf{v}_t$: $\mathbf{n}_t$ is not a separate forward-looking variable but a jump that inherits its forward-looking content from $\mathbf{v}_t$.

Next, we can build the system in $(\hat{\boldsymbol{\pi}}_t, \mathbf{v}_t)$. Flow utility on the relative subspace is $\mathbf{u}_t = -\boldsymbol{\chi}_t - \mathbf{n}_t$. Substituting $\boldsymbol{\chi}_t = \boldsymbol{\chi}_{t-1} + \hat{\boldsymbol{\pi}}_t + \Delta \hat{\mathbf{a}}_t$ and (C.2) into (C.3) and rearranging,

$$[I + \phi(I - BF)] \mathbf{v}_t - \beta F \mathbb{E}_t[\mathbf{v}_{t+1}] = -\boldsymbol{\chi}_{t-1} - \hat{\boldsymbol{\pi}}_t - B \mathbf{n}_{t-1} - \Delta \hat{\mathbf{a}}_t. \tag{C.4}$$

The demeaned sectoral Phillips curve is $\hat{\boldsymbol{\pi}}_t = M\mathbf{K}[(\gamma + \varphi)\tilde{y}_t \mathbf{1} - (1 + \varphi)\boldsymbol{\chi}_t - \varphi \mathbf{n}_t] + \beta \mathbb{E}_t \hat{\boldsymbol{\pi}}_{t+1}$.

Substituting χ_t and $\mathbf{n}_t$ as above,

$$\begin{aligned} [I + (1 + \varphi)MK] \hat{\pi}_t - \beta \mathbb{E}_t \hat{\pi}_{t+1} = & -(1 + \varphi) MK \chi_{t-1} - \varphi MK B \mathbf{n}_{t-1} \\ & - \varphi \phi MK (I - BF) \mathbf{v}_t \\ & + (\gamma + \varphi) MK \mathbf{1} \tilde{y}_t - (1 + \varphi) MK \Delta \hat{\mathbf{a}}_t. \end{aligned} \quad (\text{C.5})$$

Stack predetermined variables as $\mathbf{x}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$ and forward-looking variables as $\mathbf{z}_t = (\hat{\pi}_t, \mathbf{v}_t)$. Equations (C.4)–(C.5) together with $\chi_t = \chi_{t-1} + \hat{\pi}_t + \Delta \hat{\mathbf{a}}_t$ and (C.2) yield

$$\mathcal{A} \mathbb{E}_t \begin{bmatrix} \mathbf{x}_{t+1} \\ \mathbf{z}_{t+1} \end{bmatrix} = \mathcal{B} \begin{bmatrix} \mathbf{x}_t \\ \mathbf{z}_t \end{bmatrix} + \mathcal{C}_y \tilde{y}_t + \mathcal{C}_a \Delta \hat{\mathbf{a}}_t, \quad (\text{C.6})$$

with

$$\begin{aligned} \mathcal{A} = \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & \beta I & 0 \\ 0 & 0 & 0 & \beta F \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} I & 0 & I & 0 \\ 0 & B & 0 & \phi(I - BF) \\ (1 + \varphi)MK & \varphi MK B & I + (1 + \varphi)MK & \varphi \phi MK (I - BF) \\ I & B & I & I + \phi(I - BF) \end{bmatrix}, \\ \mathcal{C}_y = \begin{bmatrix} 0 \\ 0 \\ -(\gamma + \varphi)MK \mathbf{1} \\ 0 \end{bmatrix}, \quad \mathcal{C}_a = \begin{bmatrix} I \\ 0 \\ (1 + \varphi)MK \\ I \end{bmatrix}. \end{aligned}$$

The Blanchard–Kahn solution expresses the forward-looking variables as linear functions of the predetermined state. Write $\hat{\pi}_t = \Theta_{11} \chi_{t-1} + \Theta_{12} \mathbf{n}_{t-1} + (\text{exog})$ and $\mathbf{v}_t = \Theta_{21} \chi_{t-1} + \Theta_{22} \mathbf{n}_{t-1} + (\text{exog})$, where $\{\Theta_{kl}\}$ are determined by the stable eigenspace of $(\mathcal{B}, \mathcal{A})$. Plugging these into the state equations:

$$\begin{aligned} \chi_t &= \chi_{t-1} + \hat{\pi}_t + \Delta \hat{\mathbf{a}}_t = (I + \Theta_{11}) \chi_{t-1} + \Theta_{12} \mathbf{n}_{t-1} + (\text{exog}), \\ \mathbf{n}_t &= B \mathbf{n}_{t-1} + \phi(I - BF) \mathbf{v}_t = \phi(I - BF) \Theta_{21} \chi_{t-1} + [B + \phi(I - BF) \Theta_{22}] \mathbf{n}_{t-1} + (\text{exog}). \end{aligned}$$

Shifting forward one period, $\mathbf{x}_{t+1} = \mathbf{P} \mathbf{x}_t + \mathbf{b}_t^y + \mathbf{b}_t^a$ with

$$\mathbf{P} = \begin{bmatrix} I + \Theta_{11} & \Theta_{12} \\ \phi(I - BF) \Theta_{21} & B + \phi(I - BF) \Theta_{22} \end{bmatrix}. \quad (\text{C.7})$$

The observation equation (31) is the first row of the policy function $(\hat{\pi}_t, \mathbf{v}_t) = \Theta(\chi_{t-1}, \mathbf{n}_{t-1}) + \text{exog}$. Stability of $\mathbf{P}$ follows from the Blanchard–Kahn condition on $(\mathcal{A}, \mathcal{B})$.

Because $\mathbf{P}$ is a function of the stable eigenspace of $(\mathcal{B}, \mathcal{A})$, it inherits their parameter depen-

dence: $\mathbf{P}$ depends on $(\mathbf{K}, \varphi, \beta, \phi, F, B)$ but not on γ , which appears only in the aggregate demand loading $\mathcal{C}_y$ and therefore affects only the forcing sequences $\{\mathbf{H}_j^y\}$. $\square$

Proof of Corollary 1. If $\kappa_i = \kappa$ for all i , then $M\mathbf{K}\mathbf{1} = \kappa M\mathbf{1} = \mathbf{0}$, so $\tilde{y}_t$ drops out of the demeaned Phillips curve (C.5). The value recursion (C.4) contains no $\tilde{y}_t$ either. Therefore $\tilde{y}_{t+j}$ does not appear in any expectational equation of the relative block, and $\mathbf{H}_j^y = \mathbf{0}$ for all $j \geq 0$. $\square$

Proof of Proposition 3. The permanent productivity shock translates to $\Delta \hat{\mathbf{a}}_t = \mathbf{x}_j$ at $t = 0$ and $\mathbf{0}$ for $t \geq 1$, where $\mathbf{x}_j \equiv \mathbf{e}_j - \alpha_j \mathbf{1}$.

Part (i): Transition path. By Proposition 2 with $\tilde{y}_t = 0$ and $\mathbf{s}_0 = \mathbf{0}$, the state at $t = 1$ is

$$\mathbf{s}_1 = \mathbf{H}_0^a \mathbf{x}_j.$$

For $t \geq 1$, the state evolves as $\mathbf{s}_{t+1} = \mathbf{P} \mathbf{s}_t$. Expanding $\mathbf{s}_1$ in the eigenbasis using $\mathbf{s}_1 = \sum_h (\mathbf{v}_h^\top \mathbf{s}_1) \mathbf{u}_h = \sum_h c_h^{(j)} \mathbf{u}_h$:

$$\mathbf{s}_t = \mathbf{P}^{t-1} \mathbf{s}_1 = \sum_{h=1}^{2(N-1)} \lambda_h^{t-1} c_h^{(j)} \mathbf{u}_h.$$

Relative inflation for $t \geq 1$ follows from $\hat{\pi}_t = \Theta \mathbf{s}_t$ (with no contemporaneous forcing):

$$\hat{\pi}_t^{(j)} = \sum_h \lambda_h^{t-1} c_h^{(j)} \Theta \mathbf{u}_h = \sum_h \lambda_h^{t-1} c_h^{(j)} \mathbf{q}_h.$$

Impact inflation $\hat{\pi}_0^{(j)} = \Pi_0^a \mathbf{x}_j$ follows from (31) evaluated at $t = 0$ with $\mathbf{s}_0 = \mathbf{0}$.

Part (ii): Half-life. Along corridor h , the scalar coefficient of $\mathbf{u}_h$ in $\mathbf{s}_t$ is $\lambda_h^{t-1} c_h^{(j)}$ for $t \geq 1$. Its magnitude relative to $t = 1$ is $|\lambda_h|^{t-1}$. The half-life is the smallest $m \geq 1$ such that $|\lambda_h|^{m-1} \leq 1/2$, giving $m - 1 \geq \log(1/2) / \log |\lambda_h|$ (the inequality flips because $|\lambda_h| < 1$ makes the log negative), hence $\tau_h^{1/2} = \lceil \log(1/2) / \log |\lambda_h| \rceil$.

Part (iii): Long-run neutrality. Consider the flex-price counterfactual with the same shock. Under flexible prices, relative prices equal $\hat{p}_{it}^{\text{nat}}$ and satisfy $\hat{p}_{it}^{\text{nat}} + a_{it} = \sum_j \alpha_j a_{jt} + O(\mathbf{n}_t^{\text{nat}})$ (Lemma 1); for t large, $\mathbf{n}_t^{\text{nat}} \rightarrow \mathbf{0}$ and relative prices converge to $\hat{p}_{i\infty}^{\text{nat}} = \alpha_j - \delta_{ij}$ (the long-run flex-price value given the permanent shock).

Under sticky prices, $\hat{p}_{it} = \hat{p}_{i,t-1} + \hat{\pi}_{it}$, and from Part (i), $\hat{\pi}_t^{(j)} \rightarrow \mathbf{0}$ as $t \rightarrow \infty$ since $|\lambda_h| < 1$ for all h . Hence $\hat{p}_{it} \rightarrow \hat{p}_{i\infty}$ for some limit, and $\sum_{t=0}^{\infty} \hat{\pi}_{it}^{(j)} = \hat{p}_{i\infty} - \hat{p}_{i,-1} = \hat{p}_{i\infty}$ (steady-state initial

condition). The long-run identity $\hat{p}_{i\infty} = \alpha_j - \delta_{ij}$ then follows because relative prices must equal their flex-price long-run level: any persistent gap would violate $\hat{\pi}_t \rightarrow 0$. $\square$

Proof of Proposition 4. Set $\tilde{y}_t = 0$. From the aggregate Phillips curve (32), $\pi_t^C = u_t^\pi + \beta \mathbb{E}_t \pi_{t+1}^C$ where $u_t^\pi = -(1 + \varphi) \text{Cov}_\alpha(\kappa_i, \chi_{it}) - \varphi \text{Cov}_\alpha(\kappa_i, n_{it})$. Using $\sum_i \alpha_i \chi_{it} = \sum_i \alpha_i n_{it} = 0$, $u_t^\pi = \ell_\pi^\top \mathbf{s}_{t+1}$ with $\ell_\pi = -(\mathbf{D}_\alpha \tilde{\kappa}(1 + \varphi), \mathbf{D}_\alpha \tilde{\kappa} \varphi)$ and $\tilde{\kappa} = \kappa - \bar{\kappa} \mathbf{1}$.

From Proposition 3, $\mathbf{s}_{t+1} = \sum_h \lambda_h^t c_h^{(j)} \mathbf{u}_h$ for $t \geq 0$, so

$$u_t^\pi = \sum_h \lambda_h^t c_h^{(j)} \eta_h, \quad \eta_h \equiv \ell_\pi^\top \mathbf{u}_h = q_h^C,$$

where the identity $\ell_\pi^\top \mathbf{u}_h = q_h^C$ follows by α -weighting the sectoral Phillips curve and matching to $\mathbf{q}_h = \Theta \mathbf{u}_h$.

Iterating $\pi_t^C = u_t^\pi + \beta \mathbb{E}_t \pi_{t+1}^C$ forward: for any $t \geq 0$,

$$\pi_t^C = \pi_0^C \mathbf{1}_{\{t=0, \text{impact}\}} + \sum_{s=0}^{\infty} \beta^s u_{t+s}^\pi = \sum_h q_h^C c_h^{(j)} \frac{\lambda_h^t}{1 - \beta \lambda_h} + (\text{impact at } t = 0).$$

Summing over t and using $\sum_{t \geq 0} \lambda_h^t = 1/(1 - \lambda_h)$:

$$p_\infty^{C,(j)} - p_{-1}^C = \sum_{t=0}^{\infty} \pi_t^C = \pi_0^C + \sum_h \frac{q_h^C c_h^{(j)}}{(1 - \lambda_h)(1 - \beta \lambda_h)},$$

where π_0^C collects the impact contribution $\alpha^\top \Pi_0^a \mathbf{x}_j$.

Parts (i)–(ii): under uniform κ , $\tilde{\kappa} = 0$, so $\ell_\pi = 0$ and $q_h^C = 0$ for all h ; also $\pi_0^C = 0$ by direct computation. Under heterogeneous κ , $\tilde{\kappa} \neq 0$ and generically $\eta_h \neq 0$ for some h . $\square$

Proof of Proposition 5. Substituting $\tilde{y}_t = 0$ into the marginal cost equation (27) and the NKPC (21) gives $\pi_{it} = \kappa[-(1 + \varphi)\chi_{it} - \varphi n_{it}] + \beta \mathbb{E}_t \pi_{it+1}$. Using $\chi_{it} = \chi_{it-1} + \hat{\pi}_{it} + \Delta \hat{a}_{it}$ to eliminate $\hat{\pi}_{it}$, and noting $\Delta \hat{a}_{it} = 0$ for $t \geq 1$:

$$\beta \mathbb{E}_t[\chi_{it+1}] - [1 + \beta + \kappa(1 + \varphi)] \chi_{it} + \chi_{it-1} = \kappa \varphi n_{it}. \quad (\text{C.8})$$

Under segmentation ($\phi = 0$), $n_{it} = 0$ for all $t \geq 0$. Conjecturing $\chi_{it} = \chi_{i0} \lambda^t$ yields the characteristic equation (40) with coefficient $\Gamma_{\text{seg}} \equiv 1 + \beta + \kappa(1 + \varphi)$. Under perfect mobility ($\phi \rightarrow \infty$), $n_{it} = -\chi_{it}$ for all $t \geq 0$, so the right-hand side of (C.8) becomes $-\kappa \varphi \chi_{it}$, reducing the coefficient on χ_{it} to $\Gamma_{\text{int}} \equiv 1 + \beta + \kappa$. The same conjecture yields (41).

Both characteristic equations take the form $f(\lambda; \Gamma) = \beta\lambda^2 - \Gamma\lambda + 1 = 0$. Since $f(0; \Gamma) = 1 > 0$ and $f(1; \Gamma) = \beta + 1 - \Gamma < 0$ (which holds in both cases), there is a unique stable root $\lambda \in (0, 1)$, and the implicit function theorem gives $d\lambda/d\Gamma = \lambda/(2\beta\lambda - \Gamma) < 0$. Since $\Gamma_{\text{seg}} - \Gamma_{\text{int}} = \kappa\varphi > 0$, $\lambda_{\text{seg}} < \lambda_{\text{int}}$, establishing the persistence claims in parts (i) and (ii).

Impact. Evaluate (C.8) at $t = 0$ with $\chi_{i,-1} = 0$ and $\Delta\hat{a}_{i0} = \hat{a}_i$. Under segmentation, $n_{i0} = 0$ and we have $\chi_{it} = \chi_{i0}\lambda_{\text{seg}}^t$, so $\mathbb{E}_0[\chi_{i1}] = \lambda_{\text{seg}}\chi_{i0}$ and

$$\chi_{i0}^{\text{seg}} = \frac{\hat{a}_i}{\Gamma_{\text{seg}} - \beta\lambda_{\text{seg}}} = \hat{a}_i \lambda_{\text{seg}},$$

using the characteristic-equation identity $\Gamma - \beta\lambda = 1/\lambda$. Under perfect mobility, $n_{i0} = -\chi_{i0}$ and the impact equation uses coefficient Γ_{int} ; the analogous calculation gives $\chi_{i0}^{\text{int}} = \hat{a}_i\lambda_{\text{int}}$. Impact inflation is $\hat{\pi}_{i0} = \chi_{i0} - \hat{a}_i = \hat{a}_i(\lambda - 1)$, so $|\hat{\pi}_{i0}^{\text{seg}}| = |\hat{a}_i|(1 - \lambda_{\text{seg}}) > |\hat{a}_i|(1 - \lambda_{\text{int}}) = |\hat{\pi}_{i0}^{\text{int}}|$, establishing the impact.

For the cumulative identity, relative inflation satisfies $\hat{\pi}_{it} = \Delta\chi_{it} - \Delta\hat{a}_{it}$; summing from $t = 0$ to ∞ and telescoping with $\chi_{i,-1} = \hat{a}_{i,-1} = 0$, $\chi_{it} \rightarrow 0$, and $\hat{a}_{it} = \hat{a}_i$ for $t \geq 0$ gives $\sum_{t=0}^{\infty} \hat{\pi}_{it} = -\hat{a}_i$, independent of ϕ . $\square$

C.2 Proofs for Section 4

Proof of Lemma 2. We prove each part in turn.

Part (i): Detailed balance. Evaluating the logit transition probabilities (8) at steady state:

$$F_{ij} = \frac{\exp(\theta[\bar{V}_j - \tau^{ij}])}{\sum_k \exp(\theta[\bar{V}_k - \tau^{ik}])}.$$

Under Assumption 1, $\tau^{ij} = \tau_i^{\text{out}} + \tau_j^{\text{in}} + \hat{\tau}^{ij}$ for $i \neq j$. The origin-specific component τ_i^{out} cancels between numerator and denominator, giving

$$F_{ij} = \frac{\tilde{\theta}_j / e^{\theta\hat{\tau}^{ij}}}{\varphi_i}, \quad \tilde{\theta}_j \equiv e^{\theta(\bar{V}_j - \tau_j^{\text{in}})}, \quad \varphi_i \equiv \sum_k \frac{\tilde{\theta}_k}{e^{\theta\hat{\tau}^{ik}}}.$$

We claim $\bar{N}_i \propto \tilde{\theta}_i \varphi_i$. Computing $\sum_i \bar{N}_i F_{ij}$:

$$\sum_i \bar{N}_i F_{ij} = c \sum_i \tilde{\theta}_i \varphi_i \cdot \frac{\tilde{\theta}_j / e^{\theta\hat{\tau}^{ij}}}{\varphi_i} = c \tilde{\theta}_j \sum_i \tilde{\theta}_i e^{-\theta\hat{\tau}^{ij}} = c \tilde{\theta}_j \varphi_j = \bar{N}_j,$$

where the third equality uses $\hat{\tau}^{ij} = \hat{\tau}^{ji}$. Therefore $\bar{N}_i = \tilde{\theta}_i \varphi_i / \sum_k \tilde{\theta}_k \varphi_k$ is the stationary distribution, and detailed balance holds:

$$\frac{\bar{N}_i F_{ij}}{\bar{N}_j F_{ji}} = \frac{\tilde{\theta}_i \varphi_i}{\tilde{\theta}_j \varphi_j} \cdot \frac{\tilde{\theta}_j e^{-\theta \hat{\tau}^{ij}} / \varphi_i}{\tilde{\theta}_i e^{-\theta \hat{\tau}^{ji}} / \varphi_j} = 1.$$

By (10), $B_{ji} = (\bar{N}_i / \bar{N}_j) F_{ij} = F_{ji}$, so $B = F$.

Part (ii): Spectral properties. Define $Q \equiv A^{1/2} F A^{-1/2}$ where $A = \text{diag}(\alpha_i)$. Detailed balance in matrix form is $AF = F^\top A$, so

$$Q^\top = A^{-1/2} F^\top A^{1/2} = A^{-1/2} (A F A^{-1}) A^{1/2} = A^{1/2} F A^{-1/2} = Q,$$

and Q is real symmetric with real eigenvalues $\rho_1 \geq \dots \geq \rho_N$ and orthonormal eigenvectors $\{s_k\}$. Since Q is similar to F , they share eigenvalues. Row-stochasticity $F\mathbf{1} = \mathbf{1}$ gives $Q(A^{1/2}\mathbf{1}) = A^{1/2}\mathbf{1}$, so $\rho_1 = 1$ with eigenvector $s_1 = A^{1/2}\mathbf{1} / \|A^{1/2}\mathbf{1}\|$. Under the logit specification, $F_{ij} > 0$ for all i, j , so Perron–Frobenius gives $1 = \rho_1 > \rho_2 \geq \dots \geq \rho_N > -1$. $\square$

Proof of Lemma 3. We prove each part in turn.

Part (i): Decoupling. Define the coordinate transformation $x_{kt}^* \equiv s_k^\top A^{1/2} \hat{\mathbf{x}}_t$ for each variable $x \in \{\chi, n, v, \hat{a}\}$, where $\hat{\mathbf{x}}_t$ denotes the demeaned vector (lying in $\mathcal{R}$). Since $s_1 \propto A^{1/2}\mathbf{1}$ and $\alpha^\top \hat{\mathbf{x}}_t = 0$, we have $x_{1t}^* = 0$, so only components $k \geq 2$ are nontrivial.

For any polynomial p , the transformation diagonalizes $p(F)$:

$$s_k^\top A^{1/2} p(F) \hat{\mathbf{x}}_t = s_k^\top p(Q) A^{1/2} \hat{\mathbf{x}}_t = p(\rho_k) x_{kt}^*,$$

using $Q = A^{1/2} F A^{-1/2}$ and $Qs_k = \rho_k s_k$. In particular, F maps to ρ_k and $I - F^2$ to $1 - \rho_k^2$. We apply $s_k^\top A^{1/2}$ to each equation of the supply-side system under $B = F$ and $\kappa_i = \kappa$.

For the NKPC (C.5) with $\tilde{y}_t = 0$ (by block separation, Corollary 1), $M\mathbf{K} = \kappa M$ acts as a scalar on $\mathcal{R}$, so $s_k^\top A^{1/2} M\mathbf{K} \hat{\mathbf{x}}_t = \kappa x_{kt}^*$. Combined with the relative price identity $\chi_{kt}^* = \chi_{k,t-1}^* + \hat{\pi}_{kt}^* + \Delta \hat{a}_{kt}^*$ and eliminating $\hat{\pi}_{kt}^*$, we obtain the NKPC.

Applying $s_k^\top A^{1/2}$ to the employment law of motion $\mathbf{n}_t = F \mathbf{n}_{t-1} + \phi(I - F^2) \mathbf{v}_t$ gives $n_{kt}^* = \rho_k n_{k,t-1}^* + \phi(1 - \rho_k^2) v_{kt}^*$. Applying it to the value recursion $\mathbf{v}_t = -\chi_t - \mathbf{n}_t + \beta F \mathbb{E}_t[\mathbf{v}_{t+1}]$ gives $v_{kt}^* = -\chi_{kt}^* - n_{kt}^* + \beta \rho_k \mathbb{E}_t[v_{k,t+1}^*]$.

Each projected system for component k involves only $(\chi_{kt}^*, n_{kt}^*, v_{kt}^*)$ and the forcing $\hat{a}_{kt}^*$, with no coupling across components. The uniform- κ assumption is essential: with heterogeneous κ_i , the NKPC projection would involve $s_k^\top A^{1/2} \mathbf{K} A^{-1/2} s_{k'} \neq 0$ for $k' \neq k$, coupling the components.

Part (ii): Characteristic polynomial and limits. Conjecture geometric decay: $\chi_{kt}^* = \lambda^t C$, $n_{kt}^* = \lambda^t N$, $v_{kt}^* = \lambda^t V$. Substituting into the three equations:

$$\left[\beta \lambda^2 - (\kappa(1 + \phi) + \beta + 1)\lambda + 1 \right] C = \kappa \phi \lambda N, \quad (\text{C.9})$$

$$(\lambda - \rho_k) N = \phi(1 - \rho_k^2) \lambda V, \quad (\text{C.10})$$

$$(1 - \beta \rho_k \lambda) V = -(C + N). \quad (\text{C.11})$$

Substituting (C.11) into (C.10) to eliminate V , then using the result to express N in terms of C and substituting into (C.9), yields (42) after cancelling $C \neq 0$. The polynomial is quartic: the product of the two brackets is degree four and the additive term is degree two. The component- k system has two predetermined variables ($\chi_{k,t-1}^*$ and $n_{k,t-1}^*$) and two forward-looking variables ($\hat{\pi}_{kt}^*$ and v_{kt}^*), so Blanchard–Kahn requires exactly two stable and two unstable roots.

$\phi \rightarrow 0$. Setting $\phi = 0$ in (42):

$$(\lambda - \rho_k)(1 - \beta \rho_k \lambda) \left[\beta \lambda^2 - (\kappa(1 + \phi) + \beta + 1)\lambda + 1 \right] = 0.$$

The first factor gives $\lambda = \rho_k$ (stable). The second gives $\lambda = 1/(\beta \rho_k)$ (unstable). The quadratic is the segmented-market equation (40), whose roots have product $1/\beta > 1$: one stable root λ_{seg} and one unstable. The two stable roots are $\{\lambda_{\text{seg}}, \rho_k\}$, so $\lambda_k^*(0) = \max\{\lambda_{\text{seg}}, |\rho_k|\}$.

$\phi \rightarrow \infty$. Dividing (42) by ϕ and taking $\phi \rightarrow \infty$, the terms not proportional to ϕ vanish, leaving:

$$(1 - \rho_k^2) \lambda \left[\beta \lambda^2 - (\kappa(1 + \phi) + \beta + 1)\lambda + 1 \right] + \kappa \phi (1 - \rho_k^2) \lambda^2 = 0.$$

Dividing by $(1 - \rho_k^2) \lambda$ simplifies to $\beta \lambda^2 - (\kappa + \beta + 1)\lambda + 1 = 0$, the integrated-market equation (41) with stable root λ_{int} . The root $\lambda = 0$ reflects the collapse of one state variable: as $\phi \rightarrow \infty$, employment fully tracks relative prices ($n_{kt}^* \rightarrow -\chi_{kt}^*$), eliminating one independent degree of freedom.

Roots of a polynomial depend continuously on its coefficients. Since the coefficients of (42) are continuous in ϕ and the stable roots remain bounded away from the unit circle for all finite $\phi \geq 0$, λ_k^* is continuous on $[0, \infty)$. $\square$

Proof of Proposition 6. We proceed in three steps: we derive the effective slope, apply the implicit function theorem to find the eigenvalue shift, and finally determine the sign of the resulting coefficient.

To derive the effective slope, write $\mathbf{K} = \text{diag}(\kappa_i) = \bar{\kappa} I + \varepsilon \text{diag}(\Delta \kappa_i)$. Under uniform $\bar{\kappa}$, the NKPC projection onto component k (Lemma 3) involves the diagonal entry $s_k^\top A^{1/2} \mathbf{K} A^{-1/2} s_k = \bar{\kappa}$.

Under the perturbation, this becomes

$$s_k^\top A^{1/2} \mathbf{K} A^{-1/2} s_{k'} = \bar{\kappa} \mathbf{1}_{k=k'} + \varepsilon \sum_i \Delta \kappa_i s_{k,i} s_{k',i}.$$

Substituting $s_{k,i} = \alpha_i^{1/2} v_{k,i}$, the diagonal entry simplifies to $\bar{\kappa} + \varepsilon \Delta \kappa_k^{\text{eff}}$. The off-diagonal entries are $O(\varepsilon)$, which, by standard non-degenerate perturbation theory, only affect a simple eigenvalue at $O(\varepsilon^2)$. Therefore, to first order, the characteristic polynomial for component k (42) is obtained by simply replacing κ with $\bar{\kappa} + \varepsilon \Delta \kappa_k^{\text{eff}}$.

Next, we evaluate the eigenvalue shift. Assume λ_k^* is a simple stable root of (42) at the baseline; this holds generically throughout the parameter region of interest. Applying the implicit function theorem yields the equation

$$\lambda_k^*(\varepsilon) = \lambda_k^* + \varepsilon \Delta \kappa_k^{\text{eff}} \cdot \Gamma_k + O(\varepsilon^2),$$

with

$$\Gamma_k = - \frac{\partial_\kappa p}{\partial_\lambda p} \Big|_{(\lambda_k^*, \bar{\kappa}, \rho_k)} = \frac{d\lambda_k^*}{d\kappa} \Big|_{\bar{\kappa}}.$$

It remains to determine the sign of Γ_k , which matches the sign of $d\lambda_k^*/d\kappa$. We establish that $d\lambda_k^*/d\kappa < 0$ by examining the two limits of Lemma 3 and extending the result by continuity.

In the limit $\phi = 0$, if $\lambda_{\text{seg}} > |\rho_k|$, then $\lambda_k^*(0) = \lambda_{\text{seg}}$. Implicit differentiation of the segmented-market polynomial $\beta\lambda^2 - [\kappa(1 + \phi) + \beta + 1]\lambda + 1 = 0$ yields

$$\frac{d\lambda_{\text{seg}}}{d\kappa} = \frac{(1 + \phi)\lambda_{\text{seg}}}{2\beta\lambda_{\text{seg}} - [\kappa(1 + \phi) + \beta + 1]} < 0,$$

where the denominator is strictly negative because it equals $-(\partial_\lambda S)(\lambda_{\text{seg}}) < 0$ (following the stable-root argument from Proposition 5). If instead $\lambda_{\text{seg}} \leq |\rho_k|$, then $\lambda_k^*(0) = |\rho_k|$, which is independent of κ . A small increase in ϕ activates the coupling term $T > 0$, ensuring $d\lambda_k^*/d\kappa < 0$ by continuity.

In the limit $\phi \rightarrow \infty$, we have $\lambda_k^*(\infty) = \lambda_{\text{int}}$. Applying the same implicit differentiation to the integrated-market polynomial similarly yields $d\lambda_{\text{int}}/d\kappa < 0$.

For intermediate values of ϕ , the quantity $d\lambda_k^*/d\kappa$ varies continuously along the smooth family of simple stable roots $\lambda_k^*(\phi)$. It cannot change sign without passing through zero, which, according to $d\lambda_k^*/d\kappa = -(\partial_\kappa p)/(\partial_\lambda p)$, would require $\partial_\kappa p|_{\lambda_k^*} = 0$.

We show $\partial_\kappa p|_{\lambda_k^*} \neq 0$. Decompose $p = R \cdot S + T$, where $R(\lambda) = (\lambda - \rho_k)(1 - \beta\rho_k\lambda) + \phi(1 - \rho_k^2)\lambda$, $S(\lambda; \kappa) = \beta\lambda^2 - [\kappa(1 + \phi) + \beta + 1]\lambda + 1$, and $T(\lambda; \kappa) = \kappa\phi\phi(1 - \rho_k^2)\lambda^2$. Notice that only S

and T depend on κ . Using the fact that $p(\lambda_k^*) = 0$ implies $T(\lambda_k^*) = -R(\lambda_k^*)S(\lambda_k^*)$, we find:

$$\partial_\kappa p|_{\lambda_k^*} = -(1 + \varphi)\lambda_k^* R(\lambda_k^*) + \frac{T(\lambda_k^*)}{\kappa} = -\frac{R(\lambda_k^*)}{\kappa} [\kappa(1 + \varphi)\lambda_k^* + S(\lambda_k^*)].$$

The term in brackets simplifies to: $\kappa(1 + \varphi)\lambda_k^* + S(\lambda_k^*) = \beta(\lambda_k^*)^2 - (\beta + 1)\lambda_k^* + 1 = (\lambda_k^* - 1)(\beta\lambda_k^* - 1) > 0$, since $\lambda_k^* \in (0, 1)$ and $\beta < 1$. It then remains to show that $R(\lambda_k^*) \neq 0$. Evaluating p at λ_{seg} gives $p(\lambda_{\text{seg}}) = R(\lambda_{\text{seg}}) \cdot 0 + T(\lambda_{\text{seg}}) = T(\lambda_{\text{seg}}) > 0$ for $\phi > 0$, implying $\lambda_k^* \neq \lambda_{\text{seg}}$. Combining this with $T(\lambda_k^*) > 0$ and $R \cdot S = -T < 0$, we conclude that R and S must have opposite, nonzero signs at λ_k^* . Thus, $R(\lambda_k^*) \neq 0$, meaning $\partial_\kappa p|_{\lambda_k^*} \neq 0$ for all $\phi > 0$, and $d\lambda_k^*/d\kappa$ cannot change sign.

Combining these observations, $\Gamma_k = d\lambda_k^*/d\kappa < 0$ for all $\phi \geq 0$, with strict inequality when either $\phi > 0$ or $\lambda_{\text{seg}} > |\rho_k|$. Therefore, a negative $\Delta\kappa_k^{\text{eff}}$ strictly increases persistence. $\square$

C.3 Proofs and Derivations for Section 5

C.3.1 Proof of Proposition 9

This appendix derives the first-order conditions of the Ramsey problem (Proposition 8) and shows how they imply the targeting criterion (51).

Collect all N sectors into vectors $\boldsymbol{\pi}_t, \boldsymbol{\chi}_t, \mathbf{n}_t, \mathbf{v}_t \in \mathbb{R}^N$ and define the diagonal matrices

$$A \equiv \text{diag}(\alpha_i), \quad K \equiv \text{diag}(\kappa_i), \quad D_\pi \equiv \text{diag}\left(\frac{\alpha_i \epsilon_i}{\kappa_i}\right),$$

together with $M \equiv I - \mathbf{1}\boldsymbol{\alpha}^\top$, which removes the aggregate component: $M\boldsymbol{\pi}_t = \boldsymbol{\pi}_t - \pi_t^c \mathbf{1}$. Denote by $\lambda_t, \boldsymbol{\eta}_t, \boldsymbol{\psi}_t$, and $\boldsymbol{\xi}_t$ the multipliers on the Phillips curve, relative-price identity, employment law of motion, and value recursion, respectively. We attach $\boldsymbol{\eta}_t$ to the welfare-weighted relative-price identity $A[\boldsymbol{\chi}_t - \boldsymbol{\chi}_{t-1} - M\boldsymbol{\pi}_t - \Delta\hat{\mathbf{a}}_t] = \mathbf{0}$, so η_{it} pairs naturally with the welfare weight α_i in the targeting criterion.

The constraints (47)–(50) can be written as

$$\boldsymbol{\pi}_t - \beta \mathbb{E}_t \boldsymbol{\pi}_{t+1} - K[(\gamma + \varphi) \mathbf{1} \tilde{y}_t - (1 + \varphi) \boldsymbol{\chi}_t - \varphi \mathbf{n}_t] = \mathbf{0}, \quad [\lambda_t] \quad (\text{C.12})$$

$$A[\boldsymbol{\chi}_t - \boldsymbol{\chi}_{t-1} - M\boldsymbol{\pi}_t - \Delta\hat{\mathbf{a}}_t] = \mathbf{0}, \quad [\boldsymbol{\eta}_t] \quad (\text{C.13})$$

$$\mathbf{n}_t - B \mathbf{n}_{t-1} - \phi(I - BF) \mathbf{v}_t = \mathbf{0}, \quad [\boldsymbol{\psi}_t] \quad (\text{C.14})$$

$$\mathbf{v}_t + \boldsymbol{\chi}_t + \mathbf{n}_t - \beta F \mathbb{E}_t \mathbf{v}_{t+1} = \mathbf{0}. \quad [\boldsymbol{\xi}_t] \quad (\text{C.15})$$

Both $\mathbf{n}_t$ and $\mathbf{v}_t$ are within-period endogenous variables: $\mathbf{n}_t$ is pinned down by (C.14) given the predetermined $\mathbf{n}_{t-1}$ and the contemporaneous $\mathbf{v}_t$, and $\mathbf{v}_t$ is pinned down by (C.15) given $\boldsymbol{\chi}_t, \mathbf{n}_t$, and the expected continuation.

The planner minimizes (45) subject to (C.12)–(C.15). The date- t contribution to the Lagrangian is

$$\begin{aligned}\mathcal{L}_t = & \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \frac{1}{2} \boldsymbol{\pi}_t^\top D_\pi \boldsymbol{\pi}_t + \frac{1}{2} (1 + \varphi) \boldsymbol{\chi}_t^\top A \boldsymbol{\chi}_t + \varphi \boldsymbol{\chi}_t^\top A \mathbf{n}_t + \frac{1}{2} \varphi \mathbf{n}_t^\top A \mathbf{n}_t \\ & + \boldsymbol{\lambda}_t^\top [\boldsymbol{\pi}_t - \beta \mathbb{E}_t \boldsymbol{\pi}_{t+1} - K((\gamma + \varphi) \mathbf{1} \tilde{y}_t - (1 + \varphi) \boldsymbol{\chi}_t - \varphi \mathbf{n}_t)] \\ & + \boldsymbol{\eta}_t^\top A [\boldsymbol{\chi}_t - \boldsymbol{\chi}_{t-1} - M \boldsymbol{\pi}_t - \Delta \hat{\mathbf{a}}_t] \\ & + \boldsymbol{\psi}_t^\top [\mathbf{n}_t - B \mathbf{n}_{t-1} - \phi(I - BF) \mathbf{v}_t] \\ & + \boldsymbol{\xi}_t^\top [\mathbf{v}_t + \boldsymbol{\chi}_t + \mathbf{n}_t - \beta F \mathbb{E}_t \mathbf{v}_{t+1}].\end{aligned}$$

First-order conditions. Differentiating $\sum_{t=0}^{\infty} \beta^t \mathcal{L}_t$ with respect to each choice variable yields:

Output gap $\tilde{y}_t$. Appears in the loss and in the Phillips curve constraint:

$$\tilde{y}_t = \mathbf{1}^\top K \boldsymbol{\lambda}_t. \quad (\text{F}_{\tilde{y}})$$

Sectoral inflation $\boldsymbol{\pi}_t$. Appears in the date- t loss and Phillips curve, in the date- $(t - 1)$ Phillips curve through $\beta \mathbb{E}_{t-1} \boldsymbol{\pi}_t$, and in the date- t relative price identity:

$$D_\pi \boldsymbol{\pi}_t + \boldsymbol{\lambda}_t - \boldsymbol{\lambda}_{t-1} - M^\top A \boldsymbol{\eta}_t = \mathbf{0}. \quad (\text{F}_\pi)$$

The term $-\boldsymbol{\lambda}_{t-1}$ is the commitment term: current inflation appears in last period's Phillips curve, and the planner internalizes this dependence.

Relative prices $\boldsymbol{\chi}_t$. Appear in the loss, the Phillips curve, the relative price identities at dates t and $t + 1$, and the value recursion at t :

$$A[(1 + \varphi) \boldsymbol{\chi}_t + \varphi \mathbf{n}_t] + (1 + \varphi) K \boldsymbol{\lambda}_t + A \boldsymbol{\eta}_t - \beta \mathbb{E}_t A \boldsymbol{\eta}_{t+1} + \boldsymbol{\xi}_t = \mathbf{0}. \quad (\text{F}_\chi)$$

The first term is the direct marginal welfare cost of relative-price misalignment; the second is the cost-push effect on sectoral Phillips curves; $A[\boldsymbol{\eta}_t - \beta \mathbb{E}_t \boldsymbol{\eta}_{t+1}]$ is the net shadow value of adjusting the relative-price state; $\boldsymbol{\xi}_t$ reflects the effect on workers' reallocation incentives through the value recursion.

Employment $\mathbf{n}_t$. Appears in the loss, the Phillips curve at t , the value recursion at t , the employment law of motion at t (as the left-hand side), and the employment law of motion at $t + 1$ (as the predetermined state):

$$A\varphi(\boldsymbol{\chi}_t + \mathbf{n}_t) + \varphi K \boldsymbol{\lambda}_t + \boldsymbol{\xi}_t + \boldsymbol{\psi}_t - \beta B^\top \mathbb{E}_t \boldsymbol{\psi}_{t+1} = \mathbf{0}. \quad (\text{F}_n)$$

The term $\phi K \lambda_t$ captures the shadow value of the employment distribution through the sectoral Phillips curves: because $\mathbf{n}_t$ enters marginal costs, employment misallocation matters through pricing distortions as well as directly through the welfare loss. The net shadow value $\psi_t - \beta B^\top \mathbb{E}_t \psi_{t+1}$ captures the dynamic effect of $\mathbf{n}_t$ on tomorrow's predetermined employment state through B .

Continuation values $\mathbf{v}_t$. $\mathbf{v}_t$ is a within-period choice variable that appears in the employment law of motion at t , the value recursion at t , and the value recursion at $t - 1$ (through $\beta F \mathbb{E}_{t-1} \mathbf{v}_t$):

$$\xi_t - F^\top \xi_{t-1} - \phi(I - BF)^\top \psi_t = 0. \quad (\text{F}_v)$$

The lagged multiplier $F^\top \xi_{t-1}$ introduces history dependence: under commitment, current policy internalizes that yesterday's promised continuation values shape today's reallocation margin.

Derivation of the targeting criterion. We now derive equation (51). Pre-multiply (F $_\pi$) by $\mathbf{1}^\top K$:

$$\mathbf{1}^\top K D_\pi \pi_t + \mathbf{1}^\top K \lambda_t - \mathbf{1}^\top K \lambda_{t-1} - \mathbf{1}^\top K M^\top A \eta_t = 0. \quad (\text{C.16})$$

We evaluate each term. Since K and D_π are both diagonal,

$$\mathbf{1}^\top K D_\pi \pi_t = \sum_i \alpha_i \epsilon_i \pi_{it}.$$

From (F $_{\tilde{y}}$), $\mathbf{1}^\top K \lambda_t = \tilde{y}_t$ and $\mathbf{1}^\top K \lambda_{t-1} = \tilde{y}_{t-1}$. Finally, $\mathbf{1}^\top K M^\top = (MK\mathbf{1})^\top$ and

$$MK\mathbf{1} = K\mathbf{1} - \mathbf{1} \alpha^\top K\mathbf{1} = K\mathbf{1} - \bar{\kappa} \mathbf{1},$$

where $\bar{\kappa} \equiv \alpha^\top K\mathbf{1} = \sum_i \alpha_i \kappa_i$, so

$$\mathbf{1}^\top K M^\top A \eta_t = (MK\mathbf{1})^\top A \eta_t = \sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it}.$$

Substituting into (C.16):

$$\sum_i \alpha_i \epsilon_i \pi_{it} + \tilde{y}_t - \tilde{y}_{t-1} - \sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it} = 0,$$

which rearranges to (51).

Under the timeless perspective, the planner commits to the policy rule that would have been optimal had it been adopted in the infinite past. The lagged multipliers (λ_{-1}, ξ_{-1}) are initialized at their unconditional steady-state values, so that the targeting rule (51) holds at every date including $t = 0$. $\square$

C.3.2 Static-Discretion Derivations

This appendix derives the static-discretionary rule summarized in Section 5.3. The discretionary problem assumes (i) the central bank chooses $\tilde{y}_t$ taking inherited states $(\chi_{t-1}, \mathbf{n}_{t-1})$ and expected continuation terms $\mathbb{E}_t \pi_{i,t+1}$ and $\mathbb{E}_t \mathbf{v}_{t+1}$ as given, and (ii) workers reallocate within the period to equalize value differentials.

Static reallocation block. Under contemporaneous reallocation, the value recursion (19) reduces to a one-period condition treating expected continuation values as exogenous. From the employment law of motion (18) and the value recursion, eliminating $\mathbf{v}_t$ via $u_{it} = -\chi_{it} - n_{it}$ from (20) gives, after the within-period equilibrium reduces to a scalar condition on each component,

$$n_{it} = -\frac{\phi}{1+\phi} \chi_{it}, \quad \tilde{l}_{it} = \tilde{y}_t - \frac{1}{1+\phi} \chi_{it}. \quad (\text{C.17})$$

A fraction $\phi/(1+\phi)$ of the relative-price misalignment is absorbed by employment flows; the residual $1/(1+\phi)$ falls on hours per worker. The two limits are: $\phi = 0$ gives $n_{it} = 0$ (segmented markets, all adjustment on hours); $\phi \rightarrow \infty$ gives $n_{it} = -\chi_{it}$ (full mobility, all adjustment on employment), recovering the limits of Section 3.4.

Effective marginal cost. Substituting (C.17) into the marginal-cost equation (27):

$$\begin{aligned} mc_{it} - p_{it} &= (\gamma + \phi) \tilde{y}_t - (1 + \phi) \chi_{it} - \phi n_{it} \\ &= (\gamma + \phi) \tilde{y}_t - (1 + \phi) \chi_{it} + \frac{\phi^2}{1 + \phi} \chi_{it} \\ &= (\gamma + \phi) \tilde{y}_t - \sigma \chi_{it}, \quad \sigma \equiv 1 + \frac{\phi}{1 + \phi}. \end{aligned} \quad (\text{C.18})$$

The slope σ summarizes the static reallocation margin in marginal cost: $\sigma = 1 + \phi$ at $\phi = 0$ (segmentation) and $\sigma \rightarrow 1$ as $\phi \rightarrow \infty$ (full mobility).

Impact Phillips curve. Substituting (C.18) into the sectoral NKPC (21),

$$\pi_{it} = \kappa_i [(\gamma + \phi) \tilde{y}_t - \sigma \chi_{it}] + \beta \mathbb{E}_t \pi_{i,t+1}.$$

Using the relative-price identity $\chi_{it} = \chi_{i,t-1} + \pi_{it} - \pi_t^C + \Delta \hat{a}_{it}$ to eliminate χ_{it} on the right-hand side and solving for π_{it} :

$$(1 + \sigma \kappa_i) \pi_{it} = \kappa_i (\gamma + \phi) \tilde{y}_t - \sigma \kappa_i [\chi_{i,t-1} + \Delta \hat{a}_{it} - \pi_t^C] + \beta \mathbb{E}_t \pi_{i,t+1}. \quad (\text{C.19})$$

Define the auxiliary variables $\vartheta_i \equiv \frac{\sigma \kappa_i}{1 + \sigma \kappa_i}$ and $\vartheta^\alpha \equiv \sum_j \alpha_j \vartheta_j$. Taking the α -weighted sum of (C.19) pins down π_t^C in terms of $\tilde{y}_t$ and the inherited terms; substituting back yields

$$\pi_{it} = b_i \tilde{y}_t + u_{it}, \quad b_i \equiv \frac{(\gamma + \varphi) \kappa_i}{(1 + \sigma \kappa_i)(1 - \vartheta^\alpha)}, \quad (\text{C.20})$$

where u_{it} collects inherited misalignment $\chi_{i,t-1}$, sectoral productivity shocks $\Delta \hat{a}_{it}$, and expected continuation inflation $\mathbb{E}_t \pi_{i,t+1}$:

$$u_{it} = \frac{1}{1 + \sigma \kappa_i} [\beta \mathbb{E}_t \pi_{i,t+1} - \sigma \kappa_i (\chi_{i,t-1} + \Delta \hat{a}_{it})] + (\text{aggregate-correction term}). \quad (\text{C.21})$$

The CPI response is $b^C \equiv \sum_i \alpha_i b_i$. Therefore a unit increase in $\tilde{y}_t$ raises sector i 's inflation by b_i and changes its relative price by $b_i - b^C$.

Static welfare loss. The static welfare loss specializes the dynamic loss (46) by treating expected continuation as exogenous. Substituting $n_{it} = -\frac{\phi}{1+\phi} \chi_{it}$ from (C.17) into the cross-sector terms of (46):

$$\frac{1 + \varphi}{2} \chi_{it}^2 + \varphi \chi_{it} n_{it} + \frac{\varphi}{2} n_{it}^2 = \frac{1 + \varphi}{2} \chi_{it}^2 - \frac{\varphi^2}{1 + \phi} \chi_{it}^2 + \frac{\varphi}{2} \cdot \frac{\phi^2}{(1 + \phi)^2} \chi_{it}^2 = \frac{\sigma}{2} \chi_{it}^2,$$

where we use $\sigma = 1 + \varphi / (1 + \phi)$. The static loss is therefore

$$\mathcal{L}_t^{\text{disc}} = \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2 \kappa_i} \pi_{it}^2 + \frac{\sigma}{2} \sum_i \alpha_i \chi_{it}^2. \quad (\text{C.22})$$

First-order condition. Differentiating (C.22) with respect to $\tilde{y}_t$, using $d\pi_{it}/d\tilde{y}_t = b_i$ from (C.20) and $d\chi_{it}/d\tilde{y}_t = b_i - b^C$ from the relative-price identity:

$$0 = (\gamma + \varphi) \tilde{y}_t + \sum_i \frac{\alpha_i \epsilon_i}{\kappa_i} b_i \pi_{it} + \sigma \sum_i \alpha_i (b_i - b^C) \chi_{it}. \quad (\text{C.23})$$

Rearranging the second sum using $\chi_{it} = \chi_{i,t-1} + \pi_{it} - \pi_t^C + \Delta \hat{a}_{it}$ and collecting terms in current sectoral inflation:

$$0 = (\gamma + \varphi) \tilde{y}_t + \sum_i \alpha_i \omega_i^{\text{disc}} \pi_{it} + c_t, \quad (\text{C.24})$$

where the weights ω_i^{disc} are

$$\omega_i^{\text{disc}} = \underbrace{\frac{\epsilon_i}{\kappa_i} b_i}_{\text{within-sector dispersion}} + \underbrace{\sigma (b_i - b^C)}_{\text{relative-price transmission}}, \quad (\text{C.25})$$

and the intercept c_t collects the inherited and shock terms:

$$c_t = \sigma \sum_i \alpha_i (b_i - b^C) (\chi_{i,t-1} + \Delta \hat{a}_{it}). \quad (\text{C.26})$$

The two-component decomposition of the weights is the static analog of the targeting criterion (51): the first term is the Calvo motive for stabilizing sticky-sector inflation, and the second is the reallocation motive for using aggregate demand to tilt relative prices. Under commitment, $\sigma(b_i - b^C)$ is replaced by the present-value shadow value $(\kappa_i - \bar{\kappa})\eta_{it}$ in the targeting criterion.

C.3.3 Proof of Proposition 10

The wedge in the targeting criterion is the inner product

$$\sum_i \alpha_i (\kappa_i - \bar{\kappa}) \eta_{it} = \mathbf{b}_\kappa^\top \mathbf{z}_t, \quad \mathbf{b}_\kappa \equiv \begin{bmatrix} A(\kappa - \bar{\kappa} \mathbf{1}) \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{z}_t \equiv \begin{bmatrix} \eta_t \\ \mathbf{0} \end{bmatrix},$$

embedded into the state space $\mathcal{R} \times \mathcal{R}$ on which $\mathbf{P}$ acts. The right and left eigenvectors $\{\mathbf{u}_h, \mathbf{v}_h\}$ of $\mathbf{P}$ from Proposition 2, normalized so that $\mathbf{v}_h^\top \mathbf{u}_m = \mathbf{1}\{h = m\}$, decompose any vector pairing as $\mathbf{b}_\kappa^\top \mathbf{z}_t = \sum_h (\mathbf{v}_h^\top \mathbf{b}_\kappa) (\mathbf{u}_h^\top \mathbf{z}_t)$. Setting $d_h^P \equiv \mathbf{v}_h^\top \mathbf{b}_\kappa$ and $\eta_{ht}^P \equiv \mathbf{u}_h^\top \mathbf{z}_t$ gives (55). Complex eigenvalues are paired with their conjugates so the sum is real. $\square$

D Quantitative Section Appendix

D.1 Empirical fit of reversibility

Assumption 1 is imposed for tractability. Its observable implication, established in Lemma 2, is that the steady-state transition matrix is reversible:

$$\alpha_i F_{ij} = \alpha_j F_{ji}, \quad \text{equivalently} \quad Q \equiv A^{1/2} F A^{-1/2} \text{ is symmetric,}$$

where $A = \text{diag}(\alpha_i)$.

We assess this implication using CPS sector-to-sector transition flows. Figure D.1 shows that

reversibility is a close approximation in the data. The asymmetry measure is small,

$$\frac{\|Q - Q^\top\|}{\|Q\|} = 8 \times 10^{-4},$$

and the correlation between $\alpha_i F_{ij}$ and $\alpha_j F_{ji}$ across sector pairs is 0.9962. Thus, while Assumption 1 is not itself directly testable from observed flows, its implied symmetry restriction on labor transitions fits the data closely.

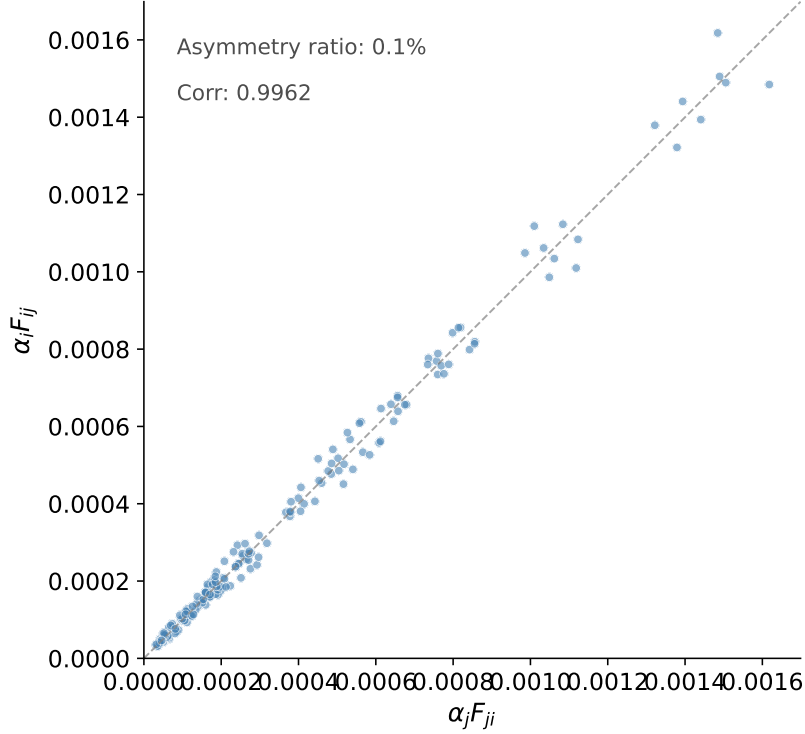


Figure D.1: Empirical fit of reversibility in sectoral labor transitions. The figure plots $\alpha_i F_{ij}$ against $\alpha_j F_{ji}$ using CPS sector-to-sector transition flows. Under reversibility, all observations lie on the 45-degree line. In the data, the asymmetry measure $\|Q - Q^\top\| / \|Q\|$ is 8×10^{-4} and the correlation between $\alpha_i F_{ij}$ and $\alpha_j F_{ji}$ is 0.9962, indicating that the symmetry implication of Lemma 2 is a close approximation empirically.

D.2 Calibrating the Mobility Elasticity

We derive the mapping between the mobility elasticity ϕ in our model and the migration dispersion parameter ν estimated by [Caliendo et al. \(2019\)](#).

Our model. The logit (8) under contemporaneous reallocation timing gives, for $i \neq j$:

$$\log(\mu_t^{ij} / \mu_t^{ii}) = \theta (V_{jt} - V_{it}) - \theta(\tau^{ij} - \tau^{ii}). \quad (\text{D.1})$$

For a temporary change in destination wage w_{jt} , the continuation value adjusts only through the contemporaneous flow utility: $\Delta V_{jt} \approx \Delta U_{jt}^{\text{flow}}$. From the flow utility and the labor supply condition (6):

$$\frac{\partial U_{jt}^{\text{flow}}}{\partial \log w_{jt}} = \zeta \varphi \bar{L}^\varphi \cdot \frac{\bar{L}}{\varphi} = \zeta \bar{L}^{1+\varphi}, \quad (\text{D.2})$$

so the semi-elasticity of relative log flows to log contemporaneous destination wages is

$$\frac{\partial \log(\mu_t^{ij} / \mu_t^{ii})}{\partial \log w_{jt}} = \theta \zeta \bar{L}^{1+\varphi} = \frac{\phi}{\varphi}, \quad (\text{D.3})$$

using $\phi = \varphi \theta \zeta \bar{L}^{1+\varphi}$.

Caliendo et al. (2019) model. In their formulation, workers choose destination at the end of period t and arrive at $t + 1$. The flow payoff is $\log(w_j / p_j)$ and idiosyncratic shocks have dispersion ν , so the logit gives:

$$\log(\mu_t^{ij} / \mu_t^{ii}) = \frac{1}{\nu} \left[\beta \mathbb{E}_t(V_{j,t+1} - V_{i,t+1}) - (\kappa^{ij} - \kappa^{ii}) \right]. \quad (\text{D.4})$$

Since $\partial V_{j,t+1} / \partial \log w_{j,t+1} = 1$ for a temporary wage change, and the destination wage workers care about when choosing μ_t^{ij} is $w_{j,t+1}$, the undiscounted semi-elasticity is $1/\nu$:

$$\frac{\partial \log(\mu_t^{ij} / \mu_t^{ii})}{\partial \log w_{j,t+1}} = \frac{\beta}{\nu}, \quad \left. \frac{\partial \log(\mu_t^{ij} / \mu_t^{ii})}{\partial \log w_{j,t+1}} \right|_{\text{undiscounted}} = \frac{1}{\nu}. \quad (\text{D.5})$$

The factor β reflects one-period discounting between the decision date and the payoff date in CDP's timing; it is absent in our model because reallocation is contemporaneous.

Mapping. Equating undiscounted semi-elasticities—the sensitivity of flow decisions to the destination wage each flow actually earns:

$$\frac{\phi}{\varphi} = \frac{1}{\nu} \implies \phi = \frac{\varphi}{\nu}. \quad (\text{D.6})$$

Caliendo et al. (2019) estimate $\nu = 5.34$ at quarterly frequency. With $\varphi = 2$, this implies $\phi = 2/5.34 \approx 0.37$.

D.3 Solving the Ramsey Problem

This appendix assembles the first-order conditions from Appendix C.3.1 into a linear rational-expectations system suitable for numerical solution via standard Blanchard–Kahn procedures (Blanchard and Kahn, 1980; Klein, 2000).

Unlike χ_t , $\mathbf{n}_t$, and $\mathbf{v}_t$ —which the primitive constraints confine to $\mathcal{R}$ —sectoral inflation π_t carries an endogenous aggregate component $\pi_t^C \equiv \boldsymbol{\alpha}^\top \pi_t$ selected implicitly by the planner through $\tilde{y}_t$. The system therefore retains π_t and its co-state λ_t in $\mathbb{R}^N$, adding three scalar dimensions and one stable eigenvalue relative to a pure relative-block system.

Variable classification. Four predetermined vectors at date t :

$$\mathbf{x}_t \equiv (\chi_{t-1}, \mathbf{n}_{t-1}, \lambda_{t-1}, \boldsymbol{\xi}_{t-1}), \quad \dim \mathbf{x}_t = 4(N-1) + 1.$$

Three live in $\mathcal{R}$ ($\chi_{t-1}, \mathbf{n}_{t-1}, \boldsymbol{\xi}_{t-1}$); the fourth, $\lambda_{t-1} \in \mathbb{R}^N$, carries the aggregate Ramsey commitment $\lambda_{t-1}^C \equiv \mathbf{1}^\top \lambda_{t-1}$.

Eight forward-looking jumps:

$$\mathbf{z}_t \equiv (\pi_t, \chi_t, \mathbf{n}_t, \mathbf{v}_t, \lambda_t, \boldsymbol{\eta}_t, \boldsymbol{\psi}_t, \boldsymbol{\xi}_t), \quad \dim \mathbf{z}_t = 8(N-1) + 2,$$

with $\pi_t, \lambda_t \in \mathbb{R}^N$ and the remaining six jumps in $\mathcal{R}$; the aggregate components are $\pi_t^C = \boldsymbol{\alpha}^\top \pi_t$ and $\lambda_t^C = \mathbf{1}^\top \lambda_t$. The output gap is then pinned down by $F_{\tilde{y}}$: $\tilde{y}_t = \mathbf{1}^\top K \lambda_t$. Stacking, $\mathbf{w}_t \equiv (\mathbf{x}_t, \mathbf{z}_t)$ has dimension $12(N-1) + 3$.

Canonical BK form. The system takes the form

$$\mathcal{A} \mathbb{E}_t \mathbf{w}_{t+1} = \mathcal{B} \mathbf{w}_t + \mathcal{C}_a \Delta \hat{\mathbf{a}}_t, \quad (\text{D.7})$$

assembled from twelve block-rows listed in Table D.1: rows 1–4 are state-identification rows, rows 5–8 are the primitive constraints (C.12)–(C.15), and rows 9–12 are the first-order conditions F_π, F_χ, F_n, F_v from Appendix C.3.1, with $\tilde{y}_t = \mathbf{1}^\top K \lambda_t$ substituted throughout.

Matrices. With $\mathbf{w}_t$ ordered as $(\chi_{t-1}, \mathbf{n}_{t-1}, \lambda_{t-1}, \boldsymbol{\xi}_{t-1}, \pi_t, \chi_t, \mathbf{n}_t, \mathbf{v}_t, \lambda_t, \boldsymbol{\eta}_t, \boldsymbol{\psi}_t, \boldsymbol{\xi}_t)$ (columns 1–12, block-row sizes $N-1, N-1, N, N-1, N, N-1, N-1, N-1, N, N-1, N-1, N-1$), the block-matrix $\mathcal{A}$ is

$$\mathcal{A} = \text{diag}(I_{N-1}, I_{N-1}, I_N, I_{N-1}, \beta I_N, \mathbf{0}, \mathbf{0}, \beta F, \mathbf{0}, \beta I_{N-1}, \beta B^\top, \mathbf{0}).$$

Row	Equation (at date t , with $\tilde{y}_t = \mathbf{1}^\top K\lambda_t$ substituted)	Source	Type	Dim
1	$\mathbb{E}_t \chi_t = \chi_t$	state	static	$N-1$
2	$\mathbb{E}_t \mathbf{n}_t = \mathbf{n}_t$	state	static	$N-1$
3	$\mathbb{E}_t \lambda_t = \lambda_t$	state	static	N
4	$\mathbb{E}_t \xi_t = \xi_t$	state	static	$N-1$
5	$\beta \mathbb{E}_t \pi_{t+1} = \pi_t + (1 + \varphi)K\chi_t + \varphi K\mathbf{n}_t - (\gamma + \varphi)K\mathbf{1}\mathbf{1}^\top K\lambda_t$	(C.12)	forward	N
6	$\mathbf{0} = \chi_t - \chi_{t-1} - M\pi_t - \Delta \hat{\mathbf{a}}_t$	(C.13)	static	$N-1$
7	$\mathbf{0} = \mathbf{n}_t - B\mathbf{n}_{t-1} - \phi(I - BF)\mathbf{v}_t$	(C.14)	static	$N-1$
8	$\beta F \mathbb{E}_t \mathbf{v}_{t+1} = \mathbf{v}_t + \chi_t + \mathbf{n}_t$	(C.15)	forward	$N-1$
9	$\mathbf{0} = D_\pi \pi_t + \lambda_t - \lambda_{t-1} - M^\top \eta_t$	(F $_\pi$)	static	N
10	$\beta \mathbb{E}_t \eta_{t+1} = MA[(1 + \varphi)\chi_t + \varphi \mathbf{n}_t] + (1 + \varphi)MK\lambda_t + \eta_t + \xi_t$	(F $_\chi$)	forward	$N-1$
11	$\beta B^\top \mathbb{E}_t \psi_{t+1} = \varphi MA(\chi_t + \mathbf{n}_t) + \varphi MK\lambda_t + \xi_t + \psi_t$	(F $_n$)	forward	$N-1$
12	$\mathbf{0} = \xi_t - F^\top \xi_{t-1} - \phi(I - BF)^\top \psi_t$	(F $_v$)	static	$N-1$

Table D.1: Rows of the Ramsey BK system (D.7). Rows 3, 5, and 9 are N -dimensional; the rest are $(N - 1)$ -dimensional. Total: $12(N - 1) + 3$.

The matrix $\mathcal{B}$ has nonzero blocks

$$\begin{aligned}
\mathcal{B}_{1,6} &= I_{N-1}, \quad \mathcal{B}_{2,7} = I_{N-1}, \quad \mathcal{B}_{3,9} = I_N, \quad \mathcal{B}_{4,12} = I_{N-1}, \\
\mathcal{B}_{5,5} &= I_N, \quad \mathcal{B}_{5,6} = (1 + \varphi)K, \quad \mathcal{B}_{5,7} = \varphi K, \quad \mathcal{B}_{5,9} = -(\gamma + \varphi)K\mathbf{1}\mathbf{1}^\top K, \\
\mathcal{B}_{6,1} &= -I_{N-1}, \quad \mathcal{B}_{6,5} = -M, \quad \mathcal{B}_{6,6} = I_{N-1}, \\
\mathcal{B}_{7,2} &= -B, \quad \mathcal{B}_{7,7} = I_{N-1}, \quad \mathcal{B}_{7,8} = -\phi(I - BF), \\
\mathcal{B}_{8,6} &= I_{N-1}, \quad \mathcal{B}_{8,7} = I_{N-1}, \quad \mathcal{B}_{8,8} = I_{N-1}, \\
\mathcal{B}_{9,3} &= -I_N, \quad \mathcal{B}_{9,5} = D_\pi, \quad \mathcal{B}_{9,9} = I_N, \quad \mathcal{B}_{9,10} = -M^\top, \\
\mathcal{B}_{10,6} &= (1 + \varphi)MA, \quad \mathcal{B}_{10,7} = \varphi MA, \quad \mathcal{B}_{10,9} = (1 + \varphi)MK, \quad \mathcal{B}_{10,10} = I_{N-1}, \quad \mathcal{B}_{10,12} = I_{N-1}, \\
\mathcal{B}_{11,6} &= \varphi MA, \quad \mathcal{B}_{11,7} = \varphi MA, \quad \mathcal{B}_{11,9} = \varphi MK, \quad \mathcal{B}_{11,11} = I_{N-1}, \quad \mathcal{B}_{11,12} = I_{N-1}, \\
\mathcal{B}_{12,4} &= -F^\top, \quad \mathcal{B}_{12,11} = -\phi(I - BF)^\top, \quad \mathcal{B}_{12,12} = I_{N-1}.
\end{aligned}$$

Blocks mapping between $\mathbb{R}^N$ and $\mathcal{R}$ are rectangular: K, D_π embed $\mathcal{R} \hookrightarrow \mathbb{R}^N$; M, MA, MK project $\mathbb{R}^N \rightarrow \mathcal{R}$. Productivity forcing enters only through row 6: $\mathcal{C}_{a,6} = I_{N-1}$, all other blocks zero.

Determinacy. Blanchard–Kahn requires $(\mathcal{B}, \mathcal{A})$ to have exactly $4(N - 1) + 1$ stable generalized eigenvalues, one per scalar predetermined entry in $\mathbf{x}_t$. When satisfied, the solution takes the

form

$$\mathbf{z}_t = \mathbf{\Theta} \mathbf{x}_t + \sum_{j=0}^{\infty} \mathbf{\Phi}_j \mathbb{E}_t[\Delta \hat{\mathbf{a}}_{t+j}], \quad \mathbf{x}_{t+1} = \mathbf{P}^{\text{Ramsey}} \mathbf{x}_t + (\text{exogenous forcing}), \quad (\text{D.8})$$

where $\mathbf{\Theta}$ is $[8(N-1)+2] \times [4(N-1)+1]$ and $\mathbf{P}^{\text{Ramsey}}$ is $[4(N-1)+1] \times [4(N-1)+1]$.

Extracting quantities of interest. Denoting blocks of $\mathbf{\Theta}$ by $\Theta_\pi, \Theta_\lambda, \Theta_\eta$, optimal sectoral inflation is $\pi_t^* = \Theta_\pi \mathbf{x}_t$ (with aggregate $\pi_t^{C,*} = \boldsymbol{\alpha}^\top \Theta_\pi \mathbf{x}_t$), the output gap is $\tilde{y}_t^* = \mathbf{1}^\top K \Theta_\lambda \mathbf{x}_t$, and the reallocation wedge $(MK\mathbf{1})^\top \Theta_\eta \mathbf{x}_t$ enters the targeting criterion (51). The implementing nominal rate follows from the IS curve (22). Generically $\pi_t^{C,*} \neq 0$; symmetric stickiness ($\kappa_i = \bar{\kappa}$, so $MK\mathbf{1} = \mathbf{0}$) kills the wedge and restores $\pi_t^{C,*} \equiv 0$.

Timeless-perspective initialization. The physical states $(\chi_{-1}, \mathbf{n}_{-1})$ are exogenous initial conditions; the commitment co-states are initialized at their steady-state values, $\boldsymbol{\lambda}_{-1} = \mathbf{0} \in \mathbb{R}^N$ and $\boldsymbol{\xi}_{-1} = \mathbf{0} \in \mathcal{R}$ (Woodford, 2003).

Numerical implementation. The system (D.7) is solved via generalized Schur decomposition (Klein, 2000). Projection of the relative-block components onto $\mathcal{R}$ uses the orthonormal basis of Lemma 2 (eigenvectors $\{s_k\}_{k=2}^N$ of $Q = A^{1/2}FA^{-1/2}$).

E Appendix: Input-Output Linkages

[in construction]